



Signals, Systems and Frequency-Domain Analysis

Lecture notes for the whole course: what a signal is, how systems are classified, why linear time-invariant systems are described by a single function, how a signal is decomposed into exponentials, and what sampling does to the result.

Covers

Chapters 1 to 7, and Appendix A

Level

Undergraduate

Assumed background

Calculus, complex numbers, basic circuits

How to read these notes

Each chapter builds on the one before it, and the seven of them are one argument: a signal is a function, a system is a map on functions, an LTI system is described by one function, and the frequency domain is what turns the operation that uses it into a product. Within a chapter, every idea is introduced in the same order: a picture, a definition, an equation, a short derivation, a worked example, and a warning about the mistake that is easiest to make. Worked examples always use five headings: Given, Find, Method, Solution, Check. Do the Check step yourself before reading it.

Two conventions apply everywhere. First, energy and power are **normalised**: the resistance is taken as $1\ \Omega$, so instantaneous power is $|x|^2$. Second, the imaginary unit is written j .

The contents below carries a third column. An entry such as **OW CH1.1–1.4** points into the course textbook, Oppenheim and Willsky, *Signals and Systems*, second edition, where the same material is developed at length. The **OW** marker is what tells the two apart, because the two numbering systems are not the same: these notes reach the continuous-time Fourier transform in chapter 5, and the textbook reaches it in chapter 4.

1	Signals	OW CH1.1–1.4
	Energy and power. Shifting, reversal and scaling. Periodicity. Impulses and steps. Complex exponentials.	
2	Systems and their properties	OW CH1.5–1.6
	Memory, invertibility, causality, stability, time invariance, linearity.	
3	Linear time-invariant systems	OW CH2.1–2.3
	Impulse response. Convolution sum and convolution integral. Properties of convolution.	
4	Fourier series	OW CH3.1–3.11
	The eigenfunction property. Analysis and synthesis. Existence. Properties. A periodic input through an LTI system.	
5	The continuous-time Fourier transform	OW CH4.1–4.7
	From series to transform. The standard pairs. Every property with its proof. Parseval. Modulation. Differential equations.	
6	The discrete-time Fourier transform	OW CH5.1–5.8
	The same construction in discrete time. Why the spectrum repeats. Periodic convolution. Difference equations.	
7	Sampling and aliasing	OW CH7.1–7.3
	Impulse-train sampling. Replication and the guard band. The sampling theorem. Reconstruction, holds, and aliasing.	

A Summary of formulas

Everything from Chapters 1 to 7 on two pages.

Signals

A signal is information written as a function. This chapter says what that means, how much energy or power a signal carries, how it changes when we shift or stretch the time axis, and when it repeats.

1.1 What a signal is

A signal is a physical quantity that varies and carries information. Mathematically, it is a function of one or more independent variables.

In this course the independent variable is almost always time. It does not have to be. An image is a signal of two space variables, and the same algebra applies.

There are two kinds of signal, and they are kept apart throughout the course.

CONTINUOUS TIME

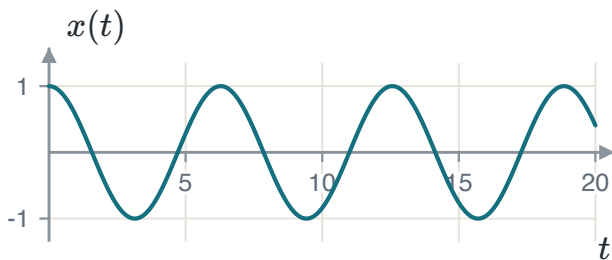
$$x(t), \quad t \in \mathbb{R}$$

Round brackets. The signal has a value at every real instant.

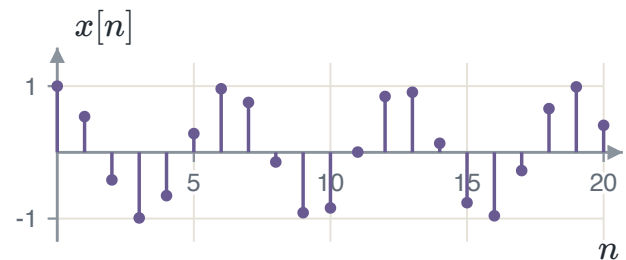
DISCRETE TIME

$$x[n], \quad n \in \mathbb{Z}$$

Square brackets. The signal has a value only at integer indices. Here n is an integer **index**, not a time in seconds.



A continuous-time signal is drawn as an unbroken curve.



A discrete-time signal is drawn with stems. The signal is the dots. There is nothing between them.

NOTATION

Writing $x[t]$ or $x(n)$ is not a small slip. It states the wrong domain, and every later statement about periodicity, convolution limits and transforms then breaks.

1.2 Signal energy and power

Start with a resistor. If $v(t)$ is the voltage across a resistance R , the instantaneous power is

$$p(t) = v(t) i(t) = v(t) \left(\frac{v(t)}{R} \right) = \frac{1}{R} v^2(t).$$

Energy is power multiplied by time. Because the power varies, we integrate:

$$E = \int_{t_1}^{t_2} p(t) dt = \int_{t_1}^{t_2} \frac{1}{R} v^2(t) dt.$$

From here on we set $R = 1 \Omega$ and work with $|x(t)|^2$. This is the normalised convention. Put the factor $1/R$ back whenever a physical number in joules or watts is required.

We use the modulus because a signal may be complex valued, and $|x(t)|^2 = x(t) x^*(t)$ is the quantity that is real and never negative. Writing $x^2(t)$ instead is correct only for real signals.

Total energy

TOTAL ENERGY OVER ALL TIME

$$E_{\infty} = \lim_{T \rightarrow \infty} \int_{-T}^T |x(t)|^2 dt = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$E_{\infty} = \lim_{N \rightarrow \infty} \sum_{n=-N}^N |x[n]|^2 = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

The integral or the sum may not converge. E_{∞} is defined as a limit, and a limit that fails to exist is not a large number. It is the absence of an answer, and that fact is useful.

Average power

If the energy is infinite, ask instead how fast energy arrives. Divide by the length of the window and let the window grow.

AVERAGE POWER OVER ALL TIME

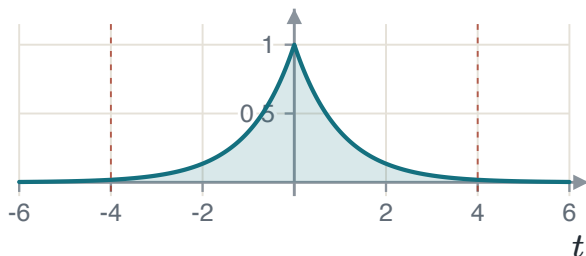
$$P_{\infty} = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt$$

$$P_{\infty} = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$$

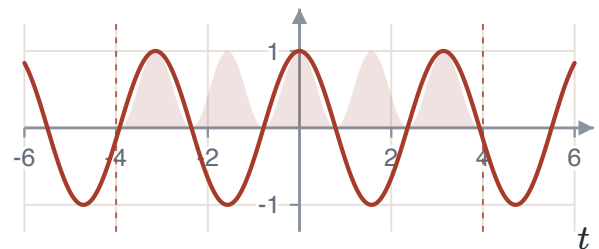
In discrete time the divisor is $2N+1$, because that is the number of samples between $-N$ and $+N$ inclusive.

READ THE DEFINITION CAREFULLY

P_{∞} is not energy divided by infinity. It is the limit of a ratio, and that ratio can settle on any value, including zero.



Converging. As the window grows, the shaded area approaches a finite limit.



Diverging. Every period adds the same area, so $E_{\infty} \rightarrow \infty$ and only the average power is meaningful.

1.3 Energy signals, power signals, and neither

TYPE	CONDITION	TYPICAL SIGNALS
Energy signal	$E_{\infty} < \infty$ and $P_{\infty} = 0$	Pulses, decaying responses, anything that ends
Power signal	$E_{\infty} \rightarrow \infty$ and $0 < P_{\infty} < \infty$	Constants, sinusoids, signals that run forever
Neither	$E_{\infty} \rightarrow \infty$ and $P_{\infty} \rightarrow \infty$	Signals that grow without bound, such as $t u(t)$

The two conditions in each row are not independent. Finite energy forces $P_{\infty} = 0$, because a finite number divided by $2T \rightarrow \infty$ goes to zero. The second condition is a consequence, not a second test.

EXAMPLE 1.1

GIVEN $x(t) = 1$ for $0 \leq t \leq 1$, and $x(t) = 0$ otherwise.

FIND Is this an energy signal or a power signal?

METHOD Find E_∞ first. If it is finite, the classification is settled.

SOLUTION

$$E_\infty = \int_0^1 1^2 dt = 1 < \infty$$

$$P_\infty = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_0^1 1 dt = \lim_{T \rightarrow \infty} \frac{1}{2T} = 0$$

So $x(t)$ is an **energy signal**, with $E_\infty = 1$ J.

CHECK Halve the amplitude. The energy must fall by a factor of four, and $\int_0^1 (1/2)^2 dt = 1/4$. The dependence is quadratic, as it must be for a squared quantity.

EXAMPLE 1.2

GIVEN $x[n] = 4$ for every integer n .

FIND Is this an energy signal or a power signal?

METHOD The energy clearly diverges, so go straight to the average power and count the samples correctly.

SOLUTION

$$E_\infty = \sum_{n=-\infty}^{\infty} |4|^2 \rightarrow \infty$$

$$P_\infty = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N 16 = \lim_{N \rightarrow \infty} \frac{(2N+1) \cdot 16}{2N+1} = 16$$

So $x[n]$ is a **power signal**, with $P_\infty = 16$.

CHECK A constant of amplitude A must have $P_\infty = A^2$. The factor $2N+1$ cancels exactly, which confirms the sample count was right.

COMMON MISTAKE

Concluding that infinite energy means infinite power. The factor $1/(2N+1)$ does the work. A sum that diverges, divided by a count that also diverges, can converge.

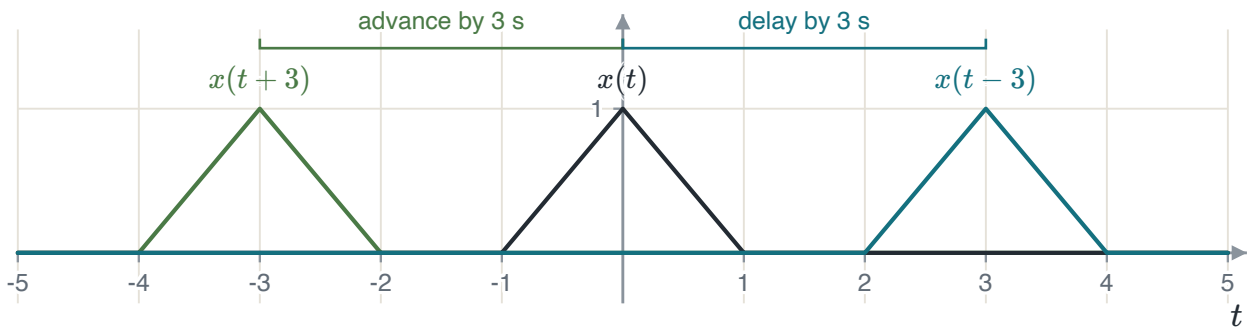
1.4 Time shifting

TIME SHIFT

$$x(t) \longrightarrow x(t - t_0)$$

If $t_0 > 0$ the signal is **delayed** and moves right. If $t_0 < 0$ it is **advanced** and moves left.

The sign feels backwards until you read it as a question about when. The output at time t shows what the input was doing at the earlier time $t - t_0$. The value has been held back by t_0 seconds.



Shifting moves the signal along the time axis. The shape is untouched.

Discrete time is identical: $x[n] \rightarrow x[n - n_0]$ with n_0 an integer. A one-sample delay, $x[n - 1]$, is the basic memory element of every difference equation in Chapter 3.

1.5 Time reversal and time scaling

TIME REVERSAL

$$x(t) \longrightarrow x(-t) \quad (x[n] \rightarrow x[-n])$$

A flip about the vertical axis.

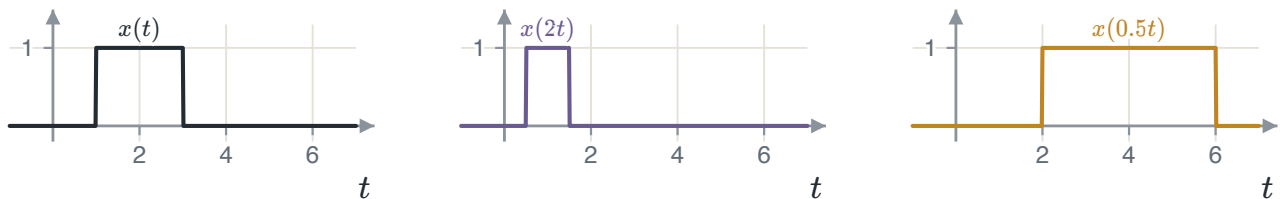
TIME SCALING

$$y(t) = x(at), \quad a > 0$$

If $a > 1$ the signal is compressed and speeded up. If $0 < a < 1$ it is stretched and slowed down.

Write the scaled signal as a new signal $y(t)$. Writing $x(t) = x(at)$ would force $a = 1$.

Scaling changes the support in a predictable way. If x is non-zero only on $[\alpha, \beta]$, then $x(at)$ is non-zero only on $[\alpha/a, \beta/a]$. The width is divided by a .



Compressed: support $[0.5, 1.5]$.

Stretched: support $[2, 6]$.

DISCRETE TIME IS NOT SYMMETRIC

$x[2n]$ throws away the odd-indexed samples. It is real decimation and it destroys information. And $x[n/2]$ is not defined at odd n without an extra rule for filling the gaps. Continuous-time scaling can be undone; discrete-time decimation cannot.

1.6 Combining a shift and a scale

To find $x(at - b)$, do the two operations in this order.

SHIFT, THEN SCALE

$$(1) \quad v(t) = x(t - b) \qquad (2) \quad y(t) = v(at) = x(at - b)$$

Shift by b first. Then scale the result by a .

WHY THE ORDER MATTERS

Scaling first gives $w(t) = x(at)$. Shifting that by b gives $w(t - b) = x(a(t - b)) = x(at - ab)$. Unless $a = 1$ this is a different signal: it is shifted by b instead of by b/a .

EXAMPLE 1.3

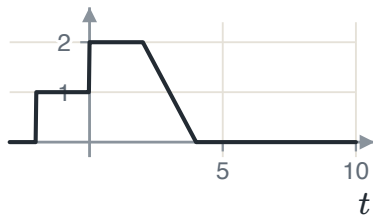
GIVEN $x(t)$ is zero for $t < -2$, equal to 1 on $[-2, 0]$, equal to 2 on $[0, 2]$, and falls linearly from 2 to 0 on $[2, 4]$.

FIND Plot $x(3t - 5)$.

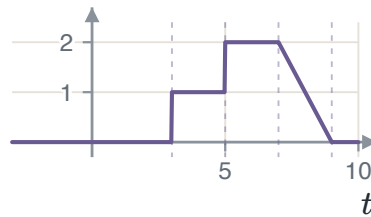
METHOD Here $a = 3$ and $b = 5$. Shift right by 5, then compress by 3.

SOLUTION $v(t) = x(t - 5)$ has corners at $t = 3, 5, 7, 9$. Then $y(t) = v(3t)$ has corners at $t = 1, 5/3, 7/3, 3$.

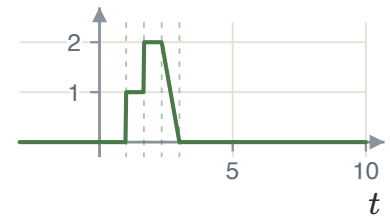
CHECK The total duration must shrink by exactly $a = 3$. The original spans $[-2, 4]$, width 6. The result spans $[1, 3]$, width 2. Also, y reproduces $x(0)$ where $at - b = 0$, that is at $t = b/a = 5/3$, which is where the level 2 begins.



$x(t)$



$v(t) = x(t - 5)$



$y(t) = x(3t - 5)$

1.7 Periodic signals

PERIODICITY

$$x(t) = x(t + T) \quad \text{for all } t \in \mathbb{R}, \text{ for some } T > 0$$

$$x[n] = x[n + N] \quad \text{for all } n \in \mathbb{Z}, \text{ for some integer } N > 0$$

A signal that is not periodic is **aperiodic**.

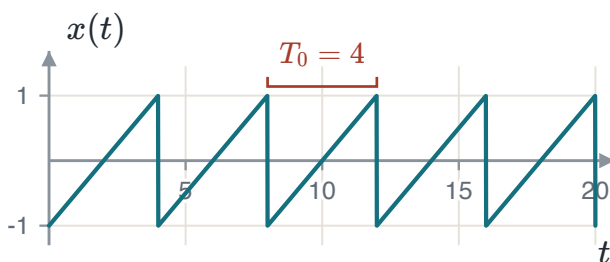
Note both quantifiers. There must **exist** a period, and it must work for **all** t or n . In continuous time T may be any positive real number. In discrete time N must be an integer. There is no period of 3.5 samples. Almost every discrete-time surprise in this course comes from that one line.

If T is a period then so is $2T$, $3T$, and so on. The **fundamental period** T_0 is the smallest positive period. The same applies to N_0 in discrete time.

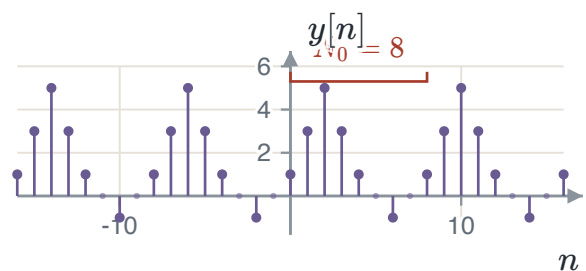
FUNDAMENTAL FREQUENCY

$$\omega_0 = \frac{2\pi}{T_0} \quad \text{and} \quad \omega_0 = \frac{2\pi}{N_0}$$

Without the word *smallest*, ω_0 would not be well defined.



Periods are 4, 8, 12, ... The fundamental period is $T_0 = 4$.



Periods are 8, 16, 24, ... The fundamental period is $N_0 = 8$.

1.8 Even and odd signals

A signal is **even** if $x(t) = x(-t)$, and **odd** if $x(t) = -x(-t)$. The same definitions apply to $x[n]$ with n in place of t .

Setting $t = 0$ in the odd condition gives $x(0) = -x(0)$, so $x(0) = 0$. A signal with $x(0) \neq 0$ cannot be odd. That is a one-second test.

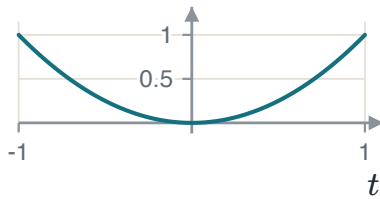
Every signal splits into an even part and an odd part, in exactly one way:

EVEN AND ODD PARTS

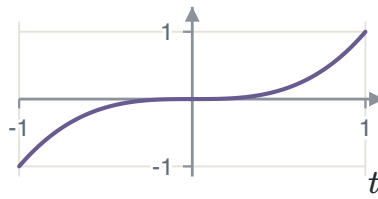
$$\mathcal{E}\{x(t)\} = \frac{1}{2}[x(t) + x(-t)], \quad \mathcal{O}\{x(t)\} = \frac{1}{2}[x(t) - x(-t)]$$

$$x(t) = \mathcal{E}\{x(t)\} + \mathcal{O}\{x(t)\}$$

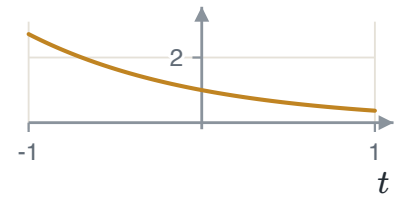
Adding the two definitions returns x exactly.



t^2 is even.



t^3 is odd.



e^{-t} is neither.

1.9 The unit impulse and the unit step in discrete time

DEFINITIONS

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & \text{otherwise} \end{cases}$$

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

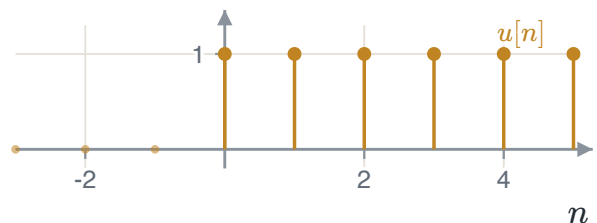
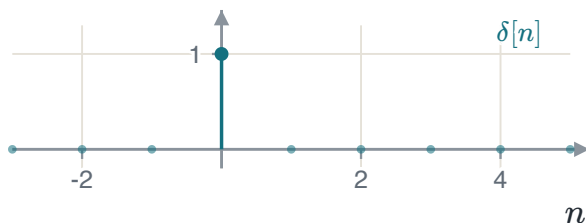
Both are ordinary sequences. Nothing infinite happens here.

They are related by a difference and by a running sum. These are the discrete versions of differentiation and integration.

FIRST DIFFERENCE AND RUNNING SUM

$$\delta[n] = u[n] - u[n-1]$$

$$u[n] = \sum_{k=0}^{\infty} \delta[n-k] = \delta[n] + \delta[n-1] + \delta[n-2] + \dots$$



Sampling and sifting

Two properties do most of the work later. They say different things, and the difference matters.

SAMPLING PROPERTY

$$x[n] \delta[n - n_0] = x[n_0] \delta[n - n_0]$$

Both sides are **sequences**. Multiplying by a shifted impulse keeps one sample and deletes the rest.

SIFTING PROPERTY

$$x[n_0] = \sum_{n=-\infty}^{\infty} x[n] \delta[n - n_0]$$

The right-hand side is a **number**. Sifting is sampling followed by a sum.

EXAMPLE 1.4

GIVEN $x[0] = 1$, $x[1] = 2$, $x[2] = 3$, and $x[n] = 0$ elsewhere. Take $n_0 = 2$.

FIND The sampled sequence and the sifted value.

SOLUTION Sampling: $x[n] \delta[n - 2] = x[2] \delta[n - 2] = 3 \delta[n - 2]$, a single stem of height 3 at $n = 2$.
Sifting: $\sum_n x[n] \delta[n - 2] = x[0] \cdot 0 + x[1] \cdot 0 + x[2] \cdot 1 = 3$.

CHECK The first answer is a signal. The second is a number. If your two answers have the same type, one of them is wrong.

Applying the sampling property at every shift and adding the results gives the identity that Chapter 3 is built on.

REPRESENTATION PROPERTY

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n - k]$$

Any sequence is a sum of weighted, shifted impulses. The weights are the sample values themselves.

1.10 The unit impulse and the unit step in continuous time

The continuous-time step is straightforward: $u(t) = 1$ for $t \geq 0$ and 0 otherwise. The impulse needs more care.

THE IMPULSE IS NOT A FUNCTION

It is often written as $\delta(t) = \infty$ at $t = 0$ and 0 elsewhere, with total area 1. No ordinary function behaves like that: a function that is zero except at one point has integral zero. δ is a **generalized function**, or distribution. It is defined by what it does inside an integral, not by its values.

DEFINING PROPERTY (SIFTING)

$$x(t_0) = \int_{-\infty}^{\infty} x(t) \delta(t - t_0) dt$$

This is the definition. Everything else about δ follows from it.

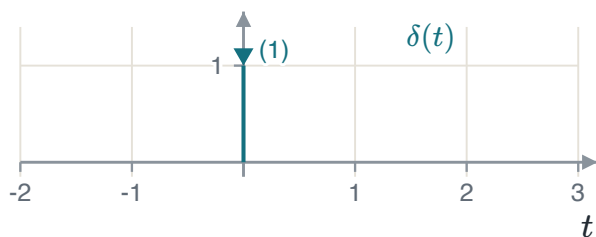
A useful picture: take a rectangle of width ε and height $1/\varepsilon$, so its area is 1 for every ε , and let $\varepsilon \rightarrow 0$. The limit does not exist point by point. It exists only under an integral sign, which is exactly the statement above.

The impulse and the step are related by differentiation and integration:

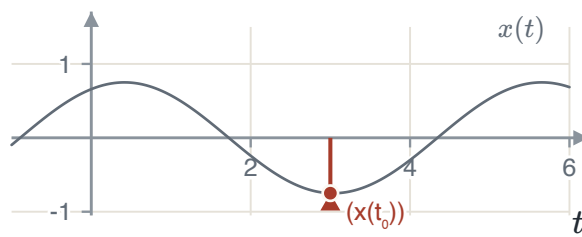
$$\delta(t) = \frac{d}{dt} u(t), \quad u(t) = \int_{-\infty}^t \delta(\tau) d\tau.$$

SAMPLING PROPERTY, CONTINUOUS TIME

$$x(t) \delta(t - t_0) = x(t_0) \delta(t - t_0)$$



The arrow height shows the **area**, never a value of a function.



Sifting: the impulse at t_0 is scaled by the value of x there, and integrating returns that number.

1.11 Complex exponentials in continuous time

DEFINITION

$$x(t) = C e^{at}, \quad C, a \in \mathbb{C}$$

The behaviour depends only on where a sits in the complex plane. There are three cases.

Case 1: C and a real

If $a < 0$ the signal decays. If $a > 0$ it grows. If $a = 0$ it is the constant C . A larger $|a|$ makes the decay or growth faster.

Case 2: $a = j\omega_0$ purely imaginary

Write $C = Ae^{j\theta}$ and use Euler's relation $e^{jx} = \cos x + j \sin x$:

$$x(t) = Ae^{j(\omega_0 t + \theta)} = A \cos(\omega_0 t + \theta) + jA \sin(\omega_0 t + \theta).$$

Here A is the amplitude, ω_0 is the angular frequency in rad/s, and θ is the phase in radians.

Is this signal periodic? Require $x(t) = x(t + T)$:

$$Ae^{j(\omega_0 t + \theta)} = Ae^{j(\omega_0(t+T) + \theta)} \implies 1 = e^{j\omega_0 T} \implies j2\pi k = j\omega_0 T \implies T = \frac{2\pi}{\omega_0} k.$$

RESULT

Taking $k = 1$ gives the fundamental period $T_0 = 2\pi/\omega_0$. Every continuous-time complex exponential with $\omega_0 \neq 0$ is periodic. There is no extra condition.

EXAMPLE 1.5

GIVEN $x(t) = e^{j0.5\pi t}$.

FIND The fundamental period.

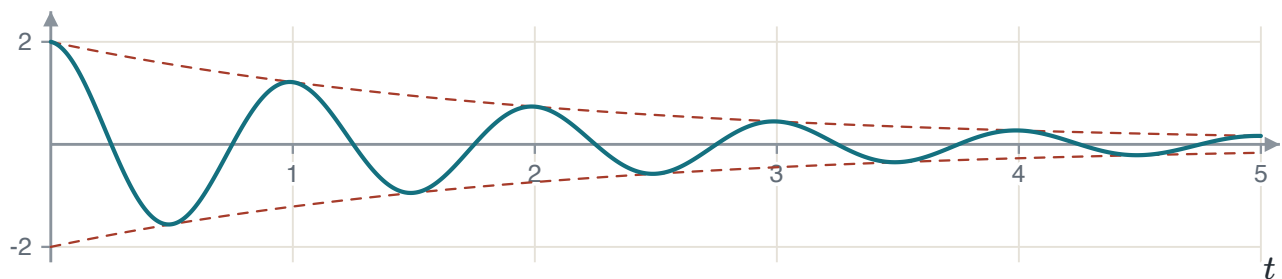
SOLUTION $T_0 = \frac{2\pi}{\omega_0} = \frac{2\pi}{0.5\pi} = 4$ seconds.

CHECK $0.5\pi \times 4 = 2\pi$, which is exactly one full turn of the phasor.

Case 3: $a = r + j\omega_0$

$$x(t) = Ae^{rt} \cos(\omega_0 t + \theta) + jAe^{rt} \sin(\omega_0 t + \theta).$$

This is a sinusoid inside an envelope $\pm Ae^{rt}$. If $r < 0$ the oscillation is damped, if $r > 0$ it grows, and if $r = 0$ it is sustained. This is the natural response of every second-order circuit.



A damped sinusoid, $\text{Re}\{2e^{-0.5t}e^{j2\pi t}\}$, with its envelope shown dashed.

1.12 Complex exponentials in discrete time

DEFINITION

$$x[n] = C e^{\beta n} = C \alpha^n, \quad \alpha = e^\beta$$

The power form is the natural one in discrete time. It is what a difference equation produces.

For real C and α : if $0 < \alpha < 1$ the sequence decreases; if $\alpha > 1$ it increases. For complex $C = |C|e^{j\theta}$ and $\alpha = |\alpha|e^{j\omega_0}$:

$$x[n] = |C| |\alpha|^n \cos(\omega_0 n + \theta) + j |C| |\alpha|^n \sin(\omega_0 n + \theta).$$

THE BOUNDARY MOVES

In continuous time the growth-decay boundary is $\text{Re}\{a\} = 0$, the imaginary axis. In discrete time it is $|\alpha| = 1$, the unit circle. The map between them is $\alpha = e^\beta$.

1.13 When is a discrete-time exponential periodic?

This is the result that has no continuous-time counterpart. Require $x[n] = x[n + N]$ for $x[n] = C e^{j\omega_0 n}$:

$$C e^{j\omega_0 n} = C e^{j\omega_0 (n+N)} \implies 1 = e^{j\omega_0 N} \implies j2\pi k = j\omega_0 N \implies N = \frac{2\pi}{\omega_0} k.$$

PERIODICITY CONDITION

$$\frac{\omega_0}{2\pi} = \frac{k}{N} \in \mathbb{Q}$$

N must be an **integer**. That is possible only when $\omega_0/2\pi$ is a rational number. If it is irrational, no integer N works and the sequence is aperiodic, no matter how sinusoidal it looks when plotted.

EXAMPLE 1.6

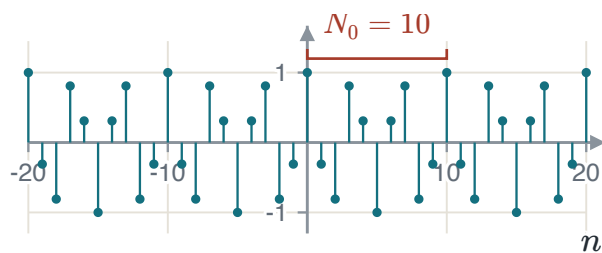
GIVEN $x[n] = e^{j(3\pi/5)n}$.

FIND The fundamental period N_0 .

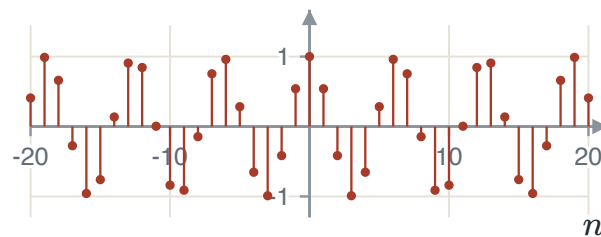
METHOD Use $N = 2\pi k/\omega_0$ and take the smallest k that makes N an integer.

SOLUTION $N = \frac{2\pi}{3\pi/5} k = \frac{10}{3} k$. The smallest positive integer k giving an integer N is $k = 3$, so $N_0 = 10$.

CHECK $\omega_0 N_0 = \frac{3\pi}{5} \times 10 = 6\pi = 2\pi \times 3$. The phasor makes exactly three full turns in ten samples, and k counts those turns.



$\text{Re}\{e^{j3\pi n/5}\}$ repeats every 10 samples.



$\cos(n)$ has $\omega_0 = 1$, so $\omega_0/2\pi$ is irrational. It never repeats. It only looks as if it should.

A SECOND DISCRETE-TIME FACT, NEEDED LATER

$e^{j(\omega_0+2\pi)n} = e^{j\omega_0 n}$ for every integer n . Discrete-time frequencies that differ by 2π cannot be told apart. The same fact makes the discrete-time Fourier transform 2π -periodic and makes aliasing unavoidable.

1.14 Chapter summary

1. E_∞ and P_∞ are limits. Finite energy forces zero average power. Finite non-zero average power forces infinite energy. Unbounded growth gives neither.
2. To build $x(at - b)$, shift by b first, then scale by a . The other order gives $x(at - ab)$.
3. T_0 and N_0 are the smallest positive periods, and $\omega_0 = 2\pi/T_0 = 2\pi/N_0$.
4. $\delta[n]$ is an ordinary sequence. $\delta(t)$ is a distribution defined by sifting.
5. Every signal is a sum of weighted, shifted impulses: $x[n] = \sum_k x[k]\delta[n - k]$.
6. A discrete-time exponential is periodic only when $\omega_0/2\pi$ is rational. A continuous-time one always is.

Exercises

1.1 Classify $x(t) = e^{-3t}u(t)$ as an energy signal, a power signal, or neither. Give the value of whichever quantity is finite.

Answer: Energy signal, $E_\infty = 1/6$.

1.2 For $x[n]$ non-zero only on $-1 \leq n \leq 1$ with values 1, 2, 1, sketch $x[n + 4]$ and $x[n - 5]$ and give the support of each.

Answer: Supports $-5 \leq n \leq -3$ and $4 \leq n \leq 6$.

1.3 A signal $x(t)$ is non-zero only on $[-1, 5]$. On what interval is $x(2t + 3)$ non-zero?

Answer: $[-2, 1]$.

1.4 Find the fundamental period of $x[n] = \cos(4\pi n/7)$, or show that it is aperiodic.

Answer: $N_0 = 7$.

1.5 Show that if $x(t)$ is odd then $\int_{-a}^a x(t) dt = 0$ for every $a > 0$.

1.6 Evaluate $\int_{-\infty}^{\infty} (t^2 + 1) \delta(t - 2) dt$ and $\sum_{n=-\infty}^{\infty} 2^{-n} \delta[n - 3]$.

Answer: 5 and $1/8$.

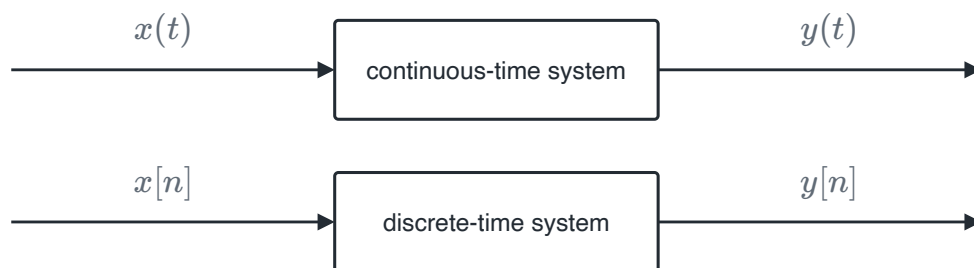
Systems and their properties

A system is judged by what it does to signals, not by what it is made of. Six properties decide almost everything that follows, and two of them are the whole of Chapter 3.

2.1 What a system is

A system is a rule that turns an input signal into an output signal. The rule is deterministic: the same input, applied twice, must give the same output. Without that, none of the tests below would even make sense.

Write it as an operator, $y = S\{x\}$. The argument is a whole function, not a number. That is why questions about memory, causality and time shifts can be asked at all.



The same picture in both domains. An amplifier, a numerical filter and a pencil-and-paper integrator can all be the same system.

2.2 Memory

CRITERION

A system is **memoryless** if the output at time t (or n) depends only on the input at that same time.

SYSTEM	VERDICT	REASON
$y(t) = [2x(t) - x^2(t)]^2$	memoryless	Only $x(t)$ appears. There is no $x(t+1)$ or $x(t-2)$ term.
$y[n] = x[n]$	memoryless	The identity system.
$y[n] = x[n-1]$	has memory	The output at n uses the sample at $n-1$.
$y[n] = x[n] + y[n-1]$	has memory	See the derivation below.

The last one is worth doing carefully, because the memory is hidden in the feedback. Substitute repeatedly:

$$\begin{aligned}
 y[n-1] &= x[n-1] + y[n-2], & y[n-2] &= x[n-2] + y[n-3], & \dots \\
 \implies y[n] &= x[n] + x[n-1] + x[n-2] + \dots = \sum_{k=0}^{\infty} x[n-k].
 \end{aligned}$$

The output depends on the whole past of the input. Feedback on the output is memory just as surely as an explicit delay on the input.

CIRCUIT READING

A resistor, $v(t) = Ri(t)$, is memoryless. A capacitor, $v(t) = \frac{1}{C} \int_{-\infty}^t i(\tau) d\tau$, is not. Memory is stored energy.

2.3 Invertibility

CRITERION

A system is **invertible** if distinct inputs always produce distinct outputs. The map must be one-to-one.

There are two ways to settle it, and they cost very different amounts of work. To prove invertibility, find an inversion formula that works for every input. To disprove it, find one pair of inputs that collide.

EXAMPLE 2.1

GIVEN $y(t) = [\cos(t) + 2]x(t)$.

FIND Is the system invertible?

SOLUTION Yes. The gain satisfies $1 \leq \cos(t) + 2 \leq 3$, so it never vanishes and

$$x(t) = \frac{y(t)}{\cos(t) + 2}.$$

CHECK Substituting back gives $[\cos t + 2] \frac{y(t)}{\cos t + 2} = y(t)$. Had the gain been $\cos(t)$ alone, the system would fail at every zero of the cosine.

EXAMPLE 2.2

GIVEN $y(t) = x^2(t)$.

FIND Is the system invertible?

SOLUTION No. Take $x_1(t) = 1$ and $x_2(t) = -1$ for all t . These are different inputs, but $y_1(t) = y_2(t) = 1$.

CHECK One counterexample is enough. The sign information is destroyed and cannot be recovered from y alone.

NOT AN INVERSION FORMULA

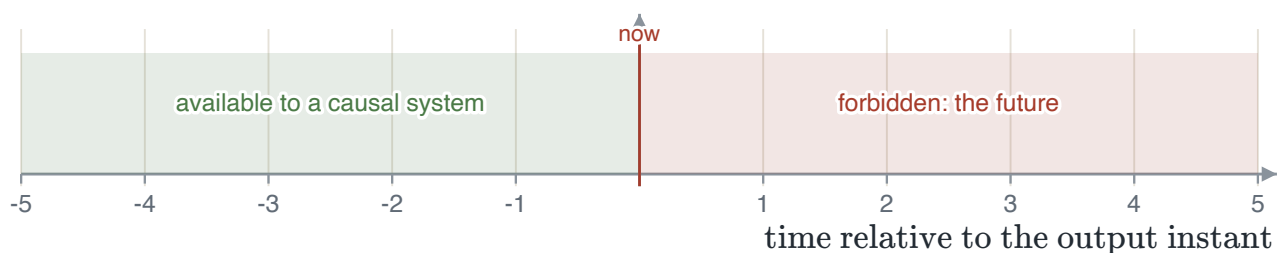
Writing $x(t) = \sqrt{y(t)}$ does not invert $y = x^2$. It picks one of two possible inputs by convention. An inversion formula must return the actual input, for every admissible input.

2.4 Causality

CRITERION

A system is **causal** if the output at time t (or n) depends only on inputs at times $\leq t$: the present and the past.

SYSTEM	VERDICT	REASON
$y[n] = x[n - 1]$	causal	Uses only a past sample.
$y[n] = x[n] + x[n + 1]$	not causal	$x[n + 1]$ is a future sample.
$y(t) = \int_{-\infty}^t x(\tau) d\tau$	causal	The upper limit is t , so only $\tau \leq t$ contributes.
$y[n] = x[-n]$	not causal	$y[-1] = x[1]$. The output at $n = -1$ needs a future input.
$y(t) = x(t) \cos(t + 1)$	causal	Uses only $x(t)$. The factor $\cos(t + 1)$ is a known function of t , fixed in advance, not a future input.



Causality restricts which part of the input axis the output is allowed to consult.

A system that must run in real time has to be causal, because the future has not happened yet. Non-causal systems are perfectly useful whenever the whole record already exists, as in offline audio or image processing.

2.5 Stability

A signal is **bounded** if there is a constant $B < \infty$ with $|x(t)| < B$ for all t .

CRITERION

A system is **BIBO stable** if every bounded input produces a bounded output. Bounded Input, Bounded Output.

THE TWO DIRECTIONS COST DIFFERENTLY

To prove stability you must bound the output for **every** bounded input. To disprove it you need **one** bounded input whose output is unbounded. Testing a single well-behaved input and concluding "stable" proves nothing.

EXAMPLE 2.3

GIVEN $y(t) = 2x^2(t-1) + x(3t)$.

FIND Is the system stable?

METHOD Assume $|x(t)| \leq B < \infty$ and bound $|y|$ with the triangle inequality.

SOLUTION

$$|y(t)| \leq |2x^2(t-1)| + |x(3t)| \leq 2B^2 + B < \infty.$$

The system is stable.

CHECK Shifting and scaling the time axis cannot change the set of values a signal takes, so $x(t-1)$ and $x(3t)$ are bounded by the same B .

EXAMPLE 2.4

GIVEN $y[n] = \sum_{k=-\infty}^n x[k]$, the accumulator.

FIND Is the system stable?

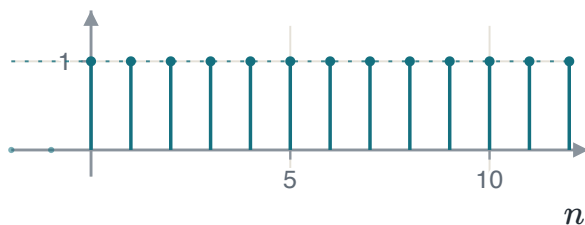
METHOD Look for one bounded input with an unbounded output.

SOLUTION Take $x[n] = u[n]$, so $|x[n]| \leq 1$. Then

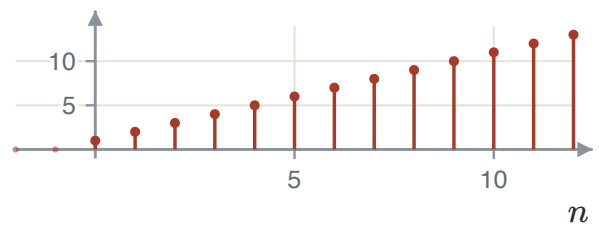
$$y[n] = \sum_{k=0}^n 1 = n + 1 \rightarrow \infty.$$

The system is **not** stable.

CHECK The input never exceeds 1 and the output grows without bound. That is exactly the failure BIBO stability forbids.



Bounded input $u[n]$.



Unbounded output $n + 1$.

CAUSAL DOES NOT MEAN STABLE

The accumulator is causal and unstable. The reversal $y[n] = x[-n]$ is stable and not causal. The six properties are independent except where a proof links them.

2.6 Time invariance

CRITERION

If $x(t) \rightarrow y(t)$, the system is **time invariant** when $x(t - t_0) \rightarrow y(t - t_0)$ for every t_0 .

The test is a comparison of two computed signals.

1. **Path 1.** Shift the input, then run the system: $y_2 = S\{x(t - t_0)\}$.
2. **Path 2.** Run the system, then shift the output: $y_1(t - t_0)$.

The system is time invariant exactly when the two agree, for every input and every t_0 .

EXAMPLE 2.5

GIVEN $y(t) = \sin(x(t))$.

FIND Is the system time invariant?

SOLUTION Path 1: with $x_2(t) = x_1(t - t_0)$ we get $y_2(t) = \sin(x_1(t - t_0))$.
Path 2: $y_1(t) = \sin(x_1(t))$, so $y_1(t - t_0) = \sin(x_1(t - t_0))$.
They agree for every t_0 , so the system is time invariant.

CHECK The rule contains no explicit t . That is usually a reliable sign.

EXAMPLE 2.6

GIVEN $y[n] = n x[n]$.

FIND Is the system time invariant?

METHOD Look for a counterexample using an impulse.

SOLUTION Take $x_1[n] = \delta[n]$. Then $y_1[n] = n \delta[n] = 0$ for every n .
Now delay by one: $x_2[n] = \delta[n - 1]$ gives $y_2[n] = n \delta[n - 1] = \delta[n - 1]$, which equals 1 at $n = 1$.
Time invariance would require $y_2[n] = y_1[n - 1] = 0$. It does not. The system is **not** time invariant.

CHECK One counterexample settles it. Note that the impulse is the cheapest test signal available, because it isolates a single index.

THE PATTERN BEHIND EVERY FAILURE

A system fails time invariance exactly when its rule contains the time variable explicitly: a coefficient n , a gain $\cos(t)$, or a scaled argument $x(3t)$. If t or n appears anywhere outside the input, be suspicious at once.

2.7 Linearity

CRITERION

$$a x_1 + b x_2 \longrightarrow a y_1 + b y_2 \quad \text{for all } a, b \in \mathbb{C}$$

where $x_1 \rightarrow y_1$ and $x_2 \rightarrow y_2$. This is additivity and homogeneity in one statement.

EXAMPLE 2.7

GIVEN $y(t) = 2\pi x(t)$.

FIND Is the system linear?

SOLUTION With $x_3 = ax_1 + bx_2$:

$$y_3 = 2\pi(ax_1 + bx_2) = \underbrace{a2\pi x_1}_{y_1} + \underbrace{b2\pi x_2}_{y_2} = ay_1 + by_2.$$

The system is linear. It is also time invariant, so it is the simplest possible LTI system.

EXAMPLE 2.8

GIVEN $y[n] = (x[2n])^2$.

FIND Is the system linear?

SOLUTION With $x_3[n] = ax_1[n] + bx_2[n]$:

$$y_3[n] = (ax_1[2n] + bx_2[2n])^2 = a^2 x_1^2[2n] + 2ab x_1[2n]x_2[2n] + b^2 x_2^2[2n],$$

while $ay_1[n] + by_2[n] = a x_1^2[2n] + b x_2^2[2n]$. The powers of a and b differ and a cross term appears, so the system is **not** linear.

CHECK A quicker route: scaling the input by a scales this output by a^2 . Homogeneity alone already fails.

LINEAR DOES NOT MEAN STRAIGHT LINE

$y(t) = x(t) + 5$ is not linear, because the zero input does not give the zero output. Any linear system must satisfy $S\{0\} = 0$. That is a five-second first test.

2.8 How to classify an unfamiliar system

1. **Look for an explicit t or n .** A coefficient like n , a gain like $\cos t$, or a scaled argument like $x(3t)$ makes time invariance the first thing to test, and usually the first thing to fail.
2. **Look for nonlinearity.** Squares, products of the input with itself, $\sin(x)$, absolute values, saturation. Test homogeneity with a single scalar; it is the cheapest disproof available.
3. **Read the arguments.** Anything other than x evaluated exactly at t or n breaks memorylessness. Future arguments break causality.
4. **Try to bound the output.** Assume $|x| \leq B$ and look for a finite bound. If the attempt fails, the failure usually shows you the counterexample.
5. **Leave invertibility to last.** It is needed least often and argued badly most often.

THE ONLY FREE IMPLICATION

Memoryless implies causal. Everything else must be established on its own. In particular, causal does not imply stable, linear does not imply time invariant, and stable does not imply memoryless.

SYSTEM	MEMORYLESS	INVERTIBLE	CAUSAL	STABLE	TIME INV.	LINEAR
$y(t) = 2\pi x(t)$	yes	yes	yes	yes	yes	yes
$y[n] = x[n - 1]$	no	yes	yes	yes	yes	yes
$y(t) = x^2(t)$	yes	no	yes	yes	yes	no
$y[n] = n x[n]$	yes	no	yes	no	no	yes
$y[n] = \sum_{k \leq n} x[k]$	no	yes	yes	no	yes	yes
$y[n] = x[-n]$	no	yes	no	yes	no	yes
$y[n] = (x[2n])^2$	no	no	no	yes	no	no

Read this table down the columns, not across the rows. No column is determined by another, which is why six separate tests are needed.

Exercises

2.1 Classify $y(t) = x(t)u(t)$ against all six properties.

Answer: Memoryless, not invertible, causal, stable, not time invariant, linear.

2.2 Show that $y[n] = x[n] - x[n - 1]$ is invertible on inputs that are zero for $n < 0$, and give the inverse.

Answer: $x[n] = \sum_{k=0}^n y[k]$.

2.3 Give a system that is linear and causal but not stable, and one that is stable and time invariant but not linear.

2.4 Is $y(t) = x(t/2)$ time invariant? Prove or give a counterexample.

Answer: No. Path 1 gives $x(t/2 - t_0)$ and path 2 gives $x((t - t_0)/2)$.

Linear time-invariant systems

Impose linearity and time invariance together, and a whole system is described by its response to a single impulse. This chapter derives that result and shows how to use it.

3.1 The impulse response

DEFINITION

The **impulse response** is the output when the input is a unit impulse: $x[n] = \delta[n]$ gives $y[n] = h[n]$. In continuous time, $x(t) = \delta(t)$ gives $y(t) = h(t)$.

For a general system this is one measurement among infinitely many. Knowing the response to $\delta[n]$ says nothing about the response to anything else.

For a linear time-invariant system it is everything. Time invariance turns one impulse response into a response for every shift. Linearity turns those into a response for every weighted sum. And Chapter 1 showed that every signal is a weighted sum of shifted impulses. Those three facts close the loop.



One experiment defines the whole system, provided the system is linear and time invariant.

3.2 Deriving the convolution sum

Start from the representation property of Chapter 1 and push it through the system.

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n-k] \quad \longrightarrow \quad S \quad \longrightarrow \quad y[n] = ?$$

1. **Time invariance.** Since $\delta[n] \rightarrow h[n]$, we also have $\delta[n-k] \rightarrow h[n-k]$.
2. **Homogeneity.** The number $x[k]$ does not depend on n , so $x[k]\delta[n-k] \rightarrow x[k]h[n-k]$.
3. **Additivity.** The response to the sum is the sum of the responses.

CONVOLUTION SUM

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k] = x[n] * h[n]$$

Substituting $m = n - k$ gives an equivalent form, which is already the statement that convolution is commutative:

$$\sum_{k=-\infty}^{\infty} x[k] h[n-k] = \sum_{m=-\infty}^{\infty} x[n-m] h[m].$$

THE PRECONDITION IS NOT OPTIONAL

Convolution applies only when the system is linear and time invariant. The derivation used time invariance once and linearity twice. Remove either property and no step survives. Applying convolution anyway produces a confident number that means nothing.

3.3 How to compute a convolution

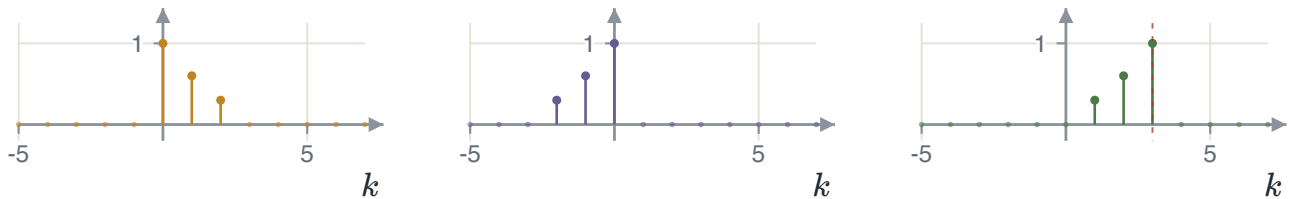
1. **Flip.** Reverse $h[k]$ to get $h[-k]$.

2. **Shift.** Move it by n to get $h[n - k]$.
3. **Multiply and add.** Form $x[k]h[n - k]$ and sum over k .
4. Repeat for every n .

Build $h[n - k]$ the same way you built $x(at - b)$ in Chapter 1. Treat k as the variable: shift $h[k]$ by n , then reverse in k .

WHAT HAPPENS IF YOU SKIP THE FLIP

Sliding an unflipped h computes $\sum_k x[k]h[k - n]$, which is the cross-correlation of x and h , not the convolution. When h is symmetric the two agree, which is why this mistake often survives until an asymmetric h appears in an exam.



$h[k]$

$h[-k]$: flip

$h[n - k]$ at $n = 3$: shift

EXAMPLE 3.1

GIVEN $x[n] = \{1, 2, 1, 2\}$ for $n = 0, 1, 2, 3$, and $h[n] = \{1, 1\}$ for $n = 0, 1$. Both zero elsewhere.

FIND $y[n] = x[n] * h[n]$.

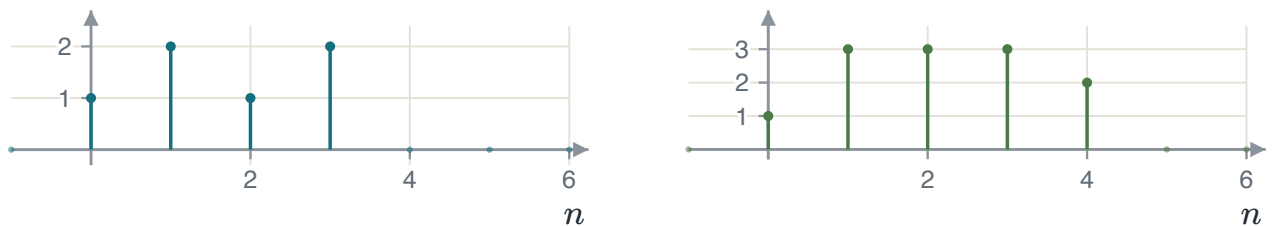
METHOD Expand the sum over the four non-zero samples of x and add the shifted copies of h .

SOLUTION

$$y[n] = x[0]h[n] + x[1]h[n - 1] + x[2]h[n - 2] + x[3]h[n - 3] = h[n] + 2h[n - 1] + h[n - 2] + 2h[n - 3]$$

Adding the four copies sample by sample gives $y = \{1, 3, 3, 3, 2\}$ on $n = 0, \dots, 4$.

CHECK Two independent checks. **Length:** $4 + 2 - 1 = 5$ samples, and the support is $[0, 4]$. **Sum:** $\sum_n y[n]$ must equal $(\sum_n x[n])(\sum_n h[n])$, and $12 = 6 \times 2$. Also note that $h = \{1, 1\}$ is a two-point moving sum, so each output is the total of an adjacent input pair, which is what the numbers show.



$x[n]$

$y[n] = x[n] * h[n]$

EXAMPLE 3.2

GIVEN $x[n] = \left(\frac{1}{2}\right)^n u[n]$ and $h[n] = u[n]$.

FIND $y[n] = x[n] * h[n]$.

METHOD $h[n - k]$ equals 1 exactly when $k \leq n$. Split on the sign of n .

SOLUTION **Case 1**, $n < 0$. The supports do not overlap: $x[k]$ needs $k \geq 0$ and $h[n - k]$ needs $k \leq n < 0$. So $y[n] = 0$.

Case 2, $n \geq 0$. The overlap is $0 \leq k \leq n$, so

$$y[n] = \sum_{k=0}^n \left(\frac{1}{2}\right)^k = \frac{1 - \left(\frac{1}{2}\right)^{n+1}}{1 - \frac{1}{2}} = 2 - \left(\frac{1}{2}\right)^n.$$

Together, $y[n] = \left(2 - \left(\frac{1}{2}\right)^n\right) u[n]$.

CHECK $y[0] = 1$, and directly $y[0] = x[0]h[0] = 1$. As $n \rightarrow \infty$, $y[n] \rightarrow 2$, which is the total sum of the input. That is what an accumulator should do.

GEOMETRIC SUM

$\sum_{k=m}^n a r^k = \frac{a(r^m - r^{n+1})}{1 - r}$. The finite sum needs only $r \neq 1$. The condition $|r| < 1$ is needed only when the sum runs to infinity.

3.4 The convolution integral

The continuous-time version follows the same three steps. Sifting gives the representation of x as a continuum of weighted impulses,

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau,$$

using $\delta(t) = \delta(-t)$. Pushing this through a linear time-invariant system gives:

CONVOLUTION INTEGRAL

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau = x(t) * h(t)$$

$$y(t) = \int_{-\infty}^{\infty} h(\tau) x(t - \tau) d\tau$$

The two forms are equal. Flip whichever signal has the simpler shape.

BUILDING THE FLIPPED, SHIFTED H

Do it by shift, then scale, in the variable τ . For example, to plot $\delta(-t + 5)$: first $v(t) = \delta(t + 5)$, then $y(t) = v(-t) = \delta(-t + 5)$, an impulse at $t = +5$, not at $t = -5$. Sign errors here are the main source of wrong integration limits.

EXAMPLE 3.3

GIVEN $x(t) = e^{2t}u(-t)$ and $h(t) = u(t - 3)$.

FIND $y(t) = x(t) * h(t)$.

METHOD $x(\tau)$ is non-zero for $\tau \leq 0$. And $h(t - \tau) = u(t - \tau - 3)$ is non-zero for $\tau \leq t - 3$. So the overlap ends at $\min(0, t - 3)$, and that is what creates two cases.

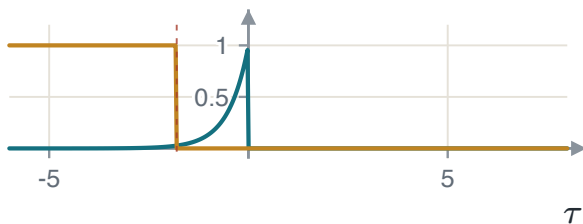
SOLUTION **Case 1, $t < 3$.** The binding limit is $t - 3$:

$$y(t) = \int_{-\infty}^{t-3} e^{2\tau} d\tau = \frac{1}{2}e^{2(t-3)}.$$

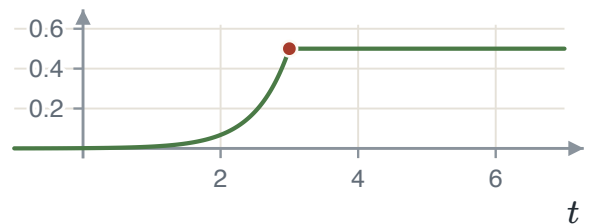
Case 2, $t > 3$. The binding limit is 0, so the whole of x is covered:

$$y(t) = \int_{-\infty}^0 e^{2\tau} d\tau = \frac{1}{2}.$$

CHECK Both branches give 0.5 at $t = 3$, so y is continuous. And $y(\infty)$ equals the total area of x , which is $1/2$. The system is a delayed integrator: it collects the area of x up to three seconds ago.



Case 1: $t < 3$. The shaded area is $y(t)$.



The result $y(t)$, continuous at $t = 3$.

EXAMPLE 3.4

GIVEN $x(t) = 1$ on $0 < t < 1$ and zero elsewhere. $h(t) = t$ on $0 < t < 2$ and zero elsewhere.

FIND $y(t) = x(t) * h(t)$.

METHOD Use $y(t) = \int h(\tau)x(t - \tau) d\tau$ and flip the rectangle, which is the simpler shape. Its support becomes $[t - 1, t]$: a window of width 1 sliding across the ramp on $[0, 2]$.

SOLUTION The window edges are $t - 1$ and t . The ramp edges are 0 and 2. Setting them equal gives $t = 0, 1, 2, 3$: four boundaries, so five cases.

$t < 0$: no overlap, $y = 0$.

$0 < t < 1$: $y = \int_0^t \tau d\tau = \frac{1}{2}t^2$.

$1 < t < 2$: $y = \int_{t-1}^t \tau d\tau = t - \frac{1}{2}$.

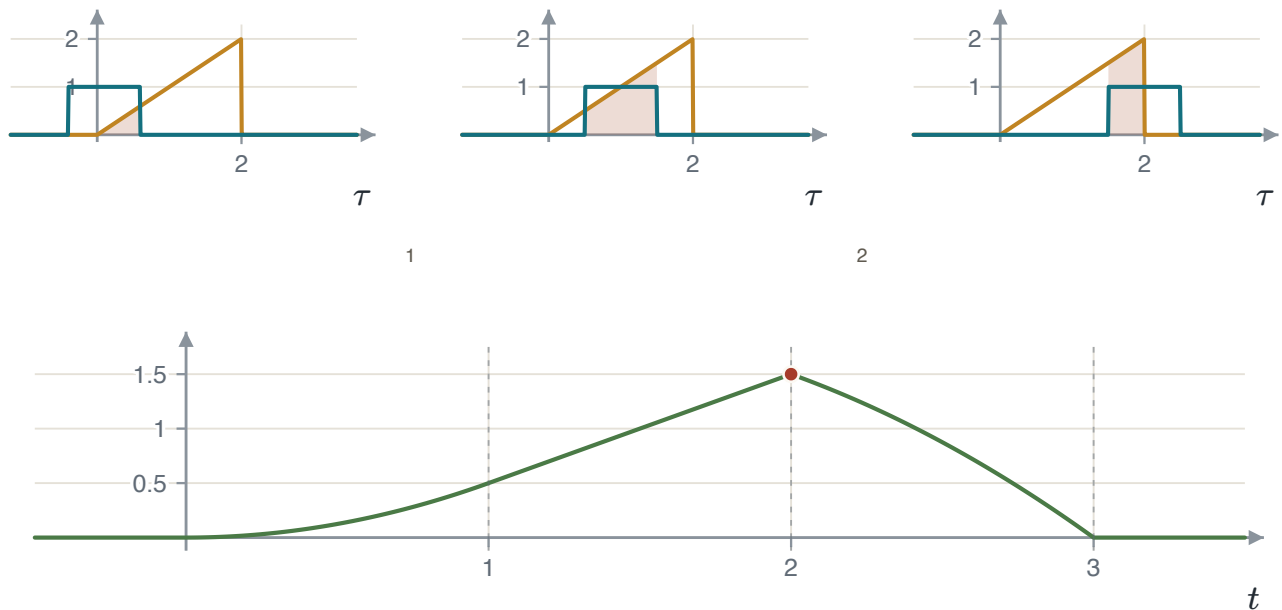
$2 < t < 3$: $y = \int_{t-1}^2 \tau d\tau = -\frac{1}{2}t^2 + t + \frac{3}{2}$.

$t > 3$: no overlap, $y = 0$.

CHECK Three checks. **Continuity:** at $t = 1$ both branches give 0.5; at $t = 2$ both give 1.5; at $t = 3$ both give 0. **Support:** $[0, 1]$ and $[0, 2]$ give $[0, 3]$, because supports add. **Area:** $\int y = (\int x)(\int h) = 1 \times 2 = 2$, and $\frac{1}{6} + 1 + \frac{5}{6} = 2$.

FIND THE BOUNDARIES BEFORE INTEGRATING

List the moving edges and the fixed edges, then set them equal. That produces the complete list of cases in one line, and no case can be lost by accident.



The complete result. The peak is 1.5 at $t = 2$. A unit-width rectangle acts as a one-second moving sum, so the output rises while the window fills, grows linearly while it is full, and falls as it leaves the ramp.

3.5 Properties of convolution

PROPERTY	STATEMENT	WHAT IT MEANS FOR INTERCONNECTIONS
Commutative	$x * h = h * x$	Input and impulse response play symmetric roles in the algebra.
Distributive	$x * (h_1 + h_2) = x * h_1 + x * h_2$	Two systems in parallel , outputs added, equal one system with impulse response $h_1 + h_2$.
Associative	$x * (h_1 * h_2) = (x * h_1) * h_2$	Two systems in cascade equal one system with impulse response $h_1 * h_2$.

Combining the last two with commutativity means the order of cascaded systems does not matter. That freedom is bought with linearity and time invariance, and it disappears the moment either is lost. A saturating amplifier followed by a filter is not the same system as the filter followed by the amplifier.

3.6 System properties in terms of h

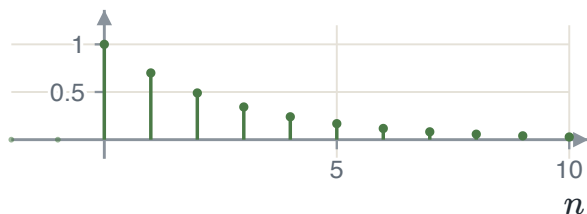
Every test of Chapter 2 becomes a statement about the single function h .

PROPERTY	CRITERION	REASON
Memoryless	$h(t) = a \delta(t)$, or $h[n] = a \delta[n]$	Then $y[n] = a \sum_k x[k] \delta[n - k] = a x[n]$, a pure gain.
Invertible	$h * g = \delta$ for some g	g is the impulse response of the inverse system, and δ is the identity system.
Causal	$h(t) = 0$ for $t < 0$, or $h[n] = 0$ for $n < 0$	The sum collapses to $y[n] = h[0]x[n] + h[1]x[n-1] + \dots$, which uses only present and past inputs.
BIBO stable	$\sum_k h[k] < \infty$, or $\int h(t) dt < \infty$	See the proof below.

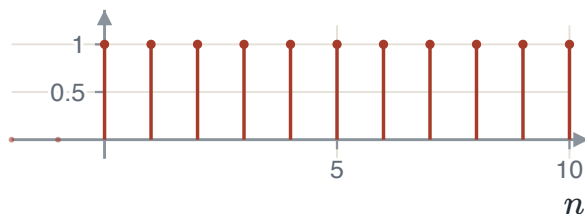
The stability condition is worth proving in one line. Assume $|x[n]| \leq B$ for all n . Then

$$|y[n]| = \left| \sum_k h[k] x[n-k] \right| \leq \sum_k |h[k]| |x[n-k]| \leq B \sum_k |h[k]| < \infty.$$

This proves that absolute summability is **sufficient** for stability. It is also necessary; that direction is stated here without proof.



$h[n] = 0.7^n u[n]$. Here $\sum |h| = 1/0.3 < \infty$: stable and causal.



$h[n] = u[n]$, the accumulator. Here $\sum |h| \rightarrow \infty$: causal but not stable.

3.7 A convolution checklist

1. Confirm the system is linear and time invariant. Nothing below is valid otherwise.
2. Choose which signal to flip. Pick the simpler one.
3. Write the support of each factor as an inequality in the dummy variable.
4. Set the moving edges equal to the fixed edges to list every case boundary, before integrating anything.
5. Integrate or sum, case by case.
6. Check continuity at every boundary, check that the supports add, and check that the total area or total sum multiplies.

WHAT THE THREE FINAL CHECKS CATCH

A discontinuity at a boundary means a limit is wrong. A support that is too wide or too narrow means a flip or a shift is wrong. A total area that does not multiply means an integrand is wrong.

Exercises

3.1 Compute $\{1, 2, 3\} * \{1, -1\}$ with both sequences starting at $n = 0$, and check your answer with the sum rule.

Answer: $\{1, 1, 1, -3\}$; the sums are $6 \times 0 = 0$.

3.2 Let $h[n] = \left(\frac{1}{3}\right)^n u[n]$. Is the system stable? Is it causal? Is it memoryless?

Answer: Stable ($\sum |h| = 3/2$), causal, not memoryless.

3.3 Find $y(t) = u(t) * u(t)$ and sketch it.

Answer: $y(t) = t u(t)$.

3.4 Two systems with $h_1[n] = \delta[n] - \delta[n - 1]$ and $h_2[n] = u[n]$ are cascaded. Find the impulse response of the cascade and identify the resulting system.

Answer: $\delta[n]$: the two systems are inverses.

3.5 $x(t) = 1$ on $0 < t < 3$ and $h(t) = t$ on $0 < t < 2$. List the case boundaries before computing anything, then find $y(t)$.

Answer: Boundaries at $t = 0, 2, 3, 5$; four cases.

Fourier series

Convolution answers every question about a linear time-invariant system, and answers it slowly. This chapter finds the one family of input signals a system cannot reshape, and shows that almost every periodic signal is a sum of them. What was an integral becomes one multiplication per frequency.

4.1 Eigenfunctions of a linear time-invariant system

DEFINITION

A signal is an **eigenfunction** of a system when the output is that same signal multiplied by a constant. The constant is the **eigenvalue**.

Put $x(t) = e^{st}$, with s any complex number, into the convolution integral. Write both limits; they are $-\infty$ and ∞ throughout this chapter.

$$y(t) = \int_{-\infty}^{\infty} h(\tau) e^{s(t-\tau)} d\tau = e^{st} \int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau.$$

The second factor does not depend on t . It is a number attached to the value of s , and it is called the transfer function.

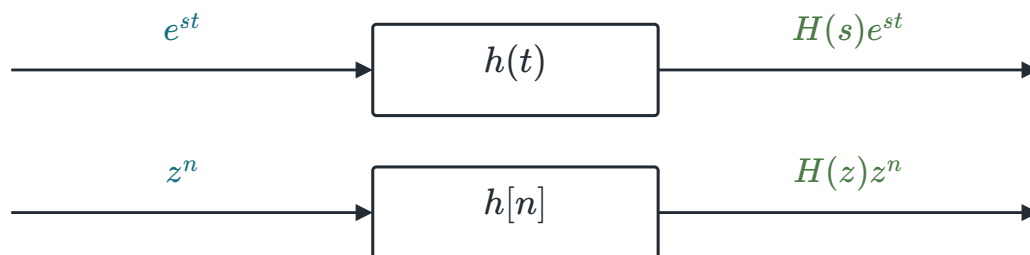
EIGENFUNCTION PROPERTY

$$e^{st} \rightarrow h(t) \rightarrow H(s)e^{st}, \quad H(s) = \int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau$$

$$z^n \rightarrow h[n] \rightarrow H(z)z^n, \quad H(z) = \sum_{k=-\infty}^{\infty} h[k] z^{-k}$$

The discrete-time output is a sequence, $y[n] = H(z)z^n$. Nothing in that line is a function of a continuous variable, and writing the continuous-time result there is the commonest slip in this section.

Setting $s = j\omega$ and $z = e^{j\omega}$ with ω real gives the two **frequency responses**, $H(j\omega)$ and $H(e^{j\omega})$. The discrete-time one repeats every 2π in ω , because $e^{j(\omega+2\pi)k} = e^{j\omega k}$ for integer k . The continuous-time one does not repeat at all.



The two statements, side by side, so that neither can borrow the other's variables.

EXAMPLE 4.1 – A PURE DELAY

GIVEN An LTI system with $y(t) = x(t - 3)$.

FIND The output for $x(t) = e^{j2t}$, and for $x(t) = \cos(4t) + \cos(7t)$, using eigenfunctions only.

METHOD Read h off the definition, integrate to get $H(s)$, then apply the eigenfunction property to each exponential.

SOLUTION $h(t) = \delta(t - 3)$, so by sifting $H(s) = \int_{-\infty}^{\infty} \delta(\tau - 3)e^{-s\tau} d\tau = e^{-3s}$. Then $e^{j2t} \rightarrow e^{-6j}e^{j2t} = e^{j2(t-3)}$. Expanding each cosine into two exponentials and repeating the step gives

$$y(t) = \cos(4(t - 3)) + \cos(7(t - 3)).$$

CHECK Apply $y(t) = x(t - 3)$ directly to both inputs. The answers agree. Note also that $|H(j\omega)| = 1$ and $\angle H(j\omega) = -3\omega$: a delay changes no amplitude and adds a phase proportional to frequency.

The point of the chapter is now in view. If $x = \sum_k a_k e^{s_k t}$, then $y = \sum_k a_k H(s_k) e^{s_k t}$, and no convolution is needed. Everything depends on being able to write x that way.

4.2 Existence: the Dirichlet conditions

Not every signal can be decomposed into complex exponentials. Periodic signals can, provided they satisfy three conditions. Let $x(t) = x(t + T)$ for every t .

1. Over one period, x is absolutely integrable: $\int_T |x(t)| dt < \infty$.
2. In any finite interval, x has only a finite number of maxima and minima.
3. In any finite interval, x has only a finite number of discontinuities, and each is finite.

SUFFICIENT, NOT NECESSARY

A signal meeting all three has a Fourier series that converges to it wherever it is continuous, and to the midpoint of the jump at a discontinuity. A signal failing one of them may still have a series. The conditions guarantee; they do not exclude.

The three are independent. The signal $1/t$ on each period fails the first while the other two hold. The signal $\sin(2\pi/t)$ has finite area but oscillates infinitely often, failing only the second. A staircase whose steps halve in height and width satisfies the first two and fails the third.

4.3 The synthesis and analysis equations

CONTINUOUS-TIME FOURIER SERIES

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}, \quad \omega_0 = \frac{2\pi}{T_0}$$
$$a_k = \frac{1}{T_0} \int_{T_0} x(t) e^{-jk\omega_0 t} dt$$

The first equation is synthesis, the second is analysis. One period symbol is used throughout: T_0 , with $\omega_0 = 2\pi/T_0$ stated once here. The sign of the exponent is the only thing that separates the pair.

Proof of the analysis equation

Multiply the synthesis equation by $e^{-jn\omega_0 t}$ and integrate over one period:

$$\int_{T_0} x(t) e^{-jn\omega_0 t} dt = \sum_{k=-\infty}^{\infty} a_k \int_{T_0} e^{j(k-n)\omega_0 t} dt.$$

Integrating from $-T_0/2$ to $T_0/2$ and using $\omega_0 T_0 = 2\pi$ gives $T_0 \sin((k-n)\pi) / ((k-n)\pi)$. For $k \neq n$ the sine of an integer multiple of π is zero. For $k = n$ the expression is $0/0$, and one application of l'Hôpital's rule in the variable $(k - n)$ gives T_0 .

ORTHOGONALITY OF COMPLEX EXPONENTIALS

$$\int_{T_0} e^{j(k-n)\omega_0 t} dt = \begin{cases} T_0, & k = n \\ 0, & k \neq n \end{cases}$$

Only the term with $k = n$ survives, leaving $\int_{T_0} x(t) e^{-jn\omega_0 t} dt = T_0 a_n$.

THE DC TERM

Setting $k = 0$ makes the exponential 1, so

$$a_0 = \frac{1}{T_0} \int_{T_0} x(t) dt,$$

the average value of the signal over one period. It is the fastest available check on any coefficient calculation, because the average of a drawn signal can be estimated by eye.

Fundamental period of a sum

A sum of periodic signals is periodic when the ratio of every pair of component periods is rational. The fundamental period of the sum is the smallest positive number that is a whole multiple of each component period. Write each period as a fraction in lowest terms:

PERIOD OF A SUM

$$T_0 = \text{LCM}\left(\frac{p_1}{q_1}, \frac{p_2}{q_2}, \dots\right) = \frac{\text{LCM}(p_1, p_2, \dots)}{\text{GCD}(q_1, q_2, \dots)}$$

The result is a least common multiple of fractions, so it need not be an integer. A constant term has no period and is left out.

WHERE THIS GOES WRONG

Putting every period over one common denominator first, then taking the least common multiple of the numerators, is the same rule in disguise — because for equal denominators $\text{LCM}(d, d) = \text{GCD}(d, d) = d$. Applying LCM to the denominators *without* that rewrite is wrong: for periods $2/9$ and $8/21$ it gives $8/63$, which is a multiple of neither. The correct answer is $\text{LCM}(2, 8) / \text{GCD}(9, 21) = 8/3$. Always divide the answer by each period and confirm both results are whole numbers.

EXAMPLE 4.2 – COEFFICIENTS OF A SUM OF SINUSOIDS

GIVEN $x(t) = 1 + \frac{1}{2} \cos(2\pi t) + \sin(3\pi t)$.

FIND Every Fourier series coefficient, with magnitude and phase.

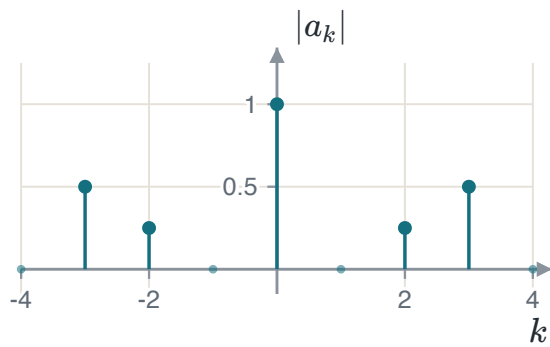
METHOD Find T_0 , hence ω_0 . Expand with Euler's relations and match each exponent to $jk\omega_0 t$.

SOLUTION The component periods are 1 and $2/3$ s, so $T_0 = \text{LCM}(1, 2) / \text{GCD}(1, 3) = 2$ s and $\omega_0 = \pi$ rad/s. Then

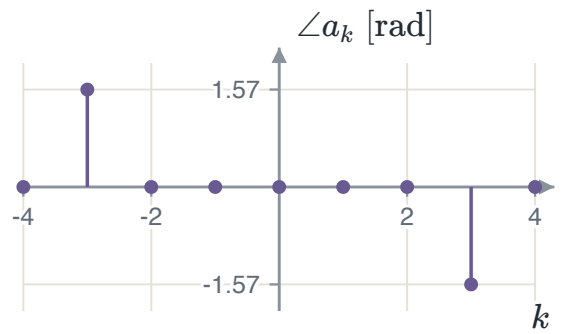
$$x(t) = 1e^{j0\omega_0 t} + \frac{1}{4}e^{j2\omega_0 t} + \frac{1}{4}e^{-j2\omega_0 t} + \frac{1}{2j}e^{j3\omega_0 t} - \frac{1}{2j}e^{-j3\omega_0 t},$$

so $a_0 = 1$, $a_{\pm 2} = 1/4$, $a_3 = 1/(2j)$, $a_{-3} = -1/(2j)$, and every other coefficient is zero.

CHECK $a_0 = 1$ is also the average of the signal: both sinusoids complete whole cycles in one period and average to zero, leaving the constant. Magnitudes: $|a_0| = 1$, $|a_{\pm 2}| = 1/4$, $|a_{\pm 3}| = 1/2$. Phases: $\angle a_3 = \angle 1 - \angle 2j = -\pi/2$ and $\angle a_{-3} = \angle(-1) - \angle 2j = +\pi/2$. The signal is real, so magnitudes are even in k and phases odd — which they are.



Magnitudes, even in k .



Phases, odd in k . A value of ± 1.57 rad is $\pm \pi/2$.

4.4 Two standard waveforms

EXAMPLE 4.3 – THE PERIODIC RECTANGULAR WAVE

GIVEN $x(t) = 1$ for $|t| < T_1$ and 0 for $T_1 < |t| < T_0/2$, repeated with period T_0 .

FIND Every coefficient.

METHOD Apply the analysis equation over $-T_0/2$ to $T_0/2$. The signal is 1 only on $[-T_1, T_1]$, so those become the limits. The case $k = 0$ is separate, because the antiderivative divides by k .

SOLUTION For $k \neq 0$,

$$a_k = \frac{1}{T_0} \int_{-T_1}^{T_1} e^{-jk\omega_0 t} dt = \frac{2 \sin(k\omega_0 T_1)}{k\omega_0 T_0} = \frac{\sin(2\pi k T_1 / T_0)}{\pi k},$$

and for $k = 0$, $a_0 = 2T_1/T_0$.

CHECK As $k \rightarrow 0$ the second branch tends to $2T_1/T_0$, because $\sin \theta / \theta \rightarrow 1$: the two branches meet, so the split is bookkeeping and not a discontinuity. The signal is real and even, so the coefficients are real and even — and they are, since $\sin$ and k are both odd. The zeros fall where k is a non-zero multiple of $T_0/(2T_1)$.

The envelope, and the samples taken from it

These coefficients are values of one continuous function of ω , read at equally spaced points. Write the function first:

ENVELOPE OF A SINGLE PULSE

$$E(\omega) = \int_{-T_1}^{T_1} e^{-j\omega t} dt = \frac{2 \sin(\omega T_1)}{\omega}$$

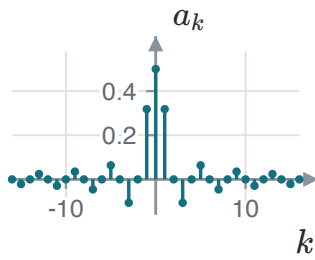
$$a_k = \frac{1}{T_0} E(k\omega_0) = \frac{\sin(2\pi k T_1 / T_0)}{\pi k}$$

E depends on the shape of one pulse and not on how often it repeats. In the unnormalised convention $\text{sinc}(\theta) = \sin \theta / \theta$, it is $E(\omega) = 2T_1 \text{sinc}(\omega T_1)$.

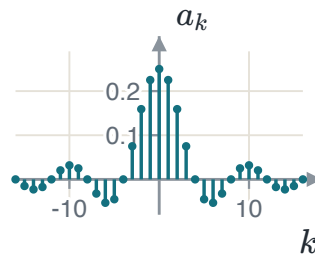
TWO OBJECTS, NOT ONE

$E(\omega)$ is a function of a continuous variable; a_k is a sequence indexed by an integer. Evaluating the envelope at the single point $\omega = \omega_0$ gives $T_0 a_1$, the first harmonic only. As a function of the harmonic index the coefficient is $T_0 a_k = T_0 \sin(2\pi k T_1 / T_0) / (\pi k)$, with k present in both places. Never join the tops of coefficient stems with a line: it claims the signal contains frequencies between the harmonics, which a periodic signal does not.

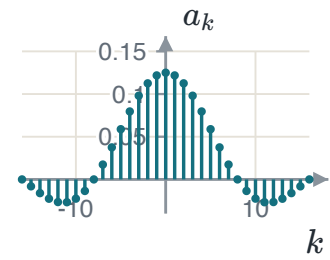
Lengthen T_0 and keep the pulse fixed. The envelope stays where it is; only the sample spacing $\omega_0 = 2\pi/T_0$ shrinks, and every coefficient falls as $1/T_0$. That observation is what produces the Fourier transform in the next chapter.



$$T_0 = 4T_1.$$



$$T_0 = 8T_1.$$



$$T_0 = 16T_1.$$

EXAMPLE 4.4 – THE SAWTOOTH WAVE

GIVEN $x(t) = t$ for $-T_0/2 < t < T_0/2$, repeated with period T_0 .

FIND Every coefficient.

METHOD The DC term first, then integration by parts, $\int_b^c t e^{at} dt = \frac{1}{a^2} [(at - 1)e^{at}]_b^c$.

SOLUTION $a_0 = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} t dt = 0$. For $k \neq 0$, using $\omega_0 T_0/2 = \pi$ and $\sin(k\pi) = 0$,

$$a_k = \frac{jT_0}{2k\pi} \cos(k\pi) = \frac{jT_0(-1)^k}{2k\pi}.$$

CHECK The signal is real and odd, so the coefficients must be purely imaginary — and they are. For $T_0 = 1$ the largest is $|a_{\pm 1}| = 1/(2\pi) \approx 0.159$. Every phase is $+\pi/2$ or $-\pi/2$, with the sign alternating in k and flipping again with the sign of k .

EXAMPLE 4.5 – THE PERIODIC IMPULSE TRAIN

GIVEN $x(t) = \sum_{m=-\infty}^{\infty} \delta(t - mT_0)$.

FIND Every coefficient.

METHOD Take the period from $-T_0/2$ to $+T_0/2$. Exactly one impulse lies inside it, so sifting finishes the integral.

SOLUTION $a_k = \frac{1}{T_0} \int_{-T_0/2}^{+T_0/2} \delta(t) e^{-jk\omega_0 t} dt = \frac{1}{T_0}$ for every k .

CHECK Both limits must be written and they must differ. An interval whose two ends are the same number has zero length, and every integral over it is zero, impulse or not. The answer is real, positive and the same at every harmonic: at $t = 0$ all harmonics add in step, and everywhere else they cancel.

4.5 How many harmonics are enough

TRUNCATION AND ITS ERROR

$$x_N(t) = \sum_{k=-N}^N a_k e^{jk\omega_0 t}, \quad e_N(t) = x(t) - x_N(t)$$

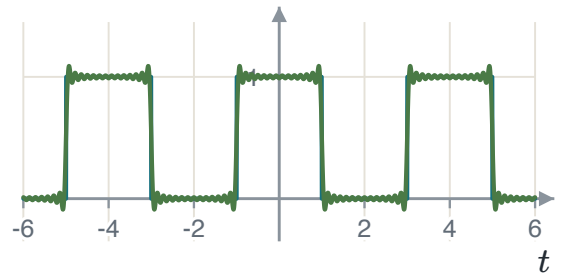
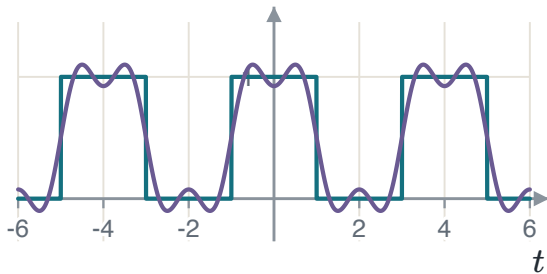
$$\text{MSE} = \frac{1}{T_0} \int_{T_0} |e_N(t)|^2 dt = \sum_{|k| > N} |a_k|^2$$

The second form follows from Parseval's relation in Section 4.7. It turns a question about a difference of signals into a question about a tail of a series.

THE GIBBS PHENOMENON

Near a jump the partial sum overshoots, and the overshoot does not shrink as N grows: its height settles at about **8.95% of the size of the jump**. What does shrink is its width. That is how the mean-square error can go to zero while the largest error does not.

N	RECTANGULAR WAVE, $T_0 = 4T_1$	SAWTOOTH, $T_0 = 1$
3	0.025	0.0144
9	0.010	0.0053
27	0.004	0.0018
81	0.001	0.0006



$N = 3$.

$N = 27$. The spike beside each jump is no shorter.

The rate at which the error falls is set by the smoothness of the signal. A jump forces the coefficients to decay like $1/k$, so the tail $\sum_{|k|>N} |a_k|^2$ behaves like $1/N$ and doubling N roughly halves the error. A signal that is continuous but has a corner — a triangular wave — has coefficients decaying like $1/k^2$, a tail like $1/N^3$, and needs far fewer harmonics.

4.6 The discrete-time Fourier series

DISCRETE-TIME FOURIER SERIES

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n}$$

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk(2\pi/N)n}$$

$\langle N \rangle$ means any N consecutive values of the index. Both sums are finite and there are exactly N distinct coefficients.

THE COEFFICIENTS REPEAT

$a_{k+N} = a_k$. Replacing k by $k + N$ in the analysis sum multiplies the summand by $(e^{-j2\pi})^n$, which is 1 because n is an integer. The same substitution in continuous time produces $e^{-j2\pi t}$, which is 1 only when t is an integer — and it is not. Continuous-time coefficients therefore do not repeat.

A FINITE, EXACT IDENTITY

A period- N sequence is described by N numbers and its series has N terms. There is no truncation, no limit, no convergence question, and no Gibbs phenomenon anywhere in discrete time.

EXAMPLE 4.6 – A SUM OF TWO SEQUENCES

GIVEN $x[n] = \sin\left(\frac{5\pi}{6}n\right) + \cos\left(\frac{3\pi}{4}n + \frac{\pi}{5}\right)$.

FIND The fundamental period and every coefficient.

METHOD A discrete-time sinusoid repeats only if $N = (2\pi/\omega)k$ is a positive integer for some integer k . Get each component period, take the least common multiple, then expand.

SOLUTION $N = 12k/5$ gives $N = 12$ at $k = 5$; $N = 8k/3$ gives $N = 8$ at $k = 3$. So $N_0 = \text{LCM}(12, 8) = 24$ and $\omega_0 = 2\pi/24$. Since $5\pi/6 = 10\omega_0$ and $3\pi/4 = 9\omega_0$,

$$a_{10} = \frac{1}{2j}, \quad a_{-10} = -\frac{1}{2j}, \quad a_9 = \frac{1}{2}e^{j\pi/5}, \quad a_{-9} = \frac{1}{2}e^{-j\pi/5},$$

with every other coefficient in one period equal to zero.

CHECK $|a_{\pm 9}| = |a_{\pm 10}| = 1/2$; $\angle a_9 = \pi/5$, $\angle a_{-9} = -\pi/5$, $\angle a_{10} = -\pi/2$, $\angle a_{-10} = +\pi/2$. The sequence is real, so magnitudes are even and phases odd — and $a_k = a_{k+24}$ extends the answer to every k .

EXAMPLE 4.7 – THE DISCRETE-TIME SQUARE WAVE

GIVEN $x[n] = 1$ for $|n| \leq N_1$ and 0 for $N_1 < |n| \leq N/2$, repeated with period N .

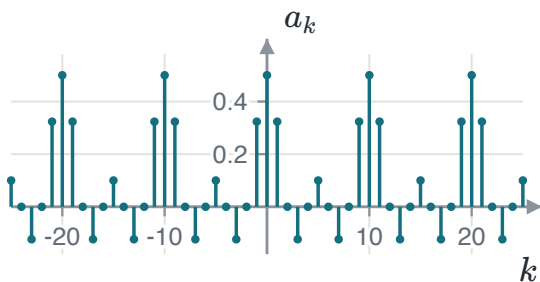
FIND Every coefficient.

METHOD The analysis sum reduces to a finite geometric sum with ratio $r = e^{-jk(2\pi/N)}$. Use $\sum_{n=m}^p r^n = (r^m - r^{p+1})/(1 - r)$, which requires only $r \neq 1$.

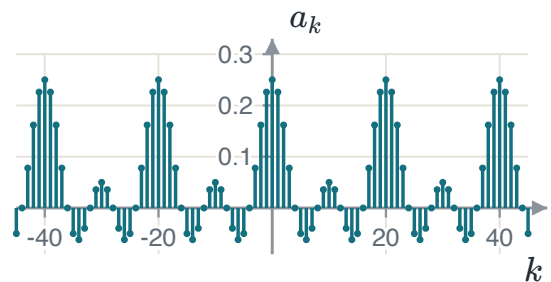
SOLUTION The lower limit is $n = -N_1$, so the first numerator term is $r^{-N_1} = e^{+jk(2\pi/N)N_1}$, with a positive exponent. Factoring $e^{-jk\pi/N}$ out of numerator and denominator turns each bracket into $2j \sin(\cdot)$, and these cancel, leaving

$$a_k = \frac{1}{N} \frac{\sin\left(\frac{2\pi k}{N} \left(N_1 + \frac{1}{2}\right)\right)}{\sin\left(\frac{\pi k}{N}\right)}, \quad a_k = \frac{2N_1 + 1}{N} \text{ when } k = 0, \pm N, \pm 2N, \dots$$

CHECK Here $|r| = 1$ at every k , so a condition $|r| < 1$ would rule the sum out altogether. What matters is $r \neq 1$, and $r = 1$ happens exactly when k is a multiple of N — which is the second branch. For $N = 10$, $N_1 = 2$ the peak is $(2N_1 + 1)/N = 0.5$, and any plot of these coefficients must be scaled so that 0.5 falls inside the labelled range.



$N = 10$, $N_1 = 2$: peak 0.5, repeating every 10.



$N = 20$: peak 0.25, repeating every 20.

EXAMPLE 4.8 – A DISCRETE SAWTOOTH

GIVEN $x[n] = n$ for $-5 \leq n \leq 5$, repeated with period $N = 11$.

FIND Every coefficient.

METHOD The sequence is odd, so pair the term at n with the term at $-n$ before expanding: $ne^{-jk\omega_0 n} + (-n)e^{+jk\omega_0 n} = -2jn \sin(k\omega_0 n)$.

SOLUTION $a_0 = \frac{1}{11} \sum_{n=-5}^5 n = 0$, and

$$a_k = -\frac{2j}{11} \sum_{m=1}^5 m \sin\left(\frac{2\pi km}{11}\right).$$

CHECK Every coefficient is purely imaginary, as a real odd sequence requires. The largest is $|a_{\pm 1}| = 1.7747$, and $a_k = a_{k+11}$, so the eleven coefficients from $k = -5$ to $k = 5$ are the whole answer.

4.7 Properties

PROPERTY	SIGNAL	COEFFICIENTS
Linearity	$Ax_1 + Bx_2$	$Aa_k + Bb_k$
Time shift, continuous	$x(t - t_0)$	$a_k e^{-jk\omega_0 t_0}$
Time shift, discrete	$x[n - n_0]$	$a_k e^{-jk\omega_0 n_0}$
Time reversal	$x(-t)$ or $x[-n]$	a_{-k}
Conjugation	$x^*(t)$ or $x^*[n]$	a_{-k}^*
Frequency shift, continuous	$e^{jM\omega_0 t} x(t)$	a_{k-M} , M an integer
Frequency shift, discrete	$e^{jM(2\pi/N)n} x[n]$	a_{k-M} , M an integer
Time scaling, continuous	$x(\alpha t)$, $\alpha > 0$	a_k , with period T_0/α
Time scaling, discrete	$x_{(m)}[n]$, m a positive integer	$\frac{1}{m} a_k$, with period mN
Periodic convolution, continuous	$\int_{T_0} x(\tau)y(t - \tau)d\tau$	$T_0 a_k b_k$
Periodic convolution, discrete	$\sum_{r=\langle N \rangle} x[r]y[n - r]$	$N a_k b_k$
Multiplication, continuous	$x(t)y(t)$	$\sum_{\ell=-\infty}^{\infty} a_\ell b_{k-\ell}$
Multiplication, discrete	$x[n]y[n]$	$\sum_{\ell=\langle N \rangle} a_\ell b_{k-\ell}$
Differentiation	$dx(t)/dt$	$jk\omega_0 a_k$
Integration	$\int_{-\infty}^t x(\tau)d\tau$	$a_k/(jk\omega_0)$, only if $a_0 = 0$
First difference	$x[n] - x[n - 1]$	$(1 - e^{-jk(2\pi/N)})a_k$
Running sum	$\sum_{r=-\infty}^n x[r]$	$a_k/(1 - e^{-jk(2\pi/N)})$, only if $a_0 = 0$

Three of these deserve a second reading. **Time scaling** in continuous time leaves the coefficients untouched and moves the frequencies they sit on, because $x(\alpha t)$ keeps every value of the signal and only changes how fast it is traversed. In discrete time there is nowhere closer to move to, so $x_{(m)}[n]$ inserts $m - 1$ zeros after every sample; nothing is discarded and nothing invented, which is why the coefficients survive, and the factor $1/m$ appears because the average is now taken over m times as many samples. **Periodic convolution** carries the factor T_0 or N in front of the product: the analysis equation divides by the period once and the convolution integrates over it once more. Dropping that factor scales every answer by the period, and the error is invisible in a sketch of the shape. **Integration** and the **running sum** both divide by a factor that vanishes at $k = 0$, which is where their condition comes from: a signal with a non-zero mean integrates to something that grows without bound, so it is neither finite-valued nor periodic. Subtract the mean, apply the rule to the rest, and add the ramp $a_0 t$ back at the end.

The time-shift factor has modulus 1, so $|b_k| = |a_k|$ at every k : a shift never changes a magnitude, only a phase, and it changes that phase by $-k\omega_0 t_0$, proportionally to the harmonic index. Two signals differing only by a delay have identical magnitude plots,

which is why a phase plot is not decoration.

THE DISCRETE-TIME PRODUCT

The multiplication property in discrete time is a **periodic** convolution: the sum runs over one period of ℓ . Discrete-time coefficients repeat, so an infinite sum would add each of the N distinct products over and over and diverge. The product $x[n]y[n]$ is itself periodic with period N and therefore has N coefficients — a number only a sum over one period can deliver.

Symmetry

- x real gives $a_{-k} = a_k^*$, so $|a_k|$ is even in k and $\angle a_k$ is odd.
- x real and even gives a_k real and even.
- x real and odd gives a_k purely imaginary and odd.
- For real x : $\mathcal{E}\mathbf{v}\{x\} \leftrightarrow \text{Re}\{a_k\}$ and $\mathcal{O}\mathbf{d}\{x\} \leftrightarrow j \text{Im}\{a_k\}$.

The last line is the general statement and the two before it are its special cases. $\mathcal{E}\mathbf{v}\{x\} = \frac{1}{2}[x(t) + x(-t)]$ has coefficients $\frac{1}{2}(a_k + a_{-k})$ by linearity and time reversal, and $a_{-k} = a_k^*$ for a real signal, so that is $\text{Re}\{a_k\}$. The odd part gives $\frac{1}{2}(a_k - a_{-k}^*) = j \text{Im}\{a_k\}$. Splitting a real signal into its even and odd parts splits its coefficients into their real and imaginary parts, and that is what tells you which half of the coefficients you actually have to compute.

The third follows in one line. Real gives $a_{-k} = a_k^*$; odd gives $a_{-k} = -a_k^*$; together $a_k^* = -a_k$, so $\text{Re}\{a_k\} = 0$. Both sawtooths of this chapter obey it.

PARSEVAL'S RELATION

$$\frac{1}{T_0} \int_{T_0} |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |a_k|^2$$

$$\frac{1}{N} \sum_{n=\langle N \rangle} |x[n]|^2 = \sum_{k=\langle N \rangle} |a_k|^2$$

The left side is the average power over one period, with the normalised convention $R = 1 \Omega$. The right side splits it among the harmonics, and $|a_k|^2$ is the power carried by the k -th harmonic alone.

4.8 A periodic signal through an LTI system

ONE MULTIPLICATION PER HARMONIC

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \longrightarrow y(t) = \sum_{k=-\infty}^{\infty} \underbrace{a_k H(jk\omega_0)}_{b_k} e^{jk\omega_0 t}$$

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_0 n} \longrightarrow y[n] = \sum_{k=\langle N \rangle} a_k H(e^{jk\omega_0}) e^{jk\omega_0 n}$$

No flipping, no shifting, no case boundaries. The whole system is the single function H .

Putting the output back together

The result above is a sum of complex terms. For real x and real h , $a_{-k} = a_k^*$ and $H(-j\omega) = H^*(j\omega)$, so $b_{-k} = b_k^*$. Pair each k with $-k$:

CONJUGATE-PAIR REASSEMBLY

$$b_k e^{jk\omega_0 t} + b_{-k} e^{-jk\omega_0 t} = 2|b_k| \cos(k\omega_0 t + \angle b_k)$$

$$y(t) = b_0 + \sum_{k=1}^{\infty} 2|b_k| \cos(k\omega_0 t + \angle b_k)$$

b_0 has no partner, so it carries no factor of two.

THE FACTOR OF TWO, AND WHOSE PHASE

The amplitude of the k -th cosine is $2|b_k|$, because both members of the pair contribute. The phase is $\angle b_k$, the one belonging to **positive** k ; taking $\angle b_{-k}$ flips the sign of that cosine's phase. Test any reassembly by setting $H = 1$: it must return the input exactly. A version missing the factor of two returns half the input plus its full average.

EXAMPLE 4.9 – LOW-PASS FILTERING

GIVEN $x(t) = 1 + \cos(\pi t) + \sin(2\pi t) + \cos(3\pi t + \frac{\pi}{3})$ into a system with $h(t) = e^{-t}u(t)$.

FIND The output signal.

METHOD $H(j\omega)$ from h ; T_0 and the a_k from x ; then $b_k = a_k H(jk\omega_0)$ and the pairing.

SOLUTION $H(j\omega) = 1/(1 + j\omega)$, so $|H| = 1/\sqrt{1 + \omega^2}$ and $\angle H = -\arctan \omega$. The component periods 2, 1 and $2/3$ s give $T_0 = 2$ s and $\omega_0 = \pi$. With $a_0 = 1$, $a_{\pm 1} = \frac{1}{2}$, $a_2 = \frac{1}{2j}$, $a_3 = \frac{1}{2}e^{j\pi/3}$:

$$b_0 = 1, \quad b_1 = 0.1517e^{-j1.2626}, \quad b_2 = 0.0786e^{-j2.9838}, \quad b_3 = 0.0528e^{-j0.4178},$$

and therefore

$$y(t) = 1 + 0.303 \cos(\pi t - 1.263) + 0.157 \cos(2\pi t - 2.984) + 0.106 \cos(3\pi t - 0.418).$$

CHECK Each amplitude is $2|b_k|$: 0.303, 0.157, 0.106. The output swings from about 0.615 to 1.417; halved amplitudes would give only 0.81 to 1.21, a visibly flatter signal. The average is untouched because $|H(0)| = 1$, and the third harmonic is cut to about a tenth.

EXAMPLE 4.10 – HIGH-PASS FILTERING

GIVEN The same $x(t)$, now into a system with $H(j\omega) = \frac{j\omega}{1 + j\omega}$.

FIND The output signal.

METHOD Unchanged. Only H is different.

SOLUTION $H(j0) = 0$, so $b_0 = 0$ and the output has zero average. Then $b_1 = 0.4764e^{+j0.3082}$, $b_2 = 0.4938e^{-j1.4130}$, $b_3 = 0.4972e^{+j1.1530}$, and

$$y(t) = 0.953 \cos(\pi t + 0.308) + 0.988 \cos(2\pi t - 1.413) + 0.994 \cos(3\pi t + 1.153).$$

CHECK $|H(jk\pi)|$ is 0.953, 0.988, 0.994 for $k = 1, 2, 3$: nearly unity, so harmonics of unit amplitude must come out with amplitudes near 1. Values near 0.5 would mean the factor of two was lost. The phases come from positive k : $+0.308$, -1.413 , $+1.153$. Reading them off b_{-k} reverses the first and third while leaving the second looking plausible, which is what makes that error hard to see.

EXAMPLE 4.11 – FILTERING A DISCRETE IMPULSE TRAIN

GIVEN $x[n] = \sum_{m=-\infty}^{\infty} \delta[n - 4m]$, and the systems $h_1[n] = 0.5\delta[n] - 0.5\delta[n - 1]$ and $h_2[n] = 0.5\delta[n] + 0.5\delta[n - 1]$.

FIND Both output signals.

METHOD a_k for the impulse train, then $b_k = a_k H(e^{jk\omega_0})$ with $\omega_0 = \pi/2$, then the pairing.

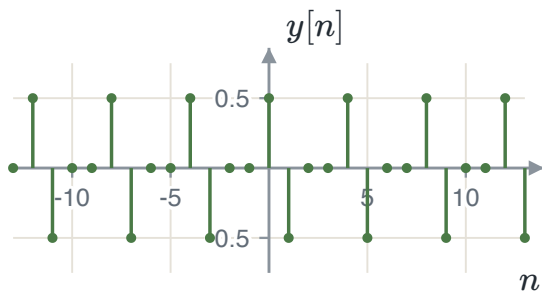
SOLUTION $a_k = \frac{1}{4}$ for every k . For h_1 , $H_1(e^{j\omega}) = 0.5(1 - e^{-j\omega})$, giving $b_0 = 0$, $b_1 = 0.1768e^{j\pi/4}$ and $b_2 = 0.25$, so

$$y[n] = 0.36 \cos\left(\frac{\pi}{2}n + \frac{\pi}{4}\right) + 0.25(-1)^n.$$

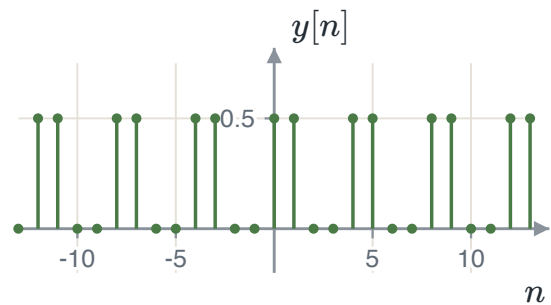
For h_2 , $H_2(e^{j\omega}) = 0.5(1 + e^{-j\omega})$, giving $b_0 = 0.25$, $b_1 = 0.1768e^{-j\pi/4}$ and $b_2 = 0$, so

$$y[n] = 0.25 + 0.36 \cos\left(\frac{\pi}{2}n - \frac{\pi}{4}\right).$$

CHECK Apply the two systems directly: $0.5x[n] \mp 0.5x[n - 1]$ reproduces both answers sample for sample. The $k = 2$ term sits at the edge of the period and has no distinct partner inside it, so it stands alone as $b_2(-1)^n$ and carries no factor of two. The high-pass system removes the average and keeps b_2 ; the low-pass system does the opposite.



First difference: zero average.



Two-point average: the average survives.

4.9 Checklist, and what comes next

1. Confirm the signal is periodic and find T_0 or N before anything else.
2. Set $\omega_0 = 2\pi/T_0$ or $2\pi/N$ and write every component frequency as a multiple of it. That multiple is k .
3. If the signal is already a sum of sinusoids, read the coefficients off with Euler's relations. Otherwise integrate or sum, treating $k = 0$ separately.
4. Check a_0 against the average of the signal by eye, and check the symmetry: real gives $a_{-k} = a_k^*$.
5. Through an LTI system, $b_k = a_k H(jk\omega_0)$ – one product per harmonic.
6. To return to a real signal, pair k with $-k$: amplitude $2|b_k|$, phase $\angle b_k$. The term b_0 stands alone.

Everything in this chapter needed the signal to repeat. A single pulse does not repeat, has no fundamental period, and has no harmonics to carry coefficients. The rectangular wave already showed the way out: its coefficients were samples of one envelope, taken every $\omega_0 = 2\pi/T_0$. Lengthen the period and the envelope stays put while the samples crowd together and shrink as $1/T_0$.

WHERE CHAPTER 5 BEGINS

$$T_0 a_k = E(k\omega_0) \xrightarrow{T_0 \rightarrow \infty} X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

In the limit the stems merge into the curve they were always sampling. That curve is the Fourier transform.

4.10 Exercises

4.1 Two periodic signals have fundamental periods $2/9$ s and $8/21$ s. Find the fundamental period of their sum, and verify it by division.

Answer: $T_0 = \text{LCM}(2, 8) / \text{GCD}(9, 21) = 8/3$ s. Check: $(8/3)/(2/9) = 12$ and $(8/3)/(8/21) = 7$, both whole, and $\text{gcd}(12, 7) = 1$.

4.2 For $x(t) = 1 + \frac{1}{2} \cos(2\pi t) + \sin(3\pi t)$, state a_0 and explain in one sentence why it is not zero.

Answer: $a_0 = 1$: the constant term is the $k = 0$ harmonic, and it is also the average of the signal, since both sinusoids complete whole cycles in one period.

4.3 A rectangular wave has $T_0 = 8T_1$. Find a_0 and a_2 .

Answer: $a_0 = 2T_1/T_0 = 1/4$ and $a_2 = \sin(\pi/2)/(2\pi) = 1/(2\pi) \approx 0.159$.

4.4 Explain why the finite geometric sum used for the discrete-time square wave requires $r \neq 1$ and not $|r| < 1$, and say what happens at the excluded value.

Answer: A finite sum always has a value; only the closed form fails, where its denominator $1 - r$ vanishes. Here $|r| = 1$ at every k , so $|r| < 1$ would exclude the sum entirely. At $r = 1$, that is k a multiple of N , every term is 1 and the sum is $2N_1 + 1$.

4.5 A filtered signal has $b_1 = 0.1517e^{-j1.263}$ and $\omega_0 = \pi$. Write the contribution of the pair $k = \pm 1$ to the real output.

Answer: $2|b_1| \cos(\pi t + \angle b_1) = 0.303 \cos(\pi t - 1.263)$.

4.6 A square wave and a triangular wave are truncated to the same number of harmonics. Which is approximated better, and which shows a Gibbs overshoot?

Answer: The triangular wave: its coefficients decay like $1/k^2$ against $1/k$, so its discarded power falls like $1/N^3$ rather than $1/N$. Only the square wave has a jump, so only it shows a Gibbs overshoot.

The continuous-time Fourier transform

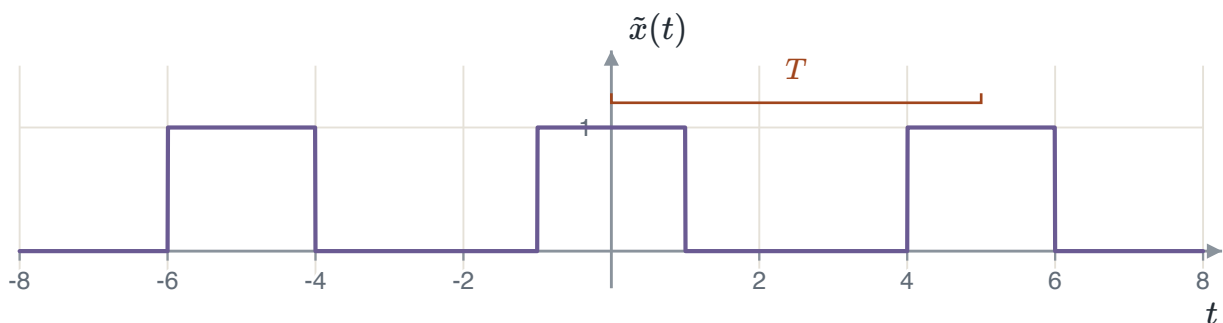
Chapter 4 needed the signal to repeat. A single pulse does not repeat, has no fundamental period, and therefore has no harmonics to carry coefficients. This chapter lets the period grow without bound and watches what the Fourier series becomes. The stems merge into a curve, the sum becomes an integral, and the result applies to almost every signal of finite energy.

5.1 From a series to a transform

Let $x(t)$ be zero for $|t| > T_1$. The number T_1 is the half-width of the support and belongs to the signal alone. Build a periodic signal $\tilde{x}$ by repeating the pulse every T seconds, with $T > 2T_1$ so that the copies do not touch:

$$\tilde{x}(t) = \sum_{m=-\infty}^{\infty} x(t - mT), \quad \tilde{x}(t) = \tilde{x}(t + T), \quad T > 2T_1.$$

Inside one period $\tilde{x}$ is the original pulse, so Chapter 4 applies to it. As T grows the pulse in the middle never changes and the copies beside it slide away, so $\tilde{x}(t) \rightarrow x(t)$ for every t .



The periodic extension with $T = 5T_1$. The condition $T > 2T_1$ is what keeps the copies apart.

The coefficients are samples of one curve

Apply the analysis equation of Chapter 4 to $\tilde{x}$ over one period. Inside that period $\tilde{x} = x$, and outside $|t| < T_1$ the integrand is zero, so the limits may be opened to all of time:

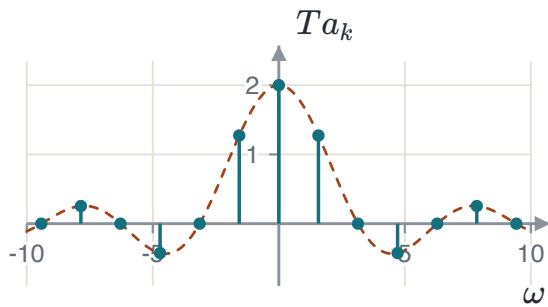
$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} \tilde{x}(t) e^{-jk\omega_0 t} dt = \frac{1}{T} \int_{-\infty}^{\infty} x(t) e^{-jk\omega_0 t} dt, \quad \omega_0 = \frac{2\pi}{T}.$$

The right-hand integral is one expression evaluated at the number $k\omega_0$. Give it a free variable and a name.

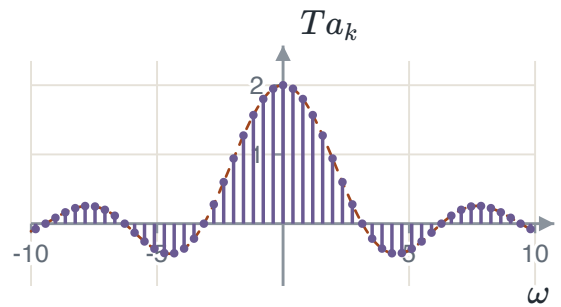
DEFINITION OF THE TRANSFORM

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

Then $a_k = \frac{1}{T} X(jk\omega_0)$, so every coefficient of the periodic extension is one point of the curve X , scaled by $1/T$. The curve was built from the pulse alone, so lengthening T cannot move it. What lengthening T does is shrink the spacing $\omega_0 = 2\pi/T$ and shrink each coefficient by the same factor.



$$T = 4T_1.$$



$$T = 16T_1: \text{ the same curve, sampled four times as finely.}$$

The sum becomes an integral

Put $a_k = \frac{1}{T} X(jk\omega_0)$ back into the synthesis equation and replace $1/T$ by $\omega_0/2\pi$. Nothing has been approximated yet.

$$\tilde{x}(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} X(jk\omega_0) e^{jk\omega_0 t} \omega_0.$$

Every term now carries the spacing ω_0 as a factor, which is the shape of a Riemann sum. Let $T \rightarrow \infty$: the left side becomes $x(t)$, the spacing becomes the differential $d\omega$, the sample points fill the axis, and the sum becomes an integral.

THE CONTINUOUS-TIME FOURIER TRANSFORM PAIR

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \quad (\text{analysis})$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega \quad (\text{synthesis})$$

Analysis integrates over t and leaves a function of ω : it takes the signal apart. Synthesis integrates over ω and leaves a function of t : it puts the signal back together. The factor $1/2\pi$ arrived as $\omega_0/2\pi$ when the spacing was substituted, and it stays on the synthesis side throughout. The pair is written $x(t) \leftrightarrow X(j\omega)$, with the argument $j\omega$ and never ω alone.

WHICH EQUATION IS WHICH

The variable of integration settles it, and nothing else needs to be remembered. Integrating time away produces a spectrum, and that is analysis. Integrating frequency away produces a signal, and that is synthesis. The two exponents differ by one minus sign, so the pair is as easy to write backwards as forwards, and the names are the only part that can be exchanged by mistake.

5.2 When the transform exists

Two conditions each guarantee that $X(j\omega)$ exists. They are alternatives, joined by "or", and neither implies the other.

1. **Finite energy.** $\int_{-\infty}^{\infty} |x(t)|^2 dt < \infty$.

2. **The Dirichlet conditions.** x is absolutely integrable, $\int_{-\infty}^{\infty} |x(t)| dt < \infty$, and has finitely many maxima, minima and finite jumps in any finite interval.

That they are independent is shown by one example each way. The signal $\sin(t)/t$ has finite energy, and its modulus decays only like $1/|t|$, so its area diverges. The signal $1/\sqrt{t}$ on $0 < t < 1$ has area 2, and its square is $1/t$, whose integral over the same interval diverges.

Neither condition is necessary. A constant, a complex exponential and every periodic signal fail both and still have spectra, in the limiting sense: the transform is an impulse, defined by what it does inside an integral, exactly as $\delta(t)$ is. Taking $e^{-a|t|} \leftrightarrow 2a/(a^2 + \omega^2)$ and letting $a \rightarrow 0$ gives $1 \leftrightarrow 2\pi\delta(\omega)$, with the curve growing tall and narrow while its area stays 2π .

5.3 Basic transform pairs

EXAMPLE 5.1 – THE IMPULSE AND THE SHIFTED IMPULSE

GIVEN $x(t) = \delta(t)$, and then $x(t) = \delta(t - t_0)$.

FIND $X(j\omega)$ in both cases, with magnitude and phase.

METHOD Substitute into the analysis equation and use the sifting property.

SOLUTION $\mathcal{F}\{\delta(t)\} = \int \delta(t)e^{-j\omega t} dt = 1$: every frequency present, equally, with no phase. For the shifted impulse the sifting happens at $t = t_0$, giving

$$\mathcal{F}\{\delta(t - t_0)\} = e^{-j\omega t_0}, \quad |X(j\omega)| = 1, \quad \angle X(j\omega) = -\omega t_0.$$

CHECK Push $e^{-j\omega t_0}$ back through the synthesis equation and the impulse returns. Reading: moving a signal in time never changes the size of any frequency component; it rotates each one by an amount proportional to its frequency.

EXAMPLE 5.2 – ONE IMPULSE IN FREQUENCY

GIVEN $X(j\omega) = 2\pi\delta(\omega - \omega_0)$.

FIND $x(t)$.

METHOD Synthesis, sifting in the variable ω .

SOLUTION $x(t) = \frac{1}{2\pi} \int 2\pi\delta(\omega - \omega_0)e^{j\omega t} d\omega = e^{j\omega_0 t}$. The 2π of the impulse weight and the $1/2\pi$ of the synthesis equation cancel exactly, which is why the weight is written as 2π and not as 1.

CHECK $|x(t)| = 1$ and $\angle x(t) = \omega_0 t$. With an impulse of weight 1 the answer would be $e^{j\omega_0 t}/2\pi$, which is not a unit-amplitude exponential.

TWO PAIRS THAT FOLLOW AT ONCE

$$e^{j\omega_0 t} \longleftrightarrow 2\pi\delta(\omega - \omega_0)$$

$$1 \longleftrightarrow 2\pi\delta(\omega)$$

The second is the first at $\omega_0 = 0$.

EXAMPLE 5.3 – THE ONE-SIDED EXPONENTIAL

GIVEN $x(t) = e^{-at}u(t)$, with a real.

FIND $X(j\omega)$, the condition on a , and the magnitude and phase.

METHOD Integrate from 0 to ∞ ; $u(t)$ removes the rest.

SOLUTION $X(j\omega) = \int_0^\infty e^{-(a+j\omega)t} dt$. The upper limit vanishes only when $a > 0$, because $|e^{-j\omega t}| = 1$ and the decay is carried by e^{-at} alone. So

$$e^{-at}u(t) \longleftrightarrow \frac{1}{a + j\omega}, \quad a > 0,$$

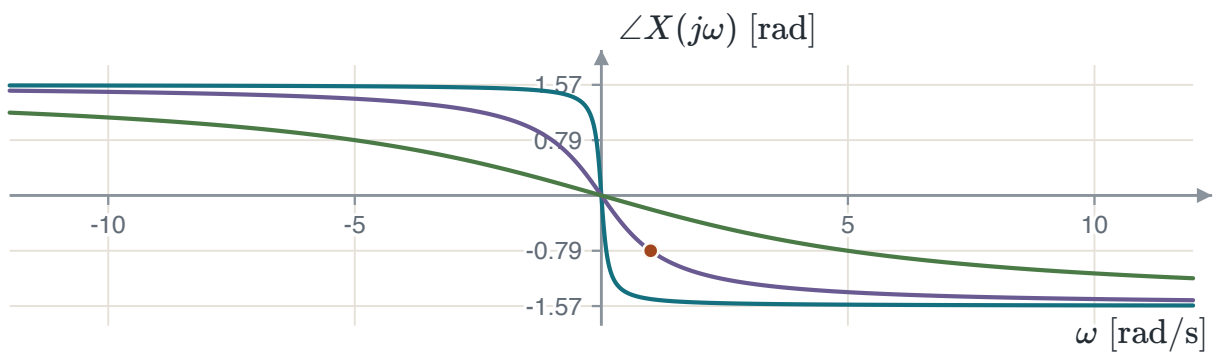
with $|X(j\omega)| = 1/\sqrt{a^2 + \omega^2}$ and, writing the angle of a quotient as a subtraction,

$$\angle X(j\omega) = \angle 1 - \angle(a + j\omega) = 0 - \tan^{-1}(\omega/a) = -\tan^{-1}(\omega/a).$$

CHECK $|X(j0)| = 1/a$ gives 10, 1 and 0.2 for $a = 0.1, 1, 5$. At $a = 1, \omega = 1$ the phase is -0.785398 rad; at $a = 0.1, \omega = 3$ it is -1.537475 ; at $a = 5, \omega = 2$ it is -0.380506 . Every one is negative, and the phase curve falls from $+\pi/2$ to $-\pi/2$. A curve that rises belongs to $+\tan^{-1}(\omega/a)$, which is the angle of the reciprocal.

WHERE THAT MINUS SIGN IS LOST

The numerator is the real number 1, so its angle is zero and it is tempting to skip that term and copy $\tan^{-1}(\omega/a)$ straight out of the denominator. What is then reported is $\angle(a + j\omega)$, the angle of the reciprocal of the answer. Write the subtraction out, including the term that is zero.



The phase of $1/(a + j\omega)$ for $a = 0.1, 1$ and 5 , with the marked value -0.785398 at $a = \omega = 1$.

EXAMPLE 5.4 – THE TWO-SIDED EXPONENTIAL

GIVEN $x(t) = e^{-a|t|}$ with $a > 0$.

FIND $X(j\omega)$.

METHOD $|t|$ means two formulas, so split at $t = 0$. On the left half the exponent is $+at$.

SOLUTION $X(j\omega) = \frac{1}{a - j\omega} + \frac{1}{a + j\omega} = \frac{2a}{a^2 + \omega^2}$. The imaginary parts cancelled when the fractions were added.

CHECK $X(j0) = 2/a$ gives 4, 2 and 0.4 for $a = 0.5, 1, 5$. The signal is real and even and the transform came out real and even, which is the general rule proved in Section 5.6. Note also that the transform is never zero: at $a = 1$ and $\omega = 10^6$ it is still 2×10^{-12} .

5.4 The rectangular pulse, the sinc, and the inverse relation

EXAMPLE 5.5 – THE RECTANGULAR PULSE

GIVEN $x(t) = 1$ for $|t| < T_1$ and 0 otherwise.

FIND $X(j\omega)$, its value at the origin, and its zeros.

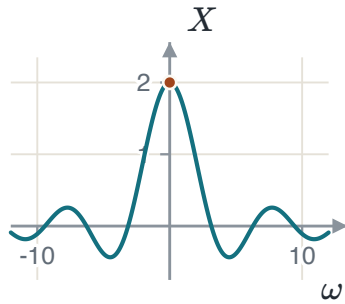
METHOD The analysis integral runs from $-T_1$ to T_1 with the integrand $e^{-j\omega t}$.

SOLUTION $X(j\omega) = \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-T_1}^{T_1} = \frac{e^{j\omega T_1} - e^{-j\omega T_1}}{j\omega} = \frac{2 \sin(\omega T_1)}{\omega}$, using $e^{j\theta} - e^{-j\theta} = 2j \sin \theta$. The j cancels, so the answer is real.

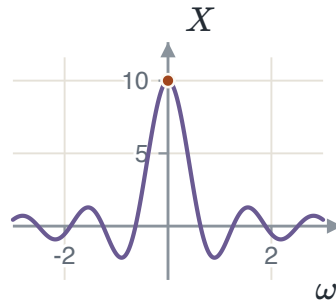
CHECK $X(j0) = 2T_1$ by l'Hôpital, which is the area under the pulse: 2, 10 and 20 for $T_1 = 1, 5, 10$. The zeros are at $\omega = \pm k\pi/T_1$ for $k = 1, 2, 3, \dots$ – the origin is excluded, because there the expression is $0/0$ and its limit is the peak.

THE SINC CONVENTION USED THROUGHOUT

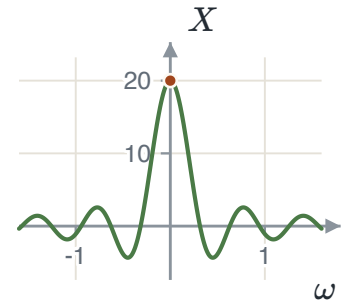
This course writes $\text{sinc}(\theta) = \frac{\sin \theta}{\theta}$, the **unnormalised** sinc, with $\text{sinc}(0) = 1$ and zeros at $\pm\pi, \pm2\pi, \dots$. In that convention the result above reads $X(j\omega) = 2T_1 \text{sinc}(\omega T_1)$. Signal-processing software and many communications texts use the **normalised** sinc, $\text{sinc}_n(\theta) = \sin(\pi\theta)/(\pi\theta)$, whose zeros are at the integers; there the same result is $2T_1 \text{sinc}_n(\omega T_1/\pi)$. The two expressions are equal, and the argument is not: it differs by a factor of π . An argument copied between the conventions agrees at the origin and is wrong everywhere else, so the convention is stated at every point of use.



$T_1 = 1$: peak 2.



$T_1 = 5$: peak 10.



$T_1 = 10$: peak 20.

EXAMPLE 5.6 – THE IDEAL LOW-PASS BAND

GIVEN $X(j\omega) = 1$ for $|\omega| < W$ and 0 otherwise.

FIND $x(t)$, its peak, and its zeros.

METHOD Synthesis from $-W$ to W , with the $1/2\pi$ in front.

SOLUTION
$$x(t) = \frac{1}{2\pi} \int_{-W}^W e^{j\omega t} d\omega = \frac{e^{jWt} - e^{-jWt}}{2\pi jt} = \frac{\sin(Wt)}{\pi t} = \frac{W}{\pi} \text{sinc}(Wt).$$

CHECK $x(0) = W/\pi$ by l'Hôpital: 0.5, 1 and 2 for $W = 0.5\pi, \pi, 2\pi$. Zeros at $t = \pm k\pi/W$, $k = 1, 2, \dots$, again with the origin excluded. This signal is not a pulse: it rings on both sides for ever, alternating in sign.

The two examples are one statement seen twice: a rectangle in either domain is a sinc in the other. Section 5.7 gives that symmetry a name and a proof.

Narrow in time, wide in frequency

The pulse of half-width T_1 has its first zero at $\omega = \pi/T_1$. Halve T_1 and that zero doubles. Taking the duration as the full width $T = 2T_1$ and the bandwidth as the distance to the first null,

$$T \times \text{BW} = 2T_1 \cdot \frac{\pi}{T_1} = 2\pi \quad \text{at every width.}$$

NOT A UNIVERSAL CONSTANT

The product is invariant **within one shape**, because scaling in time divides the duration and multiplies every frequency by the same factor. Change the shape and the number changes: a triangular pulse of the same total duration has its first null at $2\pi/T_1$, so its product is 4π . What is always true is the direction: narrowing a signal in time widens its spectrum, and the two cannot both be made small.

One implication about band limitation is a theorem and its converse is not. A signal of finite duration cannot be band-limited: the pulse above has a sinc transform, which is non-zero on stretches reaching out to every frequency. But infinite duration guarantees nothing. The signal $e^{-a|t|}$ lasts for ever and its transform $2a/(a^2 + \omega^2)$ is strictly positive at every finite frequency. Small is not zero, and Chapter 7 depends on the difference.

5.5 Periodic signals, sinusoids and the impulse train

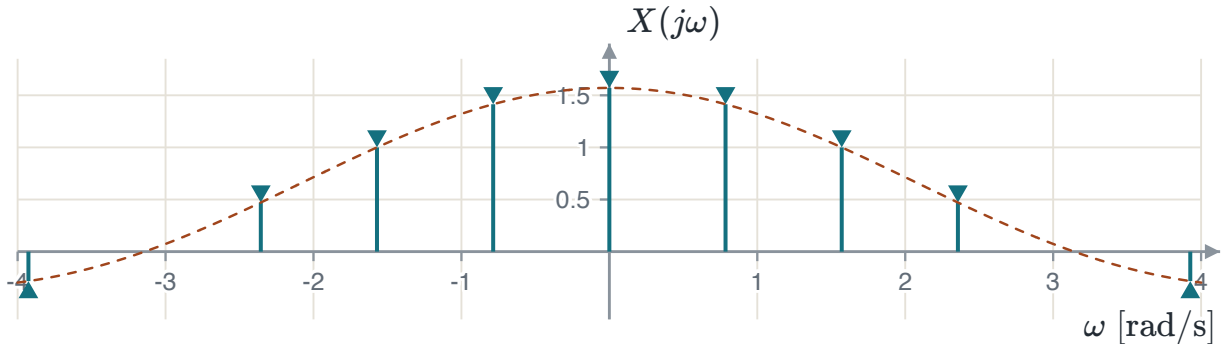
A periodic signal has a Fourier series, and the series can be transformed term by term with $e^{j\omega_0 t} \leftrightarrow 2\pi\delta(\omega - \omega_0)$.

TRANSFORM OF A PERIODIC SIGNAL

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \longleftrightarrow X(j\omega) = \sum_{k=-\infty}^{\infty} 2\pi a_k \delta(\omega - k\omega_0)$$

The spectrum is a train of impulses at the harmonics, and the impulse at $k\omega_0$ carries weight $2\pi a_k$. The coefficient and the weight are different objects: a_k multiplies a unit exponential, while $2\pi a_k$ is an area under a spectrum. Reporting the coefficients as the transform loses a factor of 2π at every harmonic.

For the rectangular wave of Chapter 4 with $T_1 = 1$, the weights are $2\pi a_k = 2 \sin(k\omega_0 T_1)/k$ and the impulse at the origin carries $2\pi a_0 = 4\pi T_1/T$. Three periods make the sequence visible: $T = 8T_1$ gives $\omega_0 = \pi/4 = 0.392699$ and weight 1.5708; $T = 16T_1$ gives $\omega_0 = \pi/8 = 0.196350$ and weight 0.7854; $T = 32T_1$ gives $\omega_0 = \pi/16 = 0.098175$ and weight 0.3927. Each doubling of the period halves the spacing and halves every weight, while the envelope does not move at all — which is the derivation of Section 5.1, carried out three steps at a time.



The square wave with $T = 8T_1$: impulses of weight $2\pi a_k$, sampling the dashed envelope. Some of the weights are negative, which a plot of $|a_k|$ would hide.

EXAMPLE 5.7 – A CONSTANT, A COSINE AND A SINE

GIVEN $x(t) = 5 + 4 \cos(3\pi t) + 6 \sin(4\pi t)$.

FIND $X(j\omega)$, drawn so that both size and phase can be read.

METHOD Expand each sinusoid with Euler's relations and transform term by term.

SOLUTION $4 \cos(3\pi t) = 2e^{j3\pi t} + 2e^{-j3\pi t}$, so its transform is $4\pi\delta(\omega - 3\pi) + 4\pi\delta(\omega + 3\pi)$: one impulse at each of $\pm 3\pi$. Likewise $6 \sin(4\pi t) = \frac{6}{2j} [e^{j4\pi t} - e^{-j4\pi t}]$ gives $\frac{6\pi}{j}\delta(\omega - 4\pi) - \frac{6\pi}{j}\delta(\omega + 4\pi)$, with imaginary weights of modulus $6\pi = 18.849556$. The constant contributes $10\pi\delta(\omega)$. Altogether

$$X(j\omega) = 10\pi\delta(\omega) + 4\pi\delta(\omega - 3\pi) + 4\pi\delta(\omega + 3\pi) + \frac{6\pi}{j}\delta(\omega - 4\pi) - \frac{6\pi}{j}\delta(\omega + 4\pi).$$

CHECK The signal is real, so the magnitude must be even in ω and the phase odd. Magnitudes: 31.4159 at 0, 12.5664 at $\pm 3\pi$, 18.8496 at $\pm 4\pi$. Phases: 0 except at $\pm 4\pi$, where they are $\mp\pi/2$ because $1/j = -j$. A spectrum with only the positive-frequency impulse of a cosine fails this test at once, and describes a complex signal.

DRAWING A COMPLEX SPECTRUM

Three of these weights are real and two are imaginary. Drawing an imaginary weight as a downward arrow on the same axis as a real one uses the vertical direction for two meanings at once, and the figure stops being readable. Draw magnitude and phase, or real part and imaginary part, and say in the caption which pair is shown.

EXAMPLE 5.8 – THE IMPULSE TRAIN

GIVEN $x(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT)$.

FIND $X(j\omega)$.

METHOD The signal is periodic with $T_0 = T$. Find a_k , then use the rule above.

SOLUTION Over one period exactly one impulse is enclosed, and sifting evaluates the exponential at $t = 0$, so $a_k = 1/T$ for every k . Hence

$$\sum_k \delta(t - kT) \longleftrightarrow \frac{2\pi}{T} \sum_k \delta\left(\omega - \frac{2\pi k}{T}\right).$$

CHECK Spacing and weight are the same number, $2\pi/T$: 6.2832 for $T = 1$ and 3.1416 for $T = 2$. Crowding the impulses in time spreads them in frequency. This one pair is the mechanism behind sampling, in Chapter 7. Note that the limits must enclose exactly one impulse; writing both as $-T/2$ encloses none.

5.6 Properties

PROPERTY	STATEMENT
Linearity	$ax_1 + bx_2 \leftrightarrow aX_1 + bX_2$
Time shift	$x(t - t_0) \leftrightarrow e^{-j\omega t_0} X(j\omega)$
Frequency shift	$e^{j\omega_0 t} x(t) \leftrightarrow X(j(\omega - \omega_0))$
Conjugation	$x^*(t) \leftrightarrow X^*(-j\omega)$
Real signal	$X(-j\omega) = X^*(j\omega)$
Time reversal	$x(-t) \leftrightarrow X(-j\omega)$
Time scaling	$x(at) \leftrightarrow \frac{1}{ a } X(j\omega/a)$
Differentiation in time	$d^n x/dt^n \leftrightarrow (j\omega)^n X(j\omega)$
Integration	$\int_{-\infty}^t x(\tau) d\tau \leftrightarrow \frac{1}{j\omega} X(j\omega) + \pi X(0)\delta(\omega)$
Differentiation in frequency	$t x(t) \leftrightarrow j dX(j\omega)/d\omega$
Duality	$X(t) \leftrightarrow 2\pi x(-\omega)$
Convolution	$x * h \leftrightarrow XH$
Multiplication	$xy \leftrightarrow \frac{1}{2\pi} X * Y$
Real and even	$X(j\omega)$ real and even
Real and odd	$X(j\omega)$ purely imaginary and odd
Even-odd parts	$\mathcal{E}\{x\} \leftrightarrow \text{Re}\{X\}, \mathcal{O}\{x\} \leftrightarrow j \text{Im}\{X\}$
Parseval	$\int x ^2 dt = \frac{1}{2\pi} \int X ^2 d\omega$

Shifts

For the time shift, substitute $\tau = t - t_0$; the limits are infinite and do not move, and $e^{-j\omega t_0}$ comes out of the integral. Since its modulus is 1, no magnitude changes at any frequency: a delay of t_0 seconds is exactly a linear phase of slope $-t_0$.

For the frequency shift, start from the expression the property states, $e^{+j\omega_0 t} x(t)$, and combine the two exponentials before anything else. The exponent becomes $-j(\omega - \omega_0)t$, which is the analysis integral read at $\omega - \omega_0$, and no substitution is needed. The operand is a complex exponential in **time** with ω_0 fixed; the time-shift kernel $e^{-j\omega t_0}$ has a fixed **time** and the opposite sign, and opening this proof with it proves a different statement.

Conjugation and the symmetry of a real signal

Conjugating the analysis integral flips the sign of j everywhere, which gives $x^*(t) \leftrightarrow X^*(-j\omega)$. If x is real then $x^* = x$, so the two sides have the same transform and $X(-j\omega) = X^*(j\omega)$. Everything follows from that one line: the real part of X is even, the imaginary part is odd, the magnitude is even and the phase is odd. A real and even signal has a real and even transform, since its imaginary part is both odd and even; a real and odd signal has a purely imaginary and odd transform.

REAL DOES NOT MEAN ZERO PHASE

A real transform can be negative. Where $X(j\omega) < 0$ the magnitude is $-X$ and the phase is π , not 0. Only a real and non-negative transform has zero phase everywhere, and the sinc of the rectangular pulse is the standing counterexample: it is real, and its side lobes are negative.

Differentiation

Differentiate the **synthesis equation**, not the signal. On the right the only factor depending on t is $e^{j\omega t}$, and ω is the variable of integration, so it is held fixed:

$$\frac{d}{dt} x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} [j\omega X(j\omega)] e^{j\omega t} d\omega \quad \implies \quad \frac{d^n x}{dt^n} \longleftrightarrow (j\omega)^n X(j\omega).$$

THE STEP TO AVOID

Pulling $j\omega$ out and writing $dx/dt = j\omega x(t)$ is false. ω is the variable being integrated away, so it is not a constant and it is not available outside the integral, and $j\omega x(t)$ is not a signal at all. One number settles it: for $x(t) = e^{-t^2}$, dx/dt at $t = 1$ is -0.735759 , a real number, while $j\omega x(t)$ at $t = 1$, $\omega = 3$ is $1.103638j$. The boxed result is correct; only the route is wrong, and the same route reappears on integration and on differential equations, where it does change the answer.

Integration, and the impulse it leaves behind

The running integral is a convolution with the unit step, $\int_{-\infty}^t x(\tau) d\tau = x(t) * u(t)$, and the step transforms to $\frac{1}{j\omega} + \pi\delta(\omega)$. The convolution property multiplies the transforms, and $X(j\omega)\delta(\omega) = X(0)\delta(\omega)$ by the sampling property of the impulse:

$$\int_{-\infty}^t x(\tau) d\tau \longleftrightarrow \frac{1}{j\omega} X(j\omega) + \pi X(0) \delta(\omega).$$

THE TERM THAT GETS DROPPED

$X(0) = \int x(t) dt$ is the total area of the signal. If the area is not zero the running integral settles at a non-zero constant instead of decaying, and a constant carries an impulse at $\omega = 0$. Writing only $X(j\omega)/(j\omega)$ describes a signal that decays, which is a different signal. A unit pulse on $0 < t < 1$ has area 1 and integrates to a unit step, so it needs the term; the same pulse followed by its negative has area 0, integrates back down to zero, and the term vanishes on its own. Compute the area first: it is one integral, and it decides half the result.

Differentiation in frequency

Differentiating the analysis integral with respect to ω gives each term a factor $-jt$, so $dX/d\omega$ is the transform of $-jt x(t)$ and therefore $t x(t) \leftrightarrow j dX/d\omega$. It is the mirror of differentiation in time, and it is what turns the pair for $e^{-at}u(t)$ into the pair for $te^{-at}u(t)$ without a second integration.

Even and odd parts

For a real signal, $\mathcal{E}\{x\} = \frac{1}{2}[x(t) + x(-t)]$ transforms to $\frac{1}{2}[X(j\omega) + X(-j\omega)] = \frac{1}{2}[X + X^*] = \text{Re}\{X\}$, and the odd part gives $j \text{Im}\{X\}$. Splitting a real signal into even and odd parts splits its spectrum into real and imaginary parts, and the two symmetry rules above are the cases in which one part is absent. On $e^{-at}u(t)$ the even part is $\frac{1}{2}e^{-a|t|}$, whose transform $a/(a^2 + \omega^2)$ is exactly $\text{Re}\{1/(a + j\omega)\}$; the odd part is $\frac{1}{2}\text{sgn}(t)e^{-a|t|}$, whose transform $-j\omega/(a^2 + \omega^2)$ is $j \text{Im}\{1/(a + j\omega)\}$.

Time scaling

Substitute $\tau = at$, so $t = \tau/a$ and $dt = d\tau/a$. For $a > 0$ the limits keep their order and the coefficient is $1/a = 1/|a|$. For $a < 0$ the substitution sends $t \rightarrow -\infty$ to $\tau \rightarrow +\infty$, so the limits arrive reversed; swapping them back costs one minus sign, and $-1/a = 1/|a|$ for negative a . That is the only sign in the calculation.

COUNTING THE FLIP TWICE

Writing the reversed limits **and** an explicit -1 in front applies the same correction twice and gives $-\frac{1}{|a|}X(j\omega/a)$, contradicting the property. Do the flip once: reverse the limits, or write the minus, never both. The case $a = -1$ is time reversal, $x(-t) \leftrightarrow X(-j\omega)$, with coefficient 1.

Worked instance. If $X(j\omega) = 1$ on $|\omega| < 2\pi$, then $x(0.5t) \leftrightarrow 2X(j2\omega)$, which is 2 on $|\omega| < \pi$, and $x(2t) \leftrightarrow 0.5X(j\omega/2)$, which is 0.5 on $|\omega| < 4\pi$. The three areas are $2 \cdot 2\pi$, $1 \cdot 4\pi$ and $0.5 \cdot 8\pi$, all 4π ; since $\int X d\omega = 2\pi x(0)$, the value at $t = 0$ is the same for all three, which is what a time scaling cannot change.

5.7 Duality

The two equations of the pair differ only by a sign and a factor, so any pair can be read a second time with the domains exchanged.

DUALITY

$$x(t) \longleftrightarrow X(j\omega) \implies X(t) \longleftrightarrow 2\pi x(-\omega)$$

The argument on the right is $-\omega$, a real number, and not $-j\omega$: the letter x names a signal and a signal takes a real argument. The j belongs to the frequency-domain function only, and duality is the point in the chapter where that distinction is most easily lost.

Proof. Rename the variables of the synthesis equation, swapping t and ω , to get $2\pi x(\omega) = \int X(jt)e^{j\omega t} dt$. Now replace ω by $-\omega$; the exponent becomes the analysis exponent and the left side becomes $2\pi x(-\omega)$, so the right side is $\mathcal{F}\{X(t)\}$.

EXAMPLE 5.9 – DUALITY ON THE RECTANGULAR PULSE

GIVEN $x_1(t) = 1$ on $|t| < W$, so $X_1(j\omega) = 2 \sin(W\omega)/\omega$.

FIND The transform of $x_2(t) = 2 \sin(Wt)/t$.

METHOD Recognise x_2 as X_1 read in time and apply duality; then confirm by an independent route.

SOLUTION Duality gives $X_2(j\omega) = 2\pi x_1(-\omega)$, and x_1 is even, so

$$X_2(j\omega) = 2\pi \text{ on } |\omega| < W, \quad 0 \text{ beyond.}$$

CHECK The second route works backwards: putting a band of height 2π on $|\omega| < W$ through the synthesis equation gives $\frac{1}{2\pi} \int_{-W}^W 2\pi e^{j\omega t} d\omega = 2 \sin(Wt)/t$, which is x_2 . The two routes use different equations, so their agreement is a real check. At the origin, $X_2(j0) = 2\pi = 6.283185$ for any W , and $x_2(0) = 2W$ by l'Hôpital.

Duality also carries properties across: the time-shift rule becomes the frequency-shift rule, and differentiation in time becomes differentiation in frequency. Every pair already derived gives a second one for nothing.

5.8 Parseval's relation

Energy and power in this course are normalised, with $R = 1 \Omega$, so the instantaneous power is $|x(t)|^2$ and the energy is its integral, in joules.

PARSEVAL'S RELATION

$$E_\infty = \int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega$$

The proof writes $|x|^2 = x x^*$, replaces x^* by the conjugate of the synthesis equation and exchanges the two integrals; the inner one is then $X(j\omega)$. The step actually used is $|x(t)|^2 = x^2(t)$ for a real signal, not $|x(t)| = x(t)$, which is a stronger claim and fails wherever a real signal is negative. The quantity $|X(j\omega)|^2$ is the energy spectral density: $\frac{1}{2\pi} |X|^2 d\omega$ is the energy in a narrow band.

EXAMPLE 5.10 – ENERGY FROM A TWO-BAND SPECTRUM

GIVEN $X_3(j\omega) = 2$ for $|\omega| < 2\pi$, 1 for $2\pi < |\omega| < 4\pi$, and 0 beyond.

FIND The total energy.

METHOD Parseval. The spectrum is piecewise constant, so square each height and multiply by its width.

SOLUTION $E_\infty = \frac{1}{2\pi} [2\pi + 16\pi + 2\pi] = \frac{20\pi}{2\pi} = 10 \text{ J}$. The inner band contributes $2^2 \cdot 4\pi = 16\pi$ and the two outer bands $1^2 \cdot 2\pi$ each.

CHECK Synthesis gives $x_3(t) = [\sin(2\pi t) + \sin(4\pi t)]/(\pi t)$, and integrating x_3^2 over all time returns 10 J as well. Note that $\frac{1}{2\pi} \int X_3 d\omega = 6$, which is the peak $x_3(0)$ and not the energy: using the heights unsquared computes the wrong quantity correctly.

The same relation on $e^{-at}u(t)$ is worth doing both ways. In time, $E_\infty = \int_0^\infty e^{-2at} dt = 1/(2a)$. In frequency, $\frac{1}{2\pi} \int d\omega/(a^2 + \omega^2) = \frac{1}{2\pi} \cdot \frac{\pi}{a} = 1/(2a)$, using $\int du/(1 + u^2) = \pi$ after $u = \omega/a$. Without the $1/2\pi$ the second route reports 2π times too much.

5.9 Convolution and multiplication

THE TWO DUAL PROPERTIES

$$y(t) = x(t) * h(t) \longleftrightarrow Y(j\omega) = X(j\omega)H(j\omega)$$

$$z(t) = x(t)y(t) \longleftrightarrow Z(j\omega) = \frac{1}{2\pi} X(j\omega) * Y(j\omega)$$

The premise is stated with one set of symbols: $x \leftrightarrow X$ is the input, $h \leftrightarrow H$ the impulse response, y the output. Convolution in time carries no factor; convolution in frequency carries $1/2\pi$. $H(j\omega)$ exists when h is absolutely integrable, which for an LTI system is exactly bounded-input bounded-output stability, so an unstable system has no frequency response to plot.

The proof of the first writes the convolution inside the analysis integral and exchanges the order; the inner bracket is then the time-shift property applied to h , and $H(j\omega)$, not depending on τ , leaves the outer integral.

EXAMPLE 5.11 – TWO EXPONENTIALS IN CASCADE

GIVEN $x(t) = e^{-at}u(t)$ and $h(t) = e^{-bt}u(t)$, with $a, b > 0$ and $a \neq b$.

FIND $y(t) = x * h$.

METHOD Multiply the transforms, split into simple fractions, invert term by term.

SOLUTION $Y = \frac{1}{(a + j\omega)(b + j\omega)} = \frac{A}{a + j\omega} + \frac{B}{b + j\omega}$ with $A = \frac{1}{b - a}$ and $B = -A$, so

$$y(t) = \frac{e^{-at} - e^{-bt}}{b - a} u(t).$$

CHECK $y(0) = 0$, as a convolution of two causal signals must be. For $a = 1$, $b = 2$ the peak is at $t = \ln 2 = 0.693147$ with value exactly $1/4$. At $\omega = 0$: $|X| = 1$, $|H| = 0.5$ and $|Y| = 0.5$, and the product of the first two is the third – the property itself at one frequency.

EXAMPLE 5.12 – IDEAL FILTERS IN CASCADE

GIVEN $X(j\omega) = 2$ on $|\omega| \leq 4\pi$; an ideal low-pass system with $H(j\omega) = 3$ on $|\omega| \leq 2\pi$.

FIND $Y(j\omega)$, $y(t)$ and the three time-domain peaks.

METHOD Multiply the spectra frequency by frequency, then invert.

SOLUTION Beyond 2π the system contributes zero, so $Y = 6$ on $|\omega| \leq 2\pi$ and zero elsewhere; the narrower band decides. Inverting, $y(t) = 6 \sin(2\pi t)/(\pi t)$.

CHECK Each peak is the area of its own band divided by 2π : $x(0) = 8$, $h(0) = 6$, $y(0) = 12$. The output peak is the largest because a peak counts area, and $6 \times 4\pi$ exceeds $2 \times 8\pi$.

5.10 Amplitude modulation

Multiplying by a cosine of frequency ω_c , the carrier, convolves the spectrum with two impulses at once. The $1/2\pi$ of the multiplication property and the π of each impulse combine into a factor $\frac{1}{2}$ on each copy.

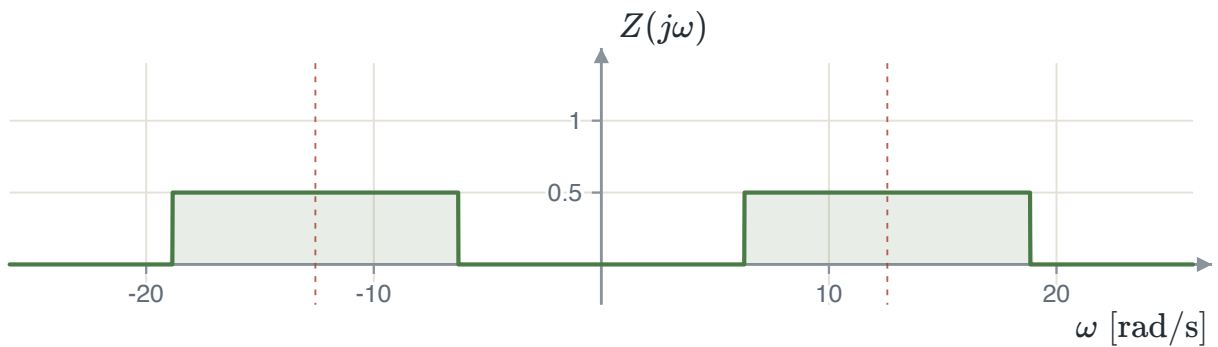
DOUBLE-SIDEBAND SUPPRESSED-CARRIER MODULATION

$$z(t) = x(t) \cos(\omega_c t) \longleftrightarrow Z(j\omega) = \frac{1}{2} X(j(\omega - \omega_c)) + \frac{1}{2} X(j(\omega + \omega_c))$$

The spectrum is not moved to ω_c ; it is **uplicated**, one copy at $+\omega_c$ and one at $-\omega_c$, each at half height. A description that says "the signal moved up to the carrier" loses the negative-frequency copy and the factor of one half in the same sentence.

With $x(t) = \cos(\pi t)$ and carrier $\cos(4\pi t)$, the product-to-sum identity gives $\frac{1}{2} \cos(5\pi t) + \frac{1}{2} \cos(3\pi t)$, whose transform has impulses of weight $\pi/2 = 1.570796$ at $\pm 3\pi$ and $\pm 5\pi$. The property gives the same four positions, $\pm(\omega_c \pm \omega_1)$. Nothing sits at $\omega_c = 4\pi$ itself, which is what "suppressed carrier" records.

With a band-limited signal the picture is cleaner. If $X = 1$ on $|\omega| < 2\pi$ and the carrier is $\cos(4\pi t)$, the copies occupy $2\pi \leq |\omega| \leq 6\pi$ at height 0.5: the same width as the original band, at half the height, and twice as much of the axis occupied because there are now two of them.



Two half-height copies on $2\pi \leq |\omega| \leq 6\pi$. The dashed lines mark $\pm\omega_c$, where nothing sits.

TWO DIFFERENT EVENTS

Copies appear whenever a signal is multiplied by a carrier. That happens at every carrier frequency and it loses nothing.

Copies overlap only when the carrier is low enough for the shifted bands to reach each other. Overlap is what destroys information, because once two copies have been added there is no way to tell what each contributed. If a band on $\pi \leq |\omega| \leq 3\pi$ is modulated by $\cos(2\pi t)$, the two half-height copies both land on $|\omega| \leq \pi$ and add to 1 there, while only one reaches $3\pi \leq |\omega| \leq 5\pi$, where the height stays 0.5. Chapter 7 asks the same question about the copies that sampling produces.

Products of two band-limited signals follow the same arithmetic. Two rectangles of height A and half-width ω_0 convolve to a triangle of apex $2A^2\omega_0$ on $|\omega| \leq 2\omega_0$; with the $1/2\pi$, the transform of $[\sin(2\pi t)/(\pi t)]^2$ is a triangle of apex 2 on $|\omega| \leq 4\pi$, and the time-domain peak is $2^2 = 4$. With unequal half-widths 2π and 4π the result is a trapezoid of height 2, flat on $|\omega| \leq 2\pi$ and zero beyond 6π , with time-domain peak $2 \times 4 = 8$. The flat top is as wide as the difference of the two half-widths, and it shrinks to a point when they are equal.

5.11 Systems described by a differential equation

Transform both sides of a linear differential equation with constant coefficients. Linearity handles the sum and the differentiation property turns each d^k/dt^k into $(j\omega)^k$, so the equation becomes algebra.

FREQUENCY RESPONSE FROM THE COEFFICIENTS

$$\sum_{k=0}^N a_k \frac{d^k y}{dt^k} = \sum_{k=0}^M b_k \frac{d^k x}{dt^k} \implies H(j\omega) = \frac{\sum_{k=0}^M b_k (j\omega)^k}{\sum_{k=0}^N a_k (j\omega)^k}$$

The frequency response is a ratio of two polynomials in $j\omega$, read straight off the coefficients. The recipe is then: read H , multiply by X , split into simple fractions and invert with the table. It is worth writing $s = j\omega$ while the algebra is done and changing back at the end; treating the two-character symbol $j\omega$ as a single variable works but invites sign slips as soon as a derivative is needed.

EXAMPLE 5.13 – SIMPLE POLES

GIVEN $\frac{d^2 y}{dt^2} + 4 \frac{dy}{dt} + 3y = \frac{dx}{dt} + 2x$, at rest.

FIND $H(j\omega)$ and $h(t)$.

METHOD Read the ratio off the coefficients, factor, split, invert.

SOLUTION $H = \frac{s+2}{s^2+4s+3} = \frac{s+2}{(s+1)(s+3)}$ with $s = j\omega$. Covering up $(s+1)$ and setting $s = -1$ gives $A = \frac{1}{2}$; covering up $(s+3)$ and setting $s = -3$ gives $B = \frac{1}{2}$. Hence $h(t) = [\frac{1}{2}e^{-t} + \frac{1}{2}e^{-3t}] u(t)$.

CHECK $H(j0) = 2/3$, and $\int_0^\infty h dt = \frac{1}{2} + \frac{1}{6} = \frac{2}{3}$. Also $h(0^+) = 1$. The poles are at $s = -1$ and $s = -3$, both with negative real part, so h is absolutely integrable and H was entitled to exist.

Repeated poles

The cover-up rule assumes each factor of the denominator appears once. When a factor is repeated, one term is needed per power, and covering up reaches only the highest of them.

REPEATED-POLE RULE

$$F(s) = \frac{N(s)}{(s - \lambda)^m Q(s)}, \quad c_{m-k} = \frac{1}{k!} \frac{d^k}{ds^k} [(s - \lambda)^m F(s)] \Big|_{s=\lambda}$$

$k = 0$ is the cover-up rule and gives c_m ; each further derivative peels off one more coefficient. The inverse transforms needed are $1/(s + a) \leftrightarrow e^{-at}u(t)$ and $1/(s + a)^2 \leftrightarrow t e^{-at}u(t)$: a repeated pole always brings a factor of t into the time domain.

EXAMPLE 5.14 – A REPEATED POLE, AND THE CHECK THAT CATCHES A SIGN

GIVEN The system of Example 5.13, with input $x(t) = e^{-t}u(t)$.

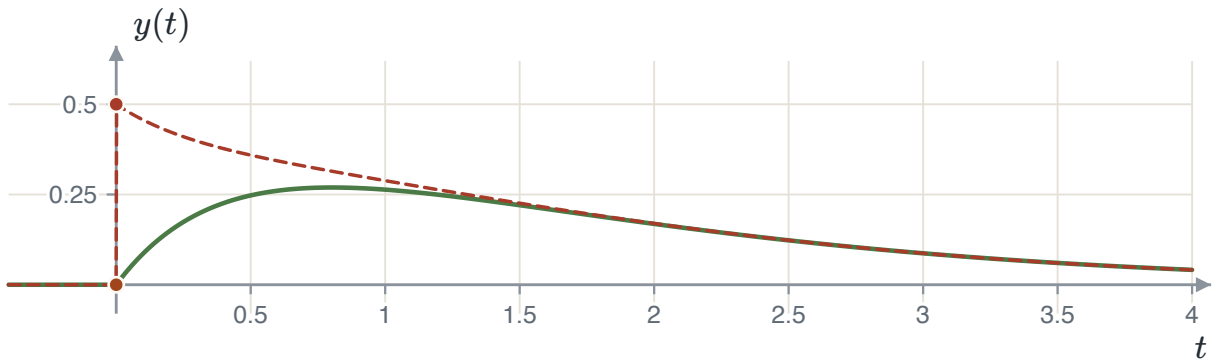
FIND $y(t)$.

METHOD $Y = XH$, then partial fractions with $s = j\omega$.

SOLUTION $Y = \frac{1}{s+1} \cdot \frac{s+2}{(s+1)(s+3)} = \frac{s+2}{(s+1)^2(s+3)}$: the input pole coincides with a system pole, so $s = -1$ is now double. Then $B = [(s+1)^2 Y]_{s=-1} = \frac{1}{2}$, $A = \frac{d}{ds} [(s+1)^2 Y]_{s=-1} = \frac{1}{(s+3)^2} \Big|_{s=-1} = \frac{1}{4}$, and $C = [(s+3)Y]_{s=-3} = -\frac{1}{4}$. Hence

$$y(t) = \left[\frac{1}{4}e^{-t} + \frac{1}{2}te^{-t} - \frac{1}{4}e^{-3t} \right] u(t).$$

CHECK $y(0) = \frac{1}{4} + 0 - \frac{1}{4} = 0$, as a convolution of two causal signals must be. Assembling C as $+\frac{1}{4}$ instead gives $y(0) = \frac{1}{2}$, which no causal convolution can. The two candidates agree to three decimals past $t = 2 - 0.168549$ against 0.169789 — so only the value at the origin separates them. Convolving directly returns the same three terms with the same signs.



The output with $C = -\frac{1}{4}$ and, dashed, the version with the sign lost. They are indistinguishable past $t = 2$ and differ by 0.5 at the origin.

5.12 Summary

- Analysis integrates over t and returns $X(j\omega)$; synthesis integrates over ω , carries $1/2\pi$, and returns $x(t)$.
- Finite energy **or** the Dirichlet conditions is enough; neither is necessary, and signals meeting neither have impulse spectra in the limiting sense.
- A periodic signal transforms to impulses of weight $2\pi a_k$ at the harmonics — not to the coefficients themselves.
- The sinc convention here is $\text{sinc}(\theta) = \sin \theta / \theta$, and it is stated at every point of use.
- A real signal has an even magnitude and an odd phase, and both halves of its spectrum are kept.
- Narrowing a signal in time widens its spectrum. Finite duration forbids band limitation; infinite duration guarantees nothing.
- Convolution in time is multiplication in frequency; multiplication in time is convolution in frequency, with $1/2\pi$.
- A carrier makes two copies at half height. Copies always appear; overlap is a separate event and is what loses information.
- A differential equation gives $H(j\omega)$ directly; a repeated pole needs the derivative rule, and a causal convolution must start at zero.

Everything here was continuous time. A computer sees a sequence, not a signal, and the same question has to be asked again for $x[n]$. One thing changes and it changes everything: $e^{-j(\omega+2\pi)n} = e^{-j\omega n}$ for every integer n , so a discrete-time spectrum repeats with period 2π . That is Chapter 6.

The discrete-time Fourier transform

A sequence that is not periodic still has a spectrum. Finding it takes one construction: make the sequence periodic by replication, apply the series of Chapter 4, and let the period grow without bound. What comes out is a continuous function of frequency that repeats every 2π , and that repetition is the one property separating this transform from the continuous-time one. Every figure in this chapter is drawn over more than one period for that reason.

6.1 From a periodic sequence to an aperiodic one

Let $x[n]$ have finite support: $x[n] = 0$ for $|n| > N_1$. The discrete-time Fourier series applies to periodic sequences only, so build a periodic sequence out of x by laying the same finite sequence down again every N samples.

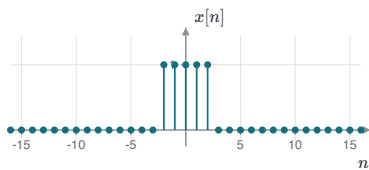
PERIODIC REPLICATION

$$\tilde{x}[n] = \sum_{r=-\infty}^{\infty} x[n - rN]$$

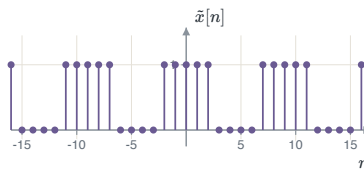
This is replication, not padding. Nothing is inserted between the samples; whole copies of x are placed side by side.

THE CONDITION THE CONSTRUCTION NEEDS

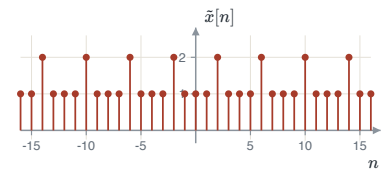
The copies must not overlap, so the period must be longer than the support: $N > 2N_1$. Only then is $\tilde{x}[n] = x[n]$ for $-N_1 \leq n \leq N_1$, a band of $2N_1 + 1$ samples centred on the origin. With $N \leq 2N_1$ the tails of neighbouring copies add and no period of $\tilde{x}$ equals x any more. Both ends of the interval point the same way; reversing the second relation would leave an interval with nothing in it.



The finite-support sequence, $N_1 = 2$.



Replication with $N = 9 > 2N_1$: the copies stand clear.



Replication with $N = 4 \leq 2N_1$: the copies add.

The coefficients are samples of one function

DISCRETE-TIME FOURIER SERIES

$$\tilde{x}[n] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_0 n}, \quad \omega_0 = \frac{2\pi}{N}$$

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} \tilde{x}[n] e^{-jk\omega_0 n}$$

The analysis exponent is negative. Synthesis builds the sequence up out of exponentials and carries $e^{+jk\omega_0 n}$; analysis takes it apart and carries the conjugate. A plus sign in both places breaks the identity derived immediately below.

Over one period $\tilde{x}[n] = x[n]$, and x vanishes outside $-N_1 \leq n \leq N_1$, so the sum may as well run over all n :

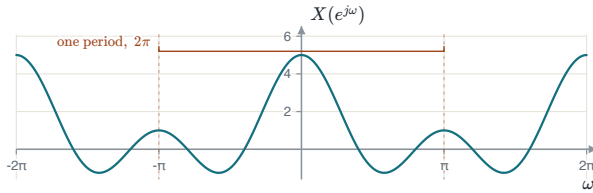
$$a_k = \frac{1}{N} \sum_{n=-N_1}^{N_1} x[n] e^{-jk\omega_0 n} = \frac{1}{N} \sum_{n=-\infty}^{\infty} x[n] e^{-jk\omega_0 n}.$$

The last sum is one function of a continuous frequency, evaluated at $\omega = k\omega_0$. Give it a name.

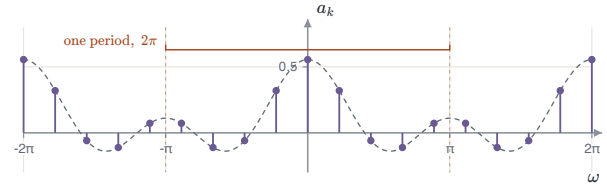
THE ENVELOPE

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \implies a_k = \frac{1}{N}X(e^{jk\omega_0})$$

The series coefficients are samples of the envelope, spaced $\omega_0 = 2\pi/N$ apart and scaled by $1/N$. The identity holds only because the two exponents match, which is the check on the sign.



The envelope of the pulse with $N_1 = 2$, over two periods. Its value at $\omega = 0$ is $2N_1 + 1 = 5$.



The coefficients for $N = 9$: the dashed envelope divided by N , sampled at $\omega = k\omega_0$.

Letting the period grow

Put $a_k = \frac{1}{N}X(e^{jk\omega_0})$ back into the synthesis equation and write $\frac{1}{N}$ as $\frac{\omega_0}{2\pi}$:

$$\tilde{x}[n] = \sum_{k \in \langle N \rangle} \frac{1}{N}X(e^{jk\omega_0})e^{jk\omega_0 n} = \frac{1}{2\pi} \sum_{k \in \langle N \rangle} X(e^{jk\omega_0})e^{jk\omega_0 n} \omega_0.$$

Read the right-hand side as a sum of rectangles: each term is a value of $X(e^{j\omega})e^{j\omega n}$ multiplied by the width ω_0 , and the N terms of one period span an interval of length $N\omega_0 = 2\pi$. Two things happen as N grows without bound. The replication stops disturbing the middle period, so $\tilde{x}[n]$ becomes $x[n]$; and the spacing $\omega_0 = 2\pi/N$ goes to zero, so the sum of rectangles becomes an integral over an interval of length 2π .

THE DISCRETE-TIME FOURIER TRANSFORM PAIR

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \quad (\text{analysis})$$

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega})e^{j\omega n} d\omega \quad (\text{synthesis})$$

Analysis takes the sequence to its spectrum; synthesis rebuilds the sequence from it. The names go by what each equation does, never by which is written first. Three details separate this pair from the continuous-time one: the factor $\frac{1}{2\pi}$ sits on the synthesis side, the integral runs over one period rather than the whole line, and the left-hand side is written $X(e^{j\omega})$ to record that it is a function of $e^{j\omega}$ and therefore repeats.

WHEN THE SUM CONVERGES

A sufficient condition is that x is absolutely summable, $\sum_n |x[n]| < \infty$. Then the analysis sum converges at every ω and $X(e^{j\omega})$ is continuous. Finite-energy sequences also have a transform, in a mean-square sense.

6.2 Why the spectrum repeats

PERIODICITY IN THE FREQUENCY VARIABLE

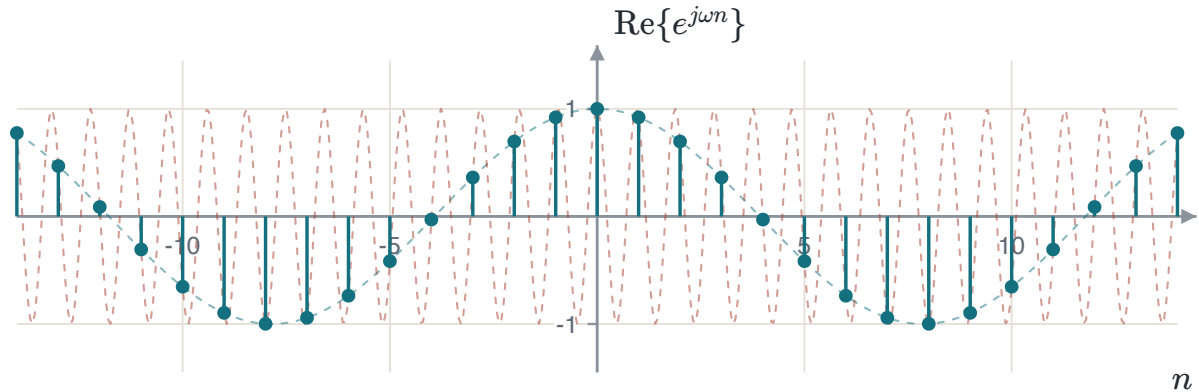
$$X(e^{j(\omega+2\pi)}) = \sum_n x[n]e^{-j(\omega+2\pi)n} = \sum_n x[n]e^{-j\omega n}e^{-j2\pi n} = X(e^{j\omega})$$

The whole argument is that n is an integer, so $e^{-j2\pi n} = 1$ in every term of the sum and the extra factor disappears term by term.

The same step in continuous time gives $\int x(t)e^{-j\omega t}e^{-j2\pi t} dt$, where t runs over the reals. There $e^{-j2\pi t}$ equals 1 only at integer t , so it cannot leave the integral, and nothing forces $X(j\omega)$ to repeat.

WHAT FREQUENCY MEANS IN DISCRETE TIME

A sequence $e^{j\omega n}$ is unchanged when ω moves by 2π , because it is only ever sampled at integers. High frequency therefore means ω near $\pm\pi$, not ω large. The fastest sequence a discrete-time signal can carry is $e^{j\pi n} = (-1)^n$.



Two continuous curves, $\omega = 0.4$ and $\omega = 0.4 + 2\pi$, and the one stem sequence they share. At the integers they agree exactly, so no sequence can tell them apart.

THE HABIT THIS CHAPTER IS BUILT TO PREVENT

A spectrum drawn on $-\pi \leq \omega \leq \pi$ alone is the same information as three periods of it, and it is also the picture a reader later mistakes for a spectrum that stops at $\pm\pi$. Every spectrum here is drawn over more than one period with the period marked.

6.3 Four transforms worked out

EXAMPLE 6.1 – A SHIFTED UNIT SAMPLE

GIVEN $x[n] = \delta[n - n_0]$.

FIND $X(e^{j\omega})$, its magnitude and its phase.

METHOD Put the sequence into the analysis sum. Every term with $n \neq n_0$ carries a factor zero.

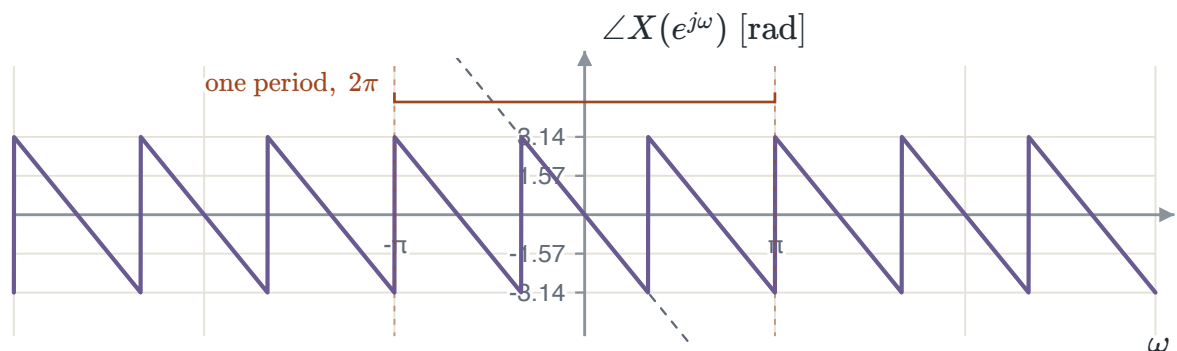
SOLUTION

$$X(e^{j\omega}) = \sum_n \delta[n - n_0] e^{-j\omega n} = e^{-j\omega n_0}.$$

So $|X(e^{j\omega})| = 1$ at every frequency and $\angle X(e^{j\omega}) = -n_0\omega$, a straight line of slope $-n_0$.

CHECK At $n_0 = 0$ the sequence is $\delta[n]$ and the transform is the constant 1, which is what the sum gives directly.

READING A straight line is not periodic, and yet the transform is. What is drawn is the principal value of the phase, which jumps by 2π whenever the line leaves $(-\pi, \pi]$. The sawtooth is that wrapping, and its period is $2\pi/|n_0|$.



The phase for $n_0 = 3$. The dashed line is the unwrapped $-n_0\omega$; the sawtooth is its principal value. Both describe the same transform.

EXAMPLE 6.2 – THE ONE-SIDED EXPONENTIAL

GIVEN $x[n] = a^n u[n]$ with $|a| < 1$.

FIND $X(e^{j\omega})$, and the extremes of its magnitude and phase.

METHOD Collect the exponentials into one ratio and sum the geometric series.

SOLUTION

$$X(e^{j\omega}) = \sum_{n=0}^{\infty} (ae^{-j\omega})^n = \frac{1}{1 - ae^{-j\omega}},$$

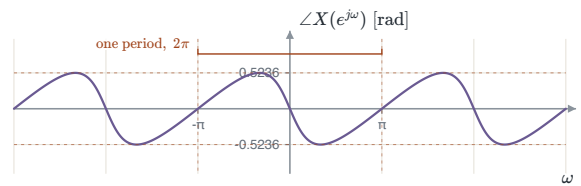
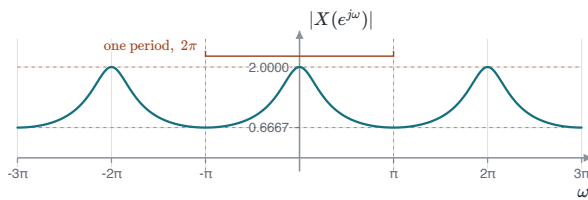
which converges because $|ae^{-j\omega}| = |a| |e^{-j\omega}| = |a| < 1$. Then

$$|X(e^{j\omega})| = \frac{1}{\sqrt{1 - 2a \cos \omega + a^2}}, \quad \angle X(e^{j\omega}) = -\arctan \frac{a \sin \omega}{1 - a \cos \omega}.$$

EXTREMES Write the denominator as $1 + ae^{j(\pi-\omega)}$: a circle of radius $|a|$ about the point 1. Its nearest point to the origin gives $|X|_{\max} = 1/(1 - |a|)$, its furthest gives $|X|_{\min} = 1/(1 + |a|)$, and the tangent from the origin gives $\max |\angle X| = \arcsin |a|$, reached where $\cos \omega = a$. For positive a the largest magnitude is at $\omega = 0$ and the smallest at $\omega = \pm\pi$; for negative a the two ends swap.

NUMBERS $a = \frac{1}{2}$: magnitude between $\frac{2}{3}$ and 2, and $\max |\angle X| = \arcsin \frac{1}{2} = \pi/6 = 0.5236$ rad exactly. $a = \frac{1}{8}$: magnitude between $\frac{8}{9} = 0.8889$ and $\frac{8}{7} = 1.1429$, and $\max |\angle X| = 0.1253$ rad.

CHECK Every one of these is a closed form, so none has to be read off the frame of a plot. An extreme that lands exactly on an axis limit is the one place a printed number is likeliest to be the limit rather than the value.



Magnitude for $a = \frac{1}{2}$, touching $1/(1 - a) = 2$ and $1/(1 + a) = 2/3$.

Phase for $a = \frac{1}{2}$, touching $\pm \arcsin \frac{1}{2} = \pm 0.5236$ rad at $\omega = \mp \pi/3$.

EXAMPLE 6.3 – THE TWO-SIDED EXPONENTIAL

GIVEN $x[n] = a^{|n|}$ with $|a| < 1$.

FIND $X(e^{j\omega})$ and its extremes.

METHOD Split the sum at $n = 0$ and sum each side as a geometric series.

SOLUTION

$$X(e^{j\omega}) = \frac{1}{1 - ae^{-j\omega}} + \frac{ae^{j\omega}}{1 - ae^{j\omega}} = \frac{1 - a^2}{1 - 2a \cos \omega + a^2}.$$

EXTREMES At $\omega = 0$ the value is $(1 + a)/(1 - a)$; at $\omega = \pm\pi$ it is $(1 - a)/(1 + a)$. For $a = \frac{1}{2}$ these are 3 and $\frac{1}{3}$; for $a = \frac{1}{4}$ they are 1.6667 and 0.6.

READING The denominator is $|1 - ae^{-j\omega}|^2$, which is positive, and the numerator is positive for $|a| < 1$. So this spectrum is real and strictly positive, and here $|X| = X$ with $\angle X = 0$ throughout. That is a property of this example, not of real spectra in general.

EXAMPLE 6.4 – THE RECTANGULAR PULSE

GIVEN $x[n] = 1$ for $-N_1 \leq n \leq N_1$ and zero elsewhere.

FIND $X(e^{j\omega})$ in closed form and its value at $\omega = 0$.

METHOD Sum the finite geometric series, then balance the exponents so both ends become sines.

SOLUTION

$$X(e^{j\omega}) = \sum_{n=-N_1}^{N_1} e^{-j\omega n} = \frac{e^{j\omega N_1} - e^{-j\omega(N_1+1)}}{1 - e^{-j\omega}}.$$

Multiplying numerator and denominator by $e^{j\omega/2}$ turns each into a difference of conjugate exponentials, $2j \sin(\omega(N_1 + \frac{1}{2}))$ over $2j \sin(\omega/2)$:

$$X(e^{j\omega}) = \frac{\sin(\omega(N_1 + \frac{1}{2}))}{\sin(\omega/2)}.$$

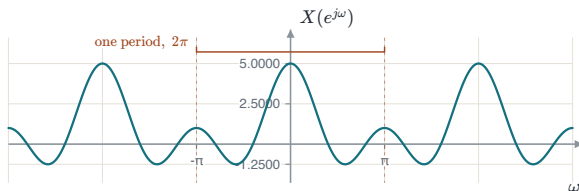
The minus in each bracket is what makes a sine; a plus would give a cosine and a different function.

EXCLUDED POINTS $r = e^{-j\omega} = 1$ at $\omega = 0, \pm 2\pi, \dots$, which is exactly where $\sin(\omega/2)$ vanishes. There every term of the sum is 1, so $X = 2N_1 + 1$: five for $N_1 = 2$, nine for $N_1 = 4$.

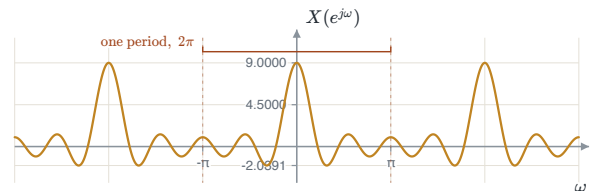
CHECK This is not a sinc. The denominator is $\sin(\omega/2)$, not $\omega/2$, and that is what makes the function periodic: a ratio of two sines repeats, while a sine over a straight line decays.

WHAT A FINITE GEOMETRIC SUM REQUIRES

$\sum_{n=p}^q r^n = (r^p - r^{q+1})/(1 - r)$ needs $r \neq 1$ and nothing else. There are finitely many terms, so nothing has to converge; only the division can fail. Here $|r| = 1$ exactly, so a condition $|r| < 1$ would exclude the sum altogether, while $|r| \leq 1$ would admit the one value that breaks it. The condition $|a| < 1$ of Example 6.2 belongs to the infinite sum, where it makes the tail vanish.



$N_1 = 2$: peak 5, least value -1.2500 .



$N_1 = 4$: peak 9, least value -2.0391 , and a main lobe half as wide.

REAL IS NOT THE SAME AS NON-NEGATIVE

The kernel above is real at every ω and negative on part of every period. Realness fixes the imaginary part and says nothing about the sign, so it gives $\angle X = 0$ where $X > 0$ and $\angle X = \pi$ where $X < 0$; and $|X| = X$ needs the extra hypothesis $X \geq 0$. A real spectrum has an even magnitude and a phase taking only the values 0 and π . Example 6.3 is the contrasting case: there the spectrum is real **and** strictly positive, so the phase really is zero, because of positivity and not because of realness.

6.4 Inverting an ideal low-pass spectrum

EXAMPLE 6.5 – THE IDEAL LOW-PASS SEQUENCE

GIVEN $X(e^{j\omega}) = 1$ for $|\omega| \leq W$ and 0 for $W < |\omega| \leq \pi$, repeated with period 2π .

FIND $x[n]$.

METHOD Integrate over the period $-\pi \leq \omega \leq \pi$, where the spectrum is 1 on a single band.

SOLUTION

$$x[n] = \frac{1}{2\pi} \int_{-W}^W e^{j\omega n} d\omega = \frac{\sin(Wn)}{\pi n}, \quad x[0] = \frac{W}{\pi}.$$

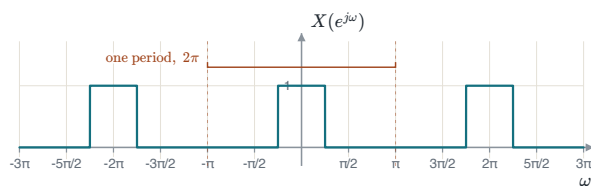
At $n = 0$ the integrand is 1 and the integral is $2W$, which is also the limit of the ratio.

NUMBERS $W = \pi/4$: 0.25, 0.225079, 0.159155, 0.075026 at $n = 0, 1, 2, 3$. $W = \pi/2$: 0.5, 0.318310, 0, -0.106103 .

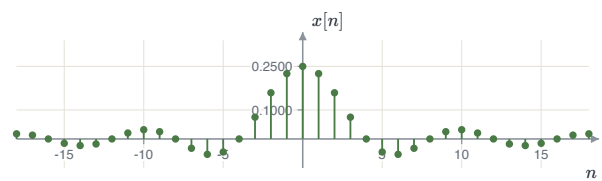
CHECK $x[0]$ must be the fraction W/π of the period that the band occupies. That single number catches both a lost 2π and a wrong prefactor.

THE SINC CONVENTION

This course uses the unnormalised sinc, $\text{sinc } \theta = \sin \theta / \theta$, with no π inside the argument. In that convention the same sequence reads $x[n] = \frac{W}{\pi} \text{sinc}(Wn)$. The other common definition puts a π inside the argument and moves every zero crossing, so the convention is restated at every point of use. The prefactor is the constant W/π : writing W/n makes the expression depend on n twice and is wrong by a factor π at every index, giving 0.707107 at $n = 1$ where the true value is 0.225079.



The spectrum for $W = \pi/4$, over three periods. The band repeats; it does not stop at $\pm\pi$.



Its inverse transform, with $x[0] = 0.25$.

6.5 Periodic sequences and impulse spectra

A complex exponential is not absolutely summable, so its analysis sum does not converge in the ordinary sense. Its transform exists as a train of impulses, and the definition is that the synthesis equation works.

THE COMPLEX EXPONENTIAL

$$e^{j\omega_0 n} \longleftrightarrow X(e^{j\omega}) = \sum_{k=-\infty}^{\infty} 2\pi \delta(\omega - \omega_0 - 2\pi k)$$

Substituting into the synthesis equation and integrating over the period containing ω_0 leaves exactly one impulse, and $\frac{1}{2\pi} \cdot 2\pi e^{j\omega_0 n} = e^{j\omega_0 n}$. The copies must be there: a single impulse is not 2π -periodic, and a sequence cannot distinguish ω_0 from $\omega_0 + 2\pi$.

WEIGHT, NOT HEIGHT

An impulse has no value at a point; it has a weight, the number that comes out when it is integrated. Every figure here draws an impulse as an arrow whose height is its weight, so the two can be read off the same axis. Arrows drawn to a fixed height with the weight written beside them hide exactly the comparison these pictures are for.

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk \frac{2\pi}{N} n} \longleftrightarrow X(e^{j\omega}) = \sum_{k=-\infty}^{\infty} 2\pi a_k \delta\left(\omega - \frac{2\pi k}{N}\right)$$

$$\cos \omega_0 n \longleftrightarrow \pi \sum_k [\delta(\omega - \omega_0 - 2\pi k) + \delta(\omega + \omega_0 - 2\pi k)]$$

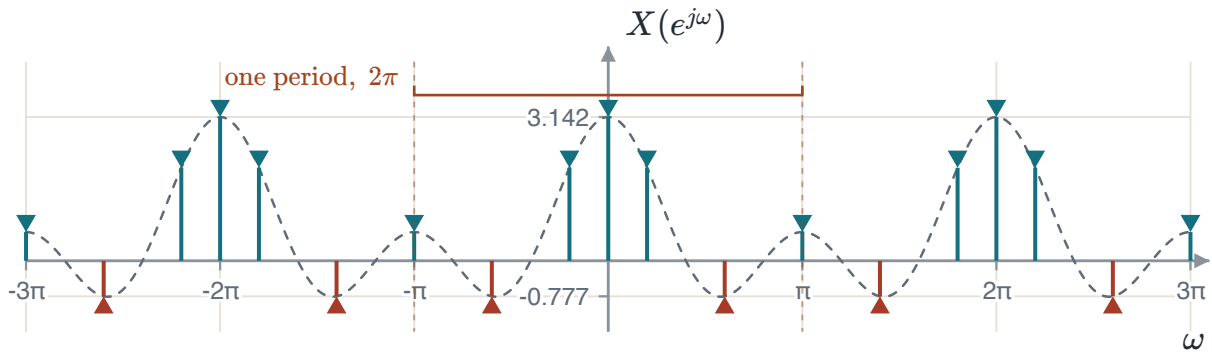
$$\sin \omega_0 n \longleftrightarrow \frac{\pi}{j} \sum_k [\delta(\omega - \omega_0 - 2\pi k) - \delta(\omega + \omega_0 - 2\pi k)]$$

The spacing is $2\pi/N$, so one period holds exactly N impulses. A real sequence always produces the pair at $+\omega_0$ and $-\omega_0$; dropping the negative one rebuilds $\frac{1}{2}e^{j\omega_0 n}$, which is complex, instead of the cosine.

For the periodic square wave, 1 on $|n| \leq N_1$ inside each period of length N , the coefficients are the same kernel again, sampled and divided by N :

$$a_k = \frac{1}{N} \frac{\sin\left(\frac{2\pi k}{N}(N_1 + \frac{1}{2})\right)}{\sin(\pi k/N)}, \quad a_k = \frac{2N_1 + 1}{N} \text{ when } k \equiv 0 \pmod{N}.$$

With $N_1 = 2$: at $N = 10$, $a_0 = 0.5000$ and $a_1 = 0.3236$; at $N = 20$, 0.2500 and 0.2260 ; at $N = 30$, 0.1667 and 0.1594 . The coefficients shrink like $1/N$ while the shape they trace stays put. Some of them are negative — at $N = 10$, $a_3 = -0.1236$ — so the impulses of weight $2\pi a_k$ are of unequal size and some point downwards. Sketching them all the same length hides both facts.



The square wave with $N = 10$, $N_1 = 2$. The dashed line is the envelope; the red arrows are the harmonics whose weight is negative.

EXAMPLE 6.6 – THE IMPULSE TRAIN

GIVEN $x[n] = \sum_{k=-\infty}^{\infty} \delta[n - kN]$: a unit sample every N indices. The free variable is n ; the index running over the copies is k , and the two must be different letters.

FIND The coefficients and the transform.

SOLUTION One period holds a single sample, at $n = 0$, so $a_k = \frac{1}{N}$ for every k , and

$$X(e^{j\omega}) = \frac{2\pi}{N} \sum_{k=-\infty}^{\infty} \delta\left(\omega - \frac{2\pi k}{N}\right).$$

NUMBERS Weights $2\pi/5 = 1.2566$, $2\pi/10 = 0.6283$, $2\pi/15 = 0.4189$ for $N = 5, 10, 15$.

READING One period always holds N impulses of weight $2\pi/N$, so their weights add to 2π whatever N is. A sparser train in time is a denser train in frequency.

Bringing a frequency into range

EXAMPLE 6.7 – TWO COSINES

GIVEN $x[n] = 2 \cos\left(\frac{5\pi}{3}n\right) + \cos\left(\frac{7\pi}{4}n\right)$.

FIND The spectrum, and the fundamental period of the sequence.

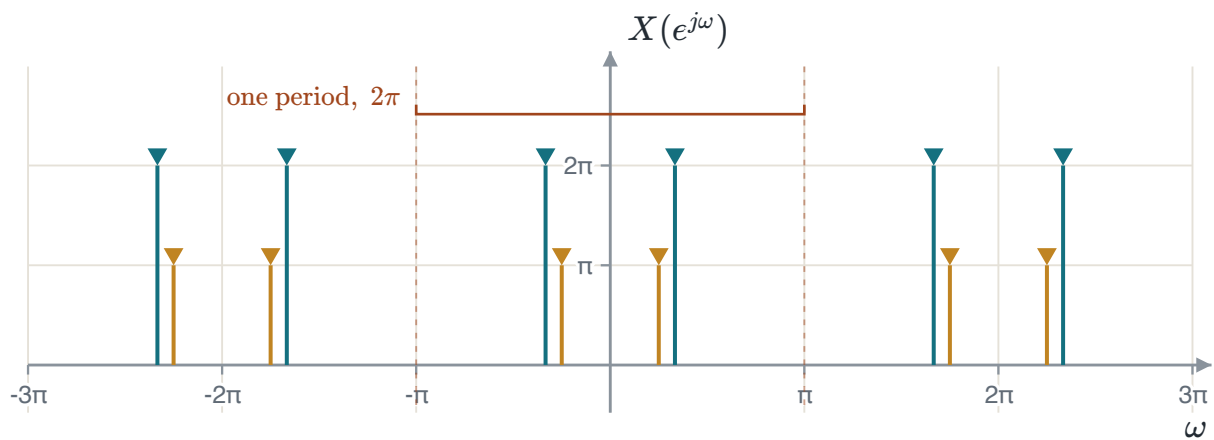
METHOD Bring each frequency into $-\pi < \omega \leq \pi$ first, then transform each term.

REDUCTION $\frac{5\pi}{3} = 2\pi - \frac{\pi}{3}$, and $2\pi n$ is a whole number of turns at every integer n , so $\cos\left(\frac{5\pi}{3}n\right) = \cos\left(\frac{\pi}{3}n\right)$. Likewise $\cos\left(\frac{7\pi}{4}n\right) = \cos\left(\frac{\pi}{4}n\right)$. These are not merely similar sequences; they take the same value at every n .

SOLUTION Inside one period there are four impulses: weight 2π at $\pm\pi/3$ and weight π at $\pm\pi/4$. The amplitude 2 doubles the first pair from π to 2π . Outside the period the same impulses reappear every 2π : weight 2π at $\pm 5\pi/3$ and $\pm 7\pi/3$, weight π at $\pm 7\pi/4$ and $\pm 9\pi/4$.

ORDER On the positive axis $\pi/4 < \pi/3 < 5\pi/3 < 7\pi/4 < 2\pi < 9\pi/4 < 7\pi/3$: the two trains interleave, and the order flips either side of 2π .

PERIOD $N_0 = (2\pi/\omega_0)m$, where m is the smallest positive integer making N_0 an integer. For $\omega_1 = 5\pi/3$: $2\pi/\omega_1 = 6/5$, $m = 5$, $N_0 = 6$. For $\omega_2 = 7\pi/4$: $8/7$, $m = 7$, $N_0 = 8$. The sum repeats after $\text{LCM}(6, 8) = 24$, and $24/6 = 4$, $24/8 = 3$ with $\text{gcd}(4, 3) = 1$, so nothing smaller works.



The spectrum over three periods. Tall arrows carry weight 2π , short ones π , and each pattern repeats every 2π .

6.6 Properties

PROPERTY	PAIR
Linearity	$a x_1[n] + b x_2[n] \leftrightarrow a X_1 + b X_2$
Time shift	$x[n - n_0] \leftrightarrow e^{-j\omega n_0} X(e^{j\omega})$
Frequency shift	$e^{j\omega_0 n} x[n] \leftrightarrow X(e^{j(\omega - \omega_0)})$
Conjugation	$x^*[n] \leftrightarrow X^*(e^{-j\omega})$
Time reversal	$x[-n] \leftrightarrow X(e^{-j\omega})$
Time expansion	$x_{(k)}[n] \leftrightarrow X(e^{jk\omega})$
Differencing in time	$x[n] - x[n - 1] \leftrightarrow (1 - e^{-j\omega})X(e^{j\omega})$
Accumulation	$\sum_{m=-\infty}^n x[m] \leftrightarrow \frac{X(e^{j\omega})}{1 - e^{-j\omega}} + \pi X(e^{j0}) \sum_k \delta(\omega - 2\pi k)$
Differentiation in frequency	$n x[n] \leftrightarrow j dX(e^{j\omega})/d\omega$
Convolution	$x[n] * h[n] \leftrightarrow X(e^{j\omega})H(e^{j\omega})$
Multiplication	$x[n]y[n] \leftrightarrow \frac{1}{2\pi} \int_{2\pi} X(e^{j\theta})Y(e^{j(\omega - \theta)})d\theta$
Real and even	$X(e^{j\omega})$ real and even
Real and odd	$X(e^{j\omega})$ purely imaginary and odd
Even-odd parts	$\mathcal{E}\mathbf{v}\{x\} \leftrightarrow \text{Re}\{X\}$, $\mathcal{O}\mathbf{d}\{x\} \leftrightarrow j \text{Im}\{X\}$
Parseval	$\sum_n x[n] ^2 = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) ^2 d\omega$

TWO NAMES THAT MUST NOT BE EXCHANGED

Discrete time has a **difference**, not a derivative: there is nothing between n and $n + 1$ over which to take a limit, and the factor is $1 - e^{-j\omega}$ rather than $j\omega$. The **differentiation** in this chapter is in frequency, and it is a genuine derivative because ω is continuous. Naming the first difference a derivative collapses the continuous and discrete cases on the one page where the distinction is being built.

For a real sequence, conjugation gives $X(e^{-j\omega}) = X^*(e^{j\omega})$, so the real part and the magnitude are even in ω and the imaginary part and the phase are odd. Half a period then determines the whole spectrum — for a real sequence only. A complex sequence needs the full period drawn.

That one line also fixes the last three rows of the table. $\mathcal{E}\mathbf{v}\{x[n]\} = \frac{1}{2}(x[n] + x[-n])$ transforms to $\frac{1}{2}[X(e^{j\omega}) + X(e^{-j\omega})] = \frac{1}{2}[X + X^*] = \text{Re}\{X\}$, and the odd part gives $j \text{Im}\{X\}$; a real and even sequence has no odd part, so its transform is real and even, and a real and odd sequence has no even part, so its transform is purely imaginary and odd. Take $x[n] = a^n u[n]$ with $0 < a < 1$. Its even part is $\frac{1}{2}a^{|n|} + \frac{1}{2}\delta[n]$ — the two halves overlap at $n = 0$ — and transforming term by term gives $(1 - a \cos \omega)/(1 - 2a \cos \omega + a^2)$, which is $\text{Re}\{1/(1 - ae^{-j\omega})\}$ exactly.

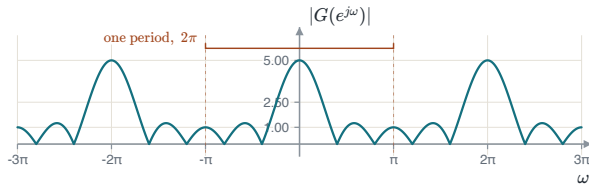
Accumulation undoes differencing, so their factors are reciprocals, and the impulse train is the part differencing destroys. A running sum is fixed only up to a constant offset, and the difference cannot see that offset; the weight $\pi X(e^{j0}) = \pi \sum_n x[n]$ is what puts it back. Where the sequence sums to zero the term vanishes on its own. It is the discrete-time counterpart of the $\pi X(0)\delta(\omega)$ in the integration property of Chapter 5, and it fails in the same place for the same reason: $1 - e^{-j\omega}$ is zero at $\omega = 0$.

Time expansion

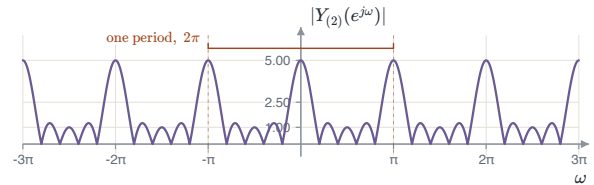
For a positive integer k , $x_{(k)}[n]$ is $x[n/k]$ when n is a multiple of k and zero otherwise: a stretch with zeros inserted, since discrete time has no operation that stretches a sequence without leaving gaps. Only the indices $n = rk$ contribute to the analysis sum, so putting $n = rk$ turns it into $\sum_r x[r]e^{-j\omega kr} = X(e^{jk\omega})$. Replacing ω by $k\omega$ compresses the frequency axis by k , so one period of length 2π now holds k copies of the old picture.

EXAMPLE 6.8 – EXPANSION OF A FIVE-POINT PULSE

GIVEN	$g[n] = 1$ on $ n \leq 2$, so $G(e^{j\omega}) = \sin(5\omega/2)/\sin(\omega/2)$.
BUILD	$y[n] = g[n-2]$, so $Y(e^{j\omega}) = e^{-j2\omega}G(e^{j\omega})$. Expanding by two gives $Y_{(2)}(e^{j\omega}) = Y(e^{j2\omega}) = e^{-j4\omega} \sin(5\omega)/\sin(\omega)$. Finally $x[n] = y_{(2)}[n] + 2y_{(2)}[n-1]$, so $X(e^{j\omega}) = (1 + 2e^{-j\omega})Y_{(2)}(e^{j\omega})$, which peaks at $3 \times 5 = 15$.
WHERE THE ARGUMENT HALVED	G carries $\sin(5\omega/2)/\sin(\omega/2)$ and $Y_{(2)}$ carries $\sin(5\omega)/\sin(\omega)$: the same expression with ω replaced by 2ω . A denominator $\sin(\omega)$ is right for the expanded sequence and wrong for the original, where it would put a pole at $\omega = \pi - 1000.0$ at $\omega = \pi - 10^{-3}$, against the true $G(e^{j\pi}) = 1$, and -0.7071 at $\pi/2$ against the true -1.0000 .



$|G(e^{j\omega})|$: peak 5, and the finite value 1 at $\omega = \pi$.



$|Y_{(2)}(e^{j\omega})|$: the same picture twice inside every 2π .

Parseval

The quantity $|X(e^{j\omega})|^2$ is the energy-density spectrum. Integrating it over one period and dividing by 2π gives the total energy; integrating over part of a period gives the energy carried by that band. Both the single period and the $\frac{1}{2\pi}$ are load-bearing, and neither shows up in the shape of a plot. Two checks: for $a^n u[n]$ with $a = \frac{1}{2}$ the time side gives $\sum_{n \geq 0} (\frac{1}{4})^n = \frac{4}{3}$ and the frequency side gives $1/(1 - a^2) = \frac{4}{3}$; for the rectangular pulse with $N_1 = 2$ both sides give 5, the number of samples.

6.7 Convolution and multiplication

CONVOLUTION

$$y[n] = x[n] * h[n] \longleftrightarrow Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega})$$

Both pairs $x \leftrightarrow X$ and $h \leftrightarrow H$ are declared before the conclusion is written, and y names the output and nothing else. A product of complex numbers gives two separate statements, $|Y| = |X| \cdot |H|$ and $\angle Y = \angle X + \angle H$. Putting modulus bars around a whole equation is not an operation on anything.

EXAMPLE 6.9 – CONVOLVING TWO EXPONENTIALS

GIVEN $x[n] = a^n u[n]$ and $h[n] = b^n u[n]$, with $|a| < 1$, $|b| < 1$ and $a \neq b$. Every one-sided exponential carries its $u[n]$: without it the sequence is defined for negative n too, where it grows without bound.

METHOD Multiply the transforms and split into partial fractions. Write $z = e^{-j\omega}$ as a named algebraic variable; then $Y = 1/((1 - az)(1 - bz))$ is an ordinary rational function of z , and substituting $z = 1/a$ is a step of algebra in z , not a claim that $e^{-j\omega}$ ever takes that value.

SOLUTION Cover-up gives $A = a/(a - b)$ and $B = -b/(a - b)$, so

$$y[n] = \frac{1}{a - b} [a^{n+1} - b^{n+1}] u[n].$$

CONDITION Both coefficients divide by $a - b$, so the route requires $a \neq b$. At $a = b$ the two poles merge and the answer takes a different form, worked out in Section 6.8.

CHECK $a = \frac{1}{2}$, $b = \frac{1}{4}$: $y[0] = 1$, $y[1] = 0.75$, $y[2] = 0.4375$, $y[3] = 0.234375$. Direct convolution gives $y[0] = x[0]h[0] = 1$ and $y[1] = \frac{1}{4} + \frac{1}{2} = 0.75$; the first value must be $x[0]h[0]$ for any two causal sequences. The magnitude $|Y|$ runs between 0.5333 and 2.6667.

The product of two ideal low-pass responses is the narrower of the two, so a cascade keeps only the band both filters pass. With cutoffs $\pi/2$ and $\pi/4$, the cascade has cutoff $\pi/4$ and impulse response $\sin(\pi n/4)/(\pi n)$. In writing such a spectrum, close one branch and leave the other open — 1 for $|\omega| \leq \pi/4$ and 0 for $\pi/4 < |\omega| \leq \pi$ — since for an ideal filter the band edge is precisely where a convention is needed rather than assumed.

EXAMPLE 6.10 – A STEPPED SPECTRUM THROUGH A FILTER

GIVEN $X(e^{j\omega}) = 2$ for $|\omega| \leq \pi/4$ and 1 for $\pi/4 < |\omega| \leq 3\pi/4$, zero to π ; H is ideal low-pass with cutoff $\pi/2$.

PRODUCT $Y(e^{j\omega}) = 2$ for $|\omega| \leq \pi/4$ and 1 for $\pi/4 < |\omega| \leq \pi/2$, zero on the rest of the period.

SOLUTION Split Y into two stacked ideal bands, one of height 1 out to $\pi/2$ and one more out to $\pi/4$:

$$y[n] = \frac{\sin(\pi n/2)}{\pi n} + \frac{\sin(\pi n/4)}{\pi n}, \quad y[0] = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}.$$

CHECK $y[0]$ must be the area of one period of Y divided by 2π . A sketch of a spectrum is not a solution; the closed form costs one line once the stack is seen.

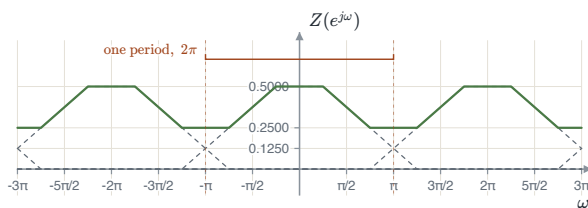
Multiplication is a periodic convolution

MULTIPLICATION

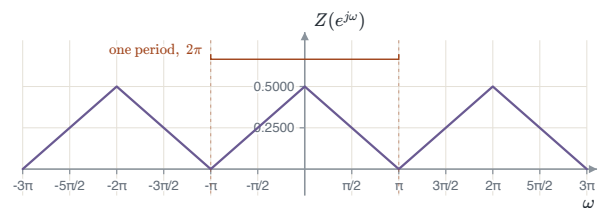
$$z[n] = x[n]y[n] \longleftrightarrow Z(e^{j\omega}) = \frac{1}{2\pi} \int_{2\pi} X(e^{j\theta}) Y(e^{j(\omega-\theta)}) d\theta$$

The integral runs over one period and both factors are 2π -periodic. That operation is a periodic convolution, and it differs from the ordinary convolution of two functions on the line whenever the result is wider than one period.

Take $X = 1$ on $|\omega| \leq 3\pi/4$ and $Y = 1$ on $|\omega| \leq \pi/2$, both repeated every 2π . Convolving the two rectangles on the line gives a trapezoid of height $\frac{1}{2\pi} \cdot 2 \cdot \frac{\pi}{2} = \frac{1}{2}$, flat on $|\omega| \leq \pi/4$ and reaching zero at $|\omega| = 5\pi/4$. Since $5\pi/4 > \pi$ the trapezoid runs past the edge of a period, so near $\omega = \pm\pi$ a second copy arrives from the other side and the two add.



The dashed trapezoids are the copies; the solid curve is their sum. At $\omega = \pi$ the value doubles from 0.125 to 0.25.



Two equal bands of half-width $\pi/2$: the triangle fills exactly one period, so nothing wraps.

The values are $Z(e^{j0}) = Z(e^{j\pi/4}) = 0.5$ on the flat top, $Z(e^{j3\pi/4}) = 0.25$ on the slope, and $Z(e^{j\pi}) = \frac{1}{8} + \frac{1}{8} = 0.25$ at the edge. The check is the synthesis equation: $\frac{1}{2\pi} \int_{2\pi} Z d\omega$ must equal $z[0] = x[0]y[0] = \frac{3}{4} \cdot \frac{1}{2} = 0.375$, and it does.

EXAMPLE 6.11 – MULTIPLYING BY A COSINE

GIVEN $X(e^{j\omega}) = 1$ for $|\omega| \leq \pi/4$, repeated every 2π , and $y[n] = \cos(\omega_0 n)$ with $\omega_0 = \pi/3$.

METHOD Write Y as its impulse train, $\pi \sum_l [\delta(\omega - \omega_0 - 2\pi l) + \delta(\omega + \omega_0 - 2\pi l)]$. The periodic convolution integrates over a single period, so choosing $-\pi < \theta \leq \pi$ leaves exactly two impulses, at $\theta = \pm\pi/3$, and only those contribute. Taking the whole train into an ordinary convolution would count every copy and diverge.

SOLUTION

$$Z(e^{j\omega}) = \frac{1}{2} X(e^{j(\omega-\omega_0)}) + \frac{1}{2} X(e^{j(\omega+\omega_0)}),$$

where the height $\frac{1}{2}$ comes from $\frac{1}{2\pi} \cdot \pi$.

BAND EDGES The band moves to $\omega_0 \mp \pi/4$, so with $\omega_0 = \pi/3$ the edges are $\pi/12$ and $7\pi/12$, and the mirror image occupies $-7\pi/12 \leq \omega \leq -\pi/12$. Both numbers are computed from ω_0 and the bandwidth, so they cannot drift apart from the drawn figure; with ω_0 below $\pi/4$ the two bands would overlap through the origin instead.

THIS OVERLAP IS NOT ALIASING

What folds back here are copies of a spectrum inside a frequency-domain convolution, and it happens because the spectrum is periodic. Chapter 7 has a different overlap, between replicas produced by sampling, and only that one is aliasing.

6.8 Difference equations

Transform both sides of a linear constant-coefficient difference equation. Linearity acts term by term and each shift brings out one factor $e^{-j\omega k}$.

FREQUENCY RESPONSE OF A DIFFERENCE EQUATION

$$\sum_{k=0}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k]$$

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{\sum_{k=0}^M b_k e^{-j\omega k}}{\sum_{k=0}^N a_k e^{-j\omega k}}$$

The impulse response of the recursion is now available without running the recursion: factor the denominator, split into partial fractions, and invert each piece with the exponential pair. Every step is algebra.

EXAMPLE 6.12 – A SECOND-ORDER SYSTEM

GIVEN $y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2] = 2x[n]$.

FIND The frequency response and the impulse response.

SOLUTION

$$H(e^{j\omega}) = \frac{2}{1 - \frac{3}{4}e^{-j\omega} + \frac{1}{8}e^{-j2\omega}} = \frac{2}{(1 - \frac{1}{2}e^{-j\omega})(1 - \frac{1}{4}e^{-j\omega})},$$

checked by expansion: $\frac{1}{2} + \frac{1}{4} = \frac{3}{4}$ and $\frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$. With $z = e^{-j\omega}$, cover-up at $z = 2$ gives $A = 4$ and at $z = 4$ gives $B = -2$, so

$$h[n] = 4\left(\frac{1}{2}\right)^n u[n] - 2\left(\frac{1}{4}\right)^n u[n].$$

CHECK $h[0] = 2$, $h[1] = 1.5$, $h[2] = 0.875$. Substituting into the equation: $h[1] - \frac{3}{4}h[0] = 0$ and $h[2] - \frac{3}{4}h[1] + \frac{1}{8}h[0] = 0$, as required for $n \geq 1$. Both poles have modulus below 1, so the system is stable, and $|H|$ runs between 1.0667 and 5.3333: this recursion is a low-pass filter.

The repeated-pole pair

Example 6.9 excluded $a = b$. That case has its own pair, and it is worth deriving rather than quoting. Start from the geometric pair and differentiate both sides with respect to a , holding ω fixed:

$$\sum_{n=1}^{\infty} n a^{n-1} e^{-j\omega n} = \frac{e^{-j\omega}}{(1 - a e^{-j\omega})^2}.$$

REPEATED POLE

$$(n+1)a^n u[n] \longleftrightarrow \frac{1}{(1 - a e^{-j\omega})^2}, \quad |a| < 1$$

Putting $m = n - 1$ and cancelling one factor $e^{-j\omega}$ gives the pair. The exponent 2 comes from the differentiation and the minus sign is inherited unchanged, so neither has to be remembered. One number settles any doubt: at $\omega = 0$ with $a = \frac{1}{4}$ the value is $(1 - \frac{1}{4})^{-2} = \frac{16}{9} = 1.7778$, while a plus sign would give $(\frac{5}{4})^{-2} = 0.6400$.

EXAMPLE 6.13 – THE OUTPUT OF THAT SYSTEM

GIVEN The system of Example 6.12 driven by $x[n] = (\frac{1}{4})^n u[n]$.

METHOD The input pole coincides with one of the system poles, so the expansion needs both a simple and a squared term for it:

$$Y = \frac{2}{(1 - \frac{1}{2}z)(1 - \frac{1}{4}z)^2} = \frac{A}{1 - \frac{1}{4}z} + \frac{B}{(1 - \frac{1}{4}z)^2} + \frac{C}{1 - \frac{1}{2}z}.$$

COEFFICIENTS Cover-up at $z = 2$ gives $C = 2/(1 - \frac{1}{2})^2 = 8$; at $z = 4$ it gives $B = 2/(1 - 2) = -2$; setting $z = 0$ in the identity gives $2 = A + B + C$, so $A = -4$.

SOLUTION

$$y[n] = -4 \left(\frac{1}{4}\right)^n u[n] - 2(n+1) \left(\frac{1}{4}\right)^n u[n] + 8 \left(\frac{1}{2}\right)^n u[n].$$

CHECK $y[0] = 2, y[1] = 2, y[2] = 1.375, y[3] = 0.8125$, and $y[0]$ must equal $h[0]x[0] = 2$ for two causal sequences. Direct convolution of h and x reproduces the whole sequence. Without the repeated-pole pair the middle term could not be inverted at all.

6.9 Duality

There is no duality inside the discrete-time transform pair. The analysis equation is a sum over an integer and the synthesis equation an integral over a continuous variable, so no relabelling turns one into the other. Two dualities do hold.

1. In the discrete-time Fourier series, where both domains are discrete and both objects are periodic with the same period N : if $x[n] \leftrightarrow a_k$, then reading the coefficients as a sequence gives $a[n] \leftrightarrow \frac{1}{N}x[-k]$. The impulse train with $N = 21$ is the example: its coefficients are all $1/21 = 0.0476$, and reading those constants back as a sequence returns an impulse train.
2. Between the discrete-time transform and the continuous-time series. $X(e^{j\omega})$ is a 2π -periodic function of a continuous variable, so it has a continuous-time Fourier series, and comparing the two synthesis equations identifies its coefficients as $x[-k]$: the spectrum of a sequence is a periodic signal whose series coefficients are the sequence itself, reversed.

A worked check on the second. Take the 2π -periodic square wave in ω , equal to 1 for $|\omega| \leq \pi/2$ and zero over the rest of the period. Its series coefficients are $a_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{-jk\omega} d\omega = \sin(k\pi/2)/(k\pi)$, that is 0.5, $1/\pi = 0.3183$, 0, $-1/(3\pi) = -0.1061$. Its inverse transform is $x[n] = \sin(\pi n/2)/(\pi n)$, which is the same list of numbers, so $a_k = x[-k]$ – and here x is even, so $a_k = x[k]$ as well.

6.10 What to carry into Chapter 7

- The pair, with analysis as the sum and synthesis as the integral over one period, carrying $\frac{1}{2\pi}$.
- The 2π -periodicity and its one-line proof, and that the same line fails in continuous time.
- Real means the phase is 0 or π , not that the phase is zero.
- Multiplication in time is a **periodic** convolution over 2π , so anything reaching across a period boundary folds back and adds.
- The unnormalised sinc, $\text{sinc } \theta = \sin \theta / \theta$, restated at every point of use.
- Discrete time has a difference, not a derivative; the derivative in this chapter is in frequency.

A sequence usually comes from somewhere: a continuous-time signal, read at regular instants. The next chapter asks what that reading costs. The answer is written in copies of a spectrum, spaced by the sampling frequency, and this chapter has already shown what happens when copies of a spectrum are allowed to meet.

Exercises

6.1 Prove that $X(e^{j\omega})$ is 2π -periodic, and say precisely where the same argument fails for $X(j\omega)$.

Answer: Substituting $\omega + 2\pi$ into the analysis sum produces a factor $e^{-j2\pi n}$, which is 1 in every term because n is an integer. In continuous time the corresponding factor is $e^{-j2\pi t}$ with t real, which equals 1 only at integer t and cannot leave the integral.

6.2 For $x[n] = a^n u[n]$ with $a = \frac{1}{8}$, give $|X|_{\max}$, $|X|_{\min}$ and $\max |\angle X|$ in closed form and as numbers.

Answer: $1/(1 - a) = 8/7 = 1.1429$, $1/(1 + a) = 8/9 = 0.8889$, and $\arcsin \frac{1}{8} = 0.1253$ rad.

6.3 Explain why the finite geometric sum behind the Dirichlet kernel needs $r \neq 1$ rather than $|r| < 1$, and give the value of the kernel at the excluded points.

Answer: A finite sum always has a value; only the closed form fails, where its denominator $1 - r$ vanishes. Here $|r| = 1$ at every ω , so $|r| < 1$ would exclude the sum entirely. At $r = 1$, that is ω a multiple of 2π , every term is 1 and the sum is $2N_1 + 1$.

6.4 The Dirichlet kernel is real at every ω . Does it follow that $|X| = X$ and $\angle X = 0$?

Answer: No. Realness fixes the imaginary part and not the sign. The kernel is negative on part of every period — least value -1.2500 for $N_1 = 2$ — and there $|X| = -X$ and $\angle X = \pi$. The conclusion needs the extra hypothesis $X \geq 0$, which holds for $a^{|n|}$ but not here.

6.5 Invert the ideal low-pass spectrum with $W = \pi/2$ and give $x[0]$ and $x[2]$. State the sinc convention you use.

Answer: $x[n] = \sin(Wn)/(\pi n)$ with $x[0] = W/\pi = 0.5$, and $x[2] = \sin(\pi)/(2\pi) = 0$. In the unnormalised convention $\text{sinc } \theta = \sin \theta/\theta$, the same sequence is $\frac{W}{\pi} \text{sinc}(Wn)$.

6.6 Where do the impulses of $2 \cos\left(\frac{5\pi}{3}n\right) + \cos\left(\frac{7\pi}{4}n\right)$ lie inside $-\pi < \omega \leq \pi$, and what is the fundamental period of the sequence?

Answer: Weight 2π at $\pm\pi/3$ and weight π at $\pm\pi/4$, after reducing $5\pi/3$ to $\pi/3$ and $7\pi/4$ to $\pi/4$. The two components have periods 6 and 8, so the sum has period $\text{LCM}(6, 8) = 24$.

6.7 Two spectra, 1 on $|\omega| \leq 3\pi/4$ and 1 on $|\omega| \leq \pi/2$, belong to sequences that are multiplied together. Why is $Z(e^{j\pi}) = \frac{1}{4}$ and not $\frac{1}{8}$?

Answer: The multiplication property gives a periodic convolution over one period. The ordinary convolution of the two rectangles is a trapezoid reaching to $|\omega| = 5\pi/4$, which is wider than one period, so near $\omega = \pi$ a second copy arrives from the other side and the two contributions of $\frac{1}{8}$ add.

6.8 Derive the pair for $(n+1)a^n u[n]$ and say what fixes the exponent and the sign.

Answer: Differentiate $\sum_{n \geq 0} a^n e^{-j\omega n} = 1/(1 - ae^{-j\omega})$ with respect to a , shift the index and cancel one factor $e^{-j\omega}$, giving $1/(1 - ae^{-j\omega})^2$. One differentiation of a first power produces the exponent 2, and the minus sign is inherited from the pair being differentiated.

6.9 Find the impulse response of $y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2] = 2x[n]$ and check it against the equation.

Answer: $h[n] = 4(\frac{1}{2})^n u[n] - 2(\frac{1}{4})^n u[n]$, so $h[0] = 2$, $h[1] = 1.5$, $h[2] = 0.875$. Substituting: $h[1] - \frac{3}{4}h[0] = 0$ and $h[2] - \frac{3}{4}h[1] + \frac{1}{8}h[0] = 0$.

6.10 Which properties of the continuous-time transform carry over unchanged, and which change form?

Answer: Linearity, the time shift, conjugate symmetry, Parseval and the convolution property carry over unchanged. The synthesis equation changes its range to one period; the multiplication property becomes a periodic convolution; the time-domain operator is a difference and not a derivative; and the transform pairs themselves differ, since the discrete-time ones are 2π -periodic and carry their own convergence conditions.

Sampling and aliasing

Keep the value of a signal only every T seconds and everything between the instants is thrown away. This chapter says exactly when that loss can be reversed, exactly what is lost when it cannot, and exactly which filter does the reversing. Two words carry the chapter and are not interchangeable: sampling always **replicates** the spectrum, and only overlapping copies are **aliasing**.

7.1 Impulse-train sampling

Sampling is modelled as a multiplication. The sampling function is an impulse train of period T , and the sampled signal is the product of the signal with it.

IMPULSE-TRAIN SAMPLING

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT), \quad x_p(t) = x(t)p(t)$$

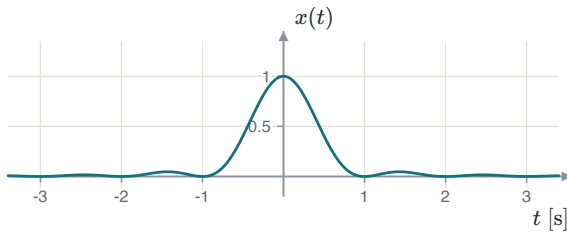
$$x_p(t) = \sum_{n=-\infty}^{\infty} x(nT) \delta(t - nT)$$

The second line follows from the sifting property $x(t)\delta(t - nT) = x(nT)\delta(t - nT)$: an impulse is zero everywhere except at its own instant, so only the value there survives.

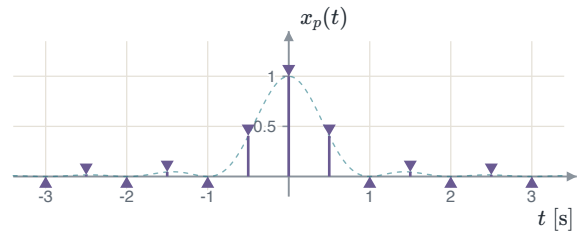
HOW AN IMPULSE IS DRAWN

An impulse is drawn as an arrow whose **height is its weight**. In a picture of $x_p(t)$ the arrow at $t = nT$ therefore reaches $x(nT)$, and the outline traced by the arrowheads is the signal itself. The arrows are weights, not values of a function.

The numbers $x(nT)$ are all that survives the sampler. Written as a sequence they are $x_p[n] = x(nT)$, and this is where a discrete-time signal comes from in practice.



A band-limited signal, $x(t) = (\sin \pi t / \pi t)^2$.



The same signal sampled with $T = 0.5$ s. Each arrow reaches $x(nT)$.

7.2 The rate, in radians per second and in hertz

One sampler, one period, two ways of naming how fast it runs. Both appear in this chapter, so both are defined here and kept apart afterwards.

SAMPLING RATE, BOTH READINGS

$$\omega_s = \frac{2\pi}{T} \left[\frac{\text{rad}}{\text{s}} \right], \quad f_s = \frac{1}{T} [\text{Hz}], \quad \omega_s = 2\pi f_s$$

ω_s is the sampling **angular** frequency. f_s is the sampling frequency in hertz, the number of samples taken per second. Every spectrum in this chapter is drawn against ω in rad/s, so ω_s is the working symbol; the hertz reading appears only where a physical rate is quoted.

THE FACTOR OF 2π IS NOT DECORATION

Calling $2\pi/T$ the sampling frequency and then reading it in hertz is the most expensive slip in this chapter. With $T = 0.25$ ms the two readings are $\omega_s = 8000\pi$ rad/s and $f_s = 4000$ Hz. They differ by 2π , and an answer that mixes them is wrong by that factor everywhere it is used. The same applies to the signal: a bandwidth of ω_M rad/s is $f_M = \omega_M/2\pi$ hertz.

One multiplication tests any pair of values: $\omega_s T = 2\pi$ always, and $f_s T = 1$ always. Every sampling period computed in this chapter is followed by that check.

7.3 The spectrum of a sampled signal

Multiplication in time is convolution in frequency. The transform of an impulse train of period T is an impulse train of spacing $\omega_s = 2\pi/T$, and convolving with a shifted impulse shifts the function. Three steps:

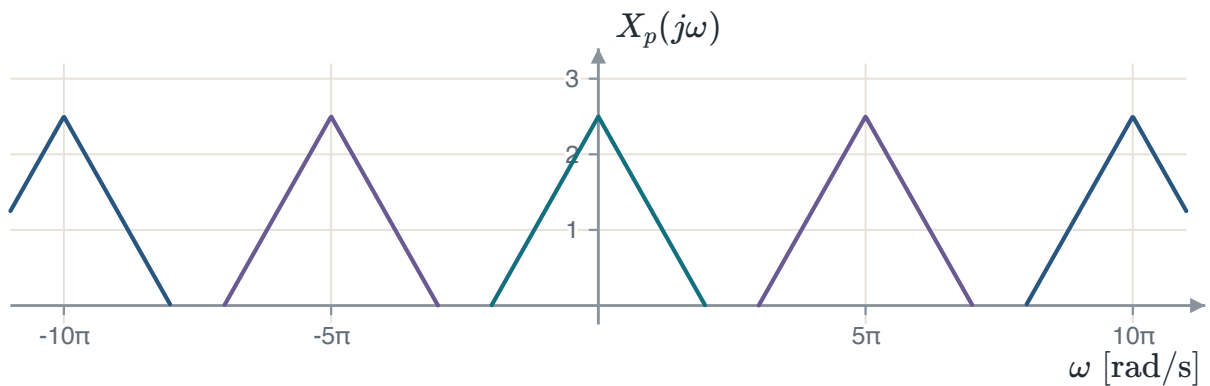
$$X_p(j\omega) = \frac{1}{2\pi} [X(j\omega) * P(j\omega)], \quad P(j\omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_s).$$

SPECTRUM OF THE SAMPLED SIGNAL

$$X_p(j\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(j(\omega - k\omega_s))$$

The $1/2\pi$ of the first step and the $2\pi/T$ of the second combine, and what is left in front is $1/T$. Every copy is $X(j\omega)$ scaled by $1/T$, not by 1. The reconstruction filter later carries gain T , exactly the inverse, so the two must be kept together.

Spacing T in time gives spacing $2\pi/T$ in frequency: the closer the samples, the further apart the copies, and the taller each one is.



The sampled spectrum of the signal above at $T = 0.4$ s, so $\omega_s = 5\pi$ rad/s. One copy per multiple of ω_s , each of height $1/T = 2.5$.

7.4 Replication and aliasing are not the same thing

The sum for $X_p(j\omega)$ runs over every integer k for every value of T . There is no rate at which the copies fail to appear and no rate at which they disappear. Two things, and only two, change with the rate: the copies sit ω_s apart, and each is scaled by $1/T$.

NAMING THE PIECES

The term $k = 0$ is $X(j\omega)/T$, the spectrum of the signal itself, sitting where it always sat. It is the **baseband** and it is not a copy of anything. The terms $k = \pm 1, \pm 2, \dots$ are the **copies**, centred at $\pm\omega_s, \pm 2\omega_s, \dots$. Calling the centre group the first replica shifts every later index by one.

The baseband occupies $|\omega| \leq \omega_M$, and the copy at $k = 1$ occupies $|\omega - \omega_s| \leq \omega_M$, so it begins at $\omega_s - \omega_M$. The gap between them is the quantity that decides everything.

GUARD BAND

$$(\omega_s - \omega_M) - \omega_M = \omega_s - 2\omega_M$$

Positive: the copies stand clear and a filter can cut the baseband out untouched. Zero: they touch. Negative: they have moved into one another and add.

REPLICATION IS UNCONDITIONAL

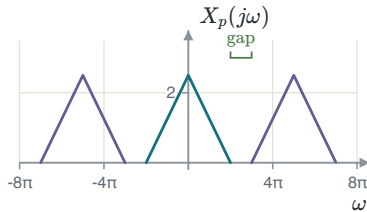
It is sometimes said that below the Nyquist rate the spectrum is no longer replicated. That is false. The formula for $X_p(j\omega)$ carries no condition, so the copies are there at every rate; what changes below the Nyquist rate is that they **overlap**.

Replication is what sampling does. Aliasing is the overlap of what sampling made.

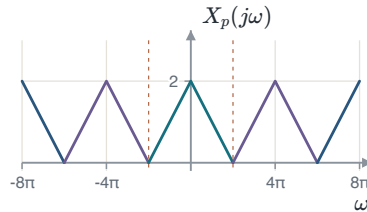
The overlap cannot be repaired afterwards. Where two copies meet, the sampler stores one number that is the sum of two contributions,

$$X_p(j\omega) = \frac{1}{T} \left[\underbrace{X(j\omega)}_{\text{wanted}} + \underbrace{X(j(\omega - \omega_s))}_{\text{intruder}} \right].$$

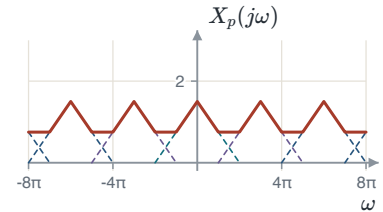
A filter multiplies $X_p(j\omega)$ frequency by frequency, including multiplying by zero. It cannot look at one value and report the two numbers that were added to make it. So the only cures are to raise the rate, or to remove the offending frequencies from $x(t)$ before the sampler.



$\omega_s = 5\pi$: a gap of π rad/s.



$\omega_s = 4\pi = 2\omega_M$: the gap is zero.



$\omega_s = 3\pi$: the copies add. The solid curve is the sum.

7.5 The sampling theorem

SAMPLING THEOREM

$$X(j\omega) = 0 \text{ for } |\omega| > \omega_M \text{ and } \omega_s > 2\omega_M$$

$$\implies x(t) \text{ is determined uniquely by } x(nT), n = 0, \pm 1, \pm 2, \dots$$

The signal is recovered by passing $x_p(t)$ through an ideal lowpass filter of gain T and cutoff ω_c with $\omega_M < \omega_c < \omega_s - \omega_M$. The quantity $2\omega_M$ is the **Nyquist rate** of the signal.

WHY THE INEQUALITY HAS TO BE STRICT

Put $\omega_s = 2\omega_M$ into the cutoff condition. It becomes $\omega_M < \omega_c < \omega_M$, an interval with nothing in it. There is no admissible cutoff at the Nyquist rate, so the theorem cannot be satisfied at its own boundary. Writing $\omega_s \geq 2\omega_M$ promises a filter that the same sentence then rules out. In practice the rate is chosen well above $2\omega_M$, so that the guard band is wide enough for a real filter, whose transition from pass to stop is not vertical.

EXAMPLE 7.1 – A SIGNAL THE NYQUIST RATE THROWS AWAY

GIVEN $x(t) = 1 + \cos(2000\pi t) + \sin(4000\pi t)$, sampled at exactly $\omega_s = 2\omega_M = 8000\pi$ rad/s.

FIND What happens to the term $\sin(4000\pi t)$.

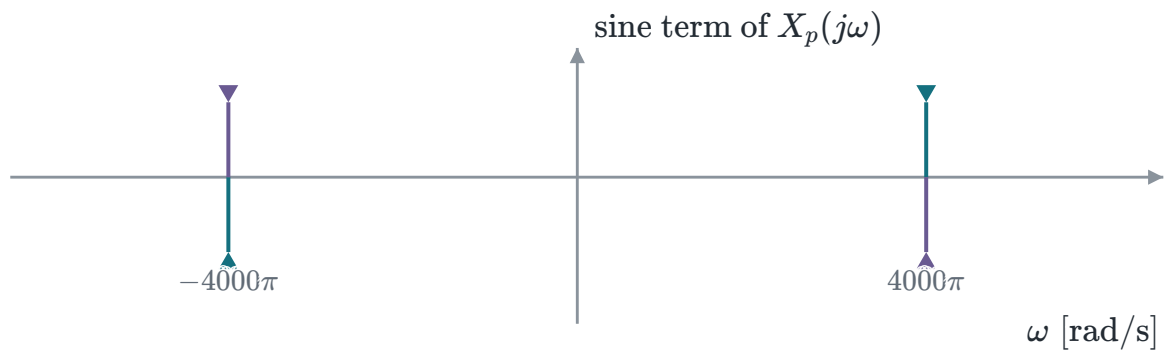
METHOD Locate the contributions to X_p at the single frequency $\omega = +4000\pi$, then check the answer in the time domain.

SOLUTION The bandwidth is $\omega_M = 4000\pi$ rad/s, so $T = 2\pi/\omega_s = 1/4000$ s = 0.25 ms. The sine transforms to $\frac{\pi}{j} [\delta(\omega - 4000\pi) - \delta(\omega + 4000\pi)]$. At $\omega = +4000\pi$ two terms of the sum contribute: the baseband gives $\frac{1}{T} \cdot \frac{\pi}{j} = +\frac{4000\pi}{j}$, and the $k = 1$ copy carries the impulse from -4000π up by ω_s and gives $-\frac{4000\pi}{j}$. Their sum is exactly zero, and the same happens at -4000π . In time,

$$\sin(4000\pi nT) = \sin(\pi n) = 0 \text{ for every integer } n.$$

CHECK Every sample of the sine term is taken at a zero crossing, so the sampler never sees it. The component is absent, not merely marginal.

REPAIR Add a guard band. With $\omega_g = 1000\pi$ rad/s the rate becomes $\omega_s = 9000\pi$ rad/s, so $T = 1/4500$ s, about 222.2 μ s, and the admissible cutoff interval $4000\pi < \omega_c < 5000\pi$ rad/s is no longer empty.



At $\omega = \pm 4000\pi$ the baseband arrow and the arrow the neighbouring copy brings there have the same length and point in opposite directions.

7.6 Worked examples on the rate

EXAMPLE 7.2 – ONE SIGNAL AT THREE RATES

GIVEN $x(t) = (\sin \pi t / \pi t)^2$, whose transform is a triangle of peak 1 reaching zero at $\pm 2\pi$. So $\omega_M = 2\pi$ rad/s and the Nyquist rate is 4π rad/s.

FIND For $T_1 = 0.40$ s, $T_2 = 0.50$ s and $T_3 = 2/3$ s: the rate, the guard band and the height of each copy.

METHOD $\omega_s = 2\pi/T$, guard band $\omega_s - 2\omega_M$, copy height $1/T$ times the original peak.

SOLUTION

$$T_1 = 0.40 : \quad \omega_s = 5\pi, \quad 5\pi > 4\pi, \quad \text{guard} = +\pi, \quad 1/T_1 = 2.5$$

$$T_2 = 0.50 : \quad \omega_s = 4\pi, \quad 4\pi = 4\pi, \quad \text{guard} = 0, \quad 1/T_2 = 2.0$$

$$T_3 = 2/3 : \quad \omega_s = 3\pi, \quad 3\pi < 4\pi, \quad \text{guard} = -\pi, \quad 1/T_3 = 1.5$$

CHECK $\omega_s T = 2\pi$ in each row: $5\pi \times 0.4 = 2\pi$, $4\pi \times 0.5 = 2\pi$, $3\pi \times 2/3 = 2\pi$. Only the first row is safe; the second sits on the boundary and the third overlaps by π rad/s on each side.

EXAMPLE 7.3 – A LINE SPECTRUM, AND ITS SAMPLING PERIOD

GIVEN $x(t) = 1 + \cos(2000\pi t) + \sin(4000\pi t)$.

FIND $X(j\omega)$, the bandwidth, and the sampling period at the Nyquist rate.

METHOD Transform each term separately, then convert the rate to a period and check it three ways.

SOLUTION

$$X(j\omega) = 2\pi\delta(\omega) + \pi\delta(\omega - 2000\pi) + \pi\delta(\omega + 2000\pi) + \frac{\pi}{j}\delta(\omega - 4000\pi) - \frac{\pi}{j}\delta(\omega + 4000\pi).$$

Five impulses, each written as a function of ω so that its position can be read. The furthest is at 4000π , so $\omega_M = 4000\pi$ rad/s, that is $f_M = 2000$ Hz, and the Nyquist rate is 8000π rad/s, that is 4000 Hz. Then

$$T = \frac{2\pi}{8000\pi} = \frac{1}{4000} \text{ s} = 2.5 \times 10^{-4} \text{ s} = 0.25 \text{ ms}.$$

CHECK Three independent tests. $\omega_s T = 8000\pi \times 2.5 \times 10^{-4} = 2\pi$. The copies are scaled by $1/T = 4000$. And doubling the rate must halve the period: at $\omega_s = 16000\pi$ the period is $125 \mu\text{s}$, and 0.25 ms divided by $125 \mu\text{s}$ is exactly 2.

WARNING The answer is a quarter of a millisecond. Cancelling the π but not the thousand gives 0.25 s, and every later number in the problem is then wrong by a factor of 1000. A quarter of a second between samples would be four samples per second for a signal carrying 2000 Hz.

EXAMPLE 7.4 – AREA, PEAK, AND THE HEIGHT OF A COPY

GIVEN $x(t) = (\sin(4000\pi t)/\pi t)^2$, the square of a signal whose transform is a rectangle of height 1 on $|\omega| \leq 4000\pi$.

FIND The peak of $X(j\omega)$, the bandwidth, the Nyquist period, and the height of one copy after sampling.

METHOD Squaring in time convolves in frequency and divides by 2π .

SOLUTION

$$A = [R * R](0) = \int_{-4000\pi}^{4000\pi} (1)(1) d\tau = 8000\pi, \quad X_{\max} = X(0) = \frac{A}{2\pi} = 4000.$$

The triangle reaches zero at twice the rectangle half-width, so $\omega_M = 8000\pi$ rad/s and $T = 2\pi/16000\pi = 1.25 \times 10^{-4}$ s, that is $125 \mu\text{s}$. The copy height is $X_{\max}/T = 4000 \times 8000 = 3.2 \times 10^7$.

CHECK This bandwidth is twice the one in Example 7.3, so this period is half of it. And $\omega_s T = 16000\pi \times 1.25 \times 10^{-4} = 2\pi$.

WARNING A is an area and $X_{\max}$ is the peak of a spectrum. They differ by 2π , so substituting one where the other belongs inflates every later height by that factor. Give the peak its own symbol.

7.7 Reconstruction and band-limited interpolation

If the copies stand clear, recovering $x(t)$ means keeping the one at the origin and discarding the rest.

IDEAL RECONSTRUCTION FILTER

$$H_r(j\omega) = \begin{cases} T, & |\omega| < \omega_c \\ 0, & |\omega| > \omega_c \end{cases} \quad \omega_M < \omega_c < \omega_s - \omega_M$$

$$X_r(j\omega) = T \cdot \frac{1}{T} X(j\omega) = X(j\omega) \implies x_r(t) = x(t)$$

The gain is T because the baseband is $X(j\omega)/T$. A filter of gain 1 returns a signal T times too small, and since T changes with the rate the error changes with it. The second line is true only when the copies do not overlap.

Filtering is convolution, and x_p is a train of impulses, so the output is a sum of shifted copies of the filter's impulse response. Each sample is replaced by a curve of its own height, and the curves are added.

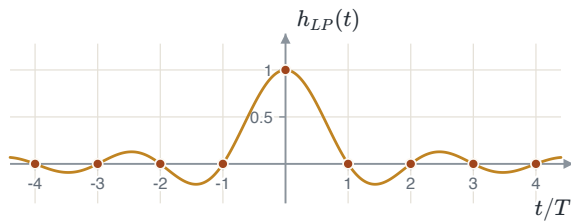
BAND-LIMITED INTERPOLATION

$$h_{LP}(t) = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} T e^{j\omega t} d\omega = \frac{T \sin(\omega_c t)}{\pi t}$$
$$x_r(t) = \sum_{n=-\infty}^{\infty} x(nT) \frac{T}{\pi} \cdot \frac{\sin(\omega_c(t - nT))}{t - nT}$$

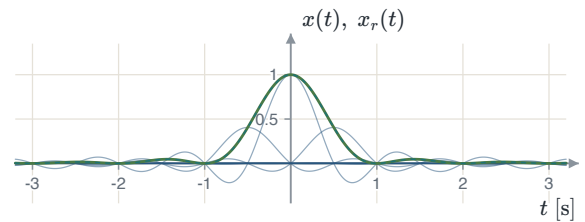
Throughout this course $\text{sinc}(\theta) = \sin \theta / \theta$, **unnormalised**. With the choice $\omega_c = \pi/T$ the kernel is $h_{LP}(t) = \sin(\pi t/T)/(\pi t/T) = \text{sinc}(\pi t/T)$. Anything written as $\text{sinc}(t/T)$ belongs to the other convention and is a different function.

THE FACTOR OF π INSIDE THE SINE IS LOAD-BEARING

The kernel must be 1 at $t = 0$ and exactly 0 at every other sample instant, so that each sample controls its own instant and no other. Drop the π from the numerator and the property is lost at once: with $T = 1$ the correct kernel at $t = 1$ is $\sin(\pi)/\pi = 0$, while $\sin(1)/\pi = 0.267849$. A kernel that does not vanish at the neighbouring instants is not an interpolation at all. Note also that $\omega_c = \pi/T$ is exactly $\omega_s/2$, the end of the admissible interval rather than a point inside it. It is a choice, and it should be stated as one.



The kernel with $\omega_c = \pi/T$: one at its own instant, zero at every other.



One kernel per sample, in grey, and their sum. It lands on the dashed original.

7.8 The zero-order and first-order holds

The ideal filter is not buildable: its impulse response starts before $t = 0$ and never ends. What hardware does instead is hold.

ZERO-ORDER HOLD

$$h_0(t) = \begin{cases} 1, & 0 \leq t < T \\ 0, & \text{otherwise} \end{cases}$$

$$H_0(j\omega) = \int_0^T e^{-j\omega t} dt = \frac{1 - e^{-j\omega T}}{j\omega} = e^{-j\omega T/2} \frac{2 \sin(\omega T/2)}{\omega}$$

The magnitude is $|2 \sin(\omega T/2)/\omega|$ and the phase is $-\omega T/2$, a delay of half a sampling period. At the origin $H_0(0) = T$, so the hold already supplies the gain the ideal filter needed.

Three differences from the ideal filter. The magnitude sags below T as ω grows, so the recovered signal is shaped as well as reconstructed. It is not zero beyond the cutoff, so parts of the neighbouring copies survive as the staircase edge. And it carries a delay. Making the hold exact needs the compensator

$$H_r(j\omega) = \frac{H(j\omega)}{H_0(j\omega)} = e^{j\omega T/2} H(j\omega) \frac{\omega}{2 \sin(\omega T/2)},$$

which has to advance in time and to grow without bound where $\sin(\omega T/2)$ vanishes. It is approximated in practice, not built.

FIRST-ORDER HOLD

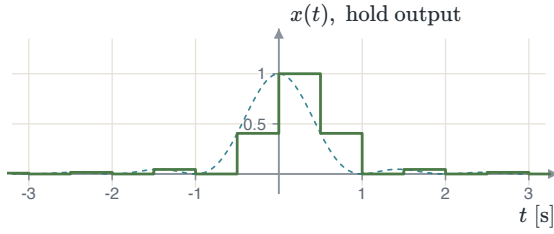
$$h_1(t) = \frac{1}{T} [g * g](t), \quad g(t) = \begin{cases} 1, & |t| \leq T/2 \\ 0, & \text{otherwise} \end{cases}$$

$$H_1(j\omega) = \frac{1}{T} G^2(j\omega) = \frac{1}{T} \left[\frac{\sin(\omega T/2)}{\omega/2} \right]^2$$

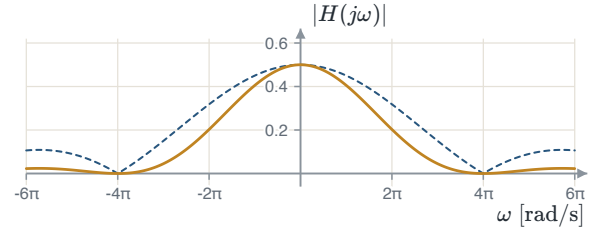
The convolution of the rectangle with itself is a triangle of peak T on $|t| \leq T$, so h_1 is a triangle of peak 1 on the same interval and the output joins consecutive samples by straight lines. Every one of these is a function of t ; the independent variable of all three sketches is time. H_1 is real and non-negative, $H_1(0) = T$, and it falls off as $1/\omega^2$ instead of $1/\omega$.

A HOLD IS NOT A RECONSTRUCTION

Both holds are filters with a real, non-flat, non-band-limited response. The staircase is a different signal from $x(t)$, and no argument about small steps makes them equal: raising the rate makes the error smaller but never zero at any finite rate. The sampling theorem guarantees that the samples *determine* $x(t)$ and names the filter that recovers it. Feeding the same samples to a different filter gives a different output.



The zero-order hold: each sample held for one period.



With $T = 0.5$ s: the first-order hold, solid, against the zero-order hold, dashed. Both start at T .

7.9 What comes back, and where from

Take $x(t) = \cos(\omega_0 t)$, so $\omega_M = \omega_0$, and reconstruct with the cutoff at $\omega_c = \omega_s/2$. Sampling puts a line at every $k\omega_s \pm \omega_0$, and the filter keeps whatever lies in $|\omega| < \omega_c$.

$$\begin{aligned} \omega_s = 6\omega_0 : \quad \omega_c = 3\omega_0, \quad \text{inside} : \pm\omega_0 &\Rightarrow x_r(t) = \cos(\omega_0 t) \\ \omega_s = 3\omega_0 : \quad \omega_c = 1.5\omega_0, \quad \text{inside} : \pm\omega_0 &\Rightarrow x_r(t) = \cos(\omega_0 t) \\ \omega_s = 1.5\omega_0 : \quad \omega_c = 0.75\omega_0, \quad \text{inside} : \pm 0.5\omega_0 &\Rightarrow x_r(t) = \cos(0.5\omega_0 t) \end{aligned}$$

THE ALIAS FREQUENCY

$$X_r(j\omega) = \pi\delta(\omega - (\omega_s - \omega_0)) + \pi\delta(\omega + (\omega_s - \omega_0)) \xrightarrow{\mathcal{F}^{-1}} x_r(t) = \cos((\omega_s - \omega_0)t)$$

In the third case the line at ω_0 lies outside the filter and is discarded, while the line the $k = 1$ copy contributes at $\omega_s - \omega_0$ lies inside and is kept. The signal takes the identity of a lower frequency, $|\omega_s - \omega_0|$. Each impulse is written as a function of ω ; an expression such as $\pi\delta(\omega_s - \omega_0)$ is a constant and locates nothing. The arrow is the inverse transform, going from a spectrum back to a signal.

EXAMPLE 7.5 – THREE PERIODS FOR ONE COSINE

GIVEN $x(t) = \cos(2\pi t)$, so $\omega_M = 2\pi$ rad/s and the Nyquist rate is 4π rad/s. Take $\omega_c = \omega_s/2$.

FIND Whether $T_1 = 1/4$ s, $T_2 = 1/3$ s and $T_3 = 2/3$ s recover the signal, and what is recovered when one does not.

METHOD Compute $\omega_s = 2\pi/T$, compare with 4π , then list the line positions and keep those inside the cutoff.

SOLUTION

$$T_1 = 1/4 : \quad \omega_s = 8\pi > 4\pi, \quad \omega_c = 4\pi, \quad x_r(t) = \cos(2\pi t)$$

$$T_2 = 1/3 : \quad \omega_s = 6\pi > 4\pi, \quad \omega_c = 3\pi, \quad x_r(t) = \cos(2\pi t)$$

$$T_3 = 2/3 : \quad \omega_s = 3\pi < 4\pi, \quad \omega_c = 1.5\pi, \quad x_r(t) = \cos(\pi t)$$

CHECK $\cos(2\pi n T_3) = \cos(4\pi n/3)$ and $\cos(\pi n T_3) = \cos(2\pi n/3)$. Since $4\pi n/3 = 2\pi n - 2\pi n/3$ and the cosine is even, the two sample sequences are identical, so no reconstruction could prefer one over the other.

WARNING The whole verdict rests on the number the rate is compared with, and for this signal that number is $2\omega_M = 4\pi$. Comparing against a neighbouring problem's bandwidth of 6π makes the middle row read $6\pi \geq 6\pi$, an equality, so a comfortable margin is turned into the boundary case the theorem excludes.

EXAMPLE 7.6 – TWO COMPONENTS, ONE RATE

GIVEN $x(t) = \cos(\pi t) + \cos(3\pi t)$, sampled with $T = 2/5$ s.

FIND $x_r(t)$, with $\omega_c = \omega_s/2$.

METHOD Test the rate against this signal's own Nyquist rate, then take the components one at a time.

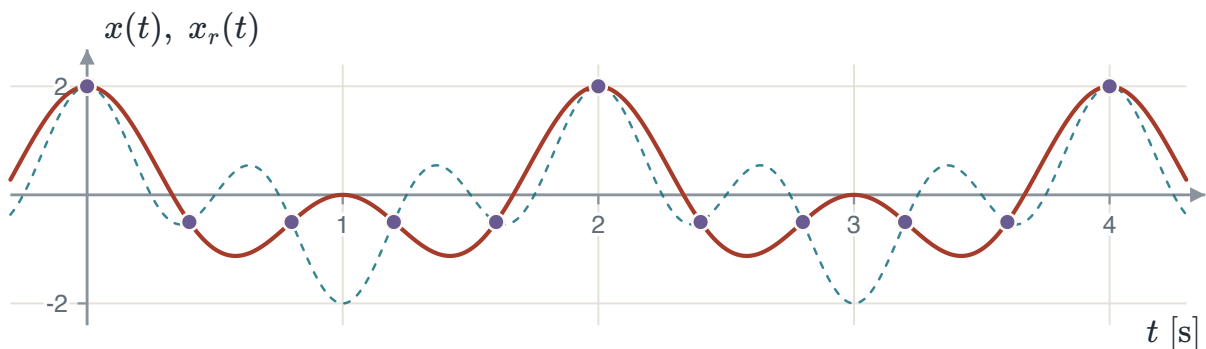
SOLUTION $\omega_M = 3\pi$, so the Nyquist rate is 6π rad/s, while $\omega_s = 2\pi/T = 5\pi$ rad/s and $\omega_c = 2.5\pi$ rad/s. Since $5\pi < 6\pi$, aliasing is expected. From the component at π : the baseband line at π is inside and survives, the copy line at 4π is outside. From the component at 3π : the baseband line is outside, the copy line at $5\pi - 3\pi = 2\pi$ is inside. Hence

$$X_r(j\omega) = \pi[\delta(\omega - \pi) + \delta(\omega + \pi)] + \pi[\delta(\omega - 2\pi) + \delta(\omega + 2\pi)]$$

and $x_r(t) = \cos(\pi t) + \cos(2\pi t)$.

CHECK $\cos(3\pi nT) = \cos(6\pi n/5)$ and $\cos(2\pi nT) = \cos(4\pi n/5)$, and $6\pi n/5 = 2\pi n - 4\pi n/5$, so the two sequences agree.

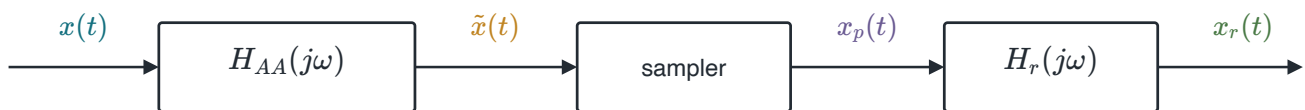
READING Aliasing is not all-or-nothing. One component came through untouched and the other left at 3π and returned at 2π , a frequency the signal never contained.



Both curves pass through every sample. The solid one is what the filter returns.

7.10 The anti-aliasing filter

A real signal is not band-limited, and a sampler at a fixed rate cannot be helped after the fact. The repair is a lowpass filter placed **before** the sampler, which removes everything above $\omega_s/2$ while it can still be removed cleanly.



The chain in the only order that works. The sampler output is $x_p(t)$; the name $x_r(t)$ belongs to the output of the reconstruction filter and to nothing before it.

THE ORDER OF THE STAGES IS THE WHOLE POINT

Put the same filter after the sampler and it does nothing useful: by then the high frequencies have folded into the band and been added to the wanted content, and a filter cannot separate two numbers that were summed before it saw them. Anti-aliasing filtering is only anti-aliasing while it is upstream of the sampler.

The comparison is quantitative. Take $x(t) = \cos(\pi t) + \cos(3\pi t)$ at $T = 2/5$ s, as in Example 7.6. Without the filter, $x_r(t) = \cos(\pi t) + \cos(2\pi t)$ and the error is $e(t) = \cos(3\pi t) - \cos(2\pi t)$. With a lowpass at $\omega_c = 2.5\pi$ ahead of the sampler, the component at 3π is removed first, what is sampled is $\cos(\pi t)$ alone, it is recovered exactly, and the error is $e(t) = \cos(3\pi t)$.

$$\overline{e^2}\Big|_{\text{no filter}} = \frac{1}{2} + \frac{1}{2} = 1, \quad \overline{e^2}\Big|_{\text{filtered}} = \frac{1}{2}$$

The two cosines in the unfiltered error sit at different harmonics of π rad/s, so their cross term averages to zero and the powers add.

Filtering first halves the error power. It costs the component at 3π , which was lost either way; what it buys is the removal of an impostor at 2π , a frequency the signal never contained. Losing content is a known, bounded error. Adding content that was never there is not, because nothing downstream can tell it apart from the signal.

NO FINITE-DURATION SIGNAL IS BAND-LIMITED

A rectangular pulse of width $2T_1$ has transform $2 \sin(\omega T_1)/\omega$, which crosses zero again and again but is never zero on a whole interval. There is no ω_M beyond which it vanishes, so no rate makes the copies disjoint and some aliasing is present at every rate. What is done in practice is to impose the band limit with a filter, and then to reconstruct the *filtered* signal exactly. The difference from the original is a design decision with a known size, not an accident.

7.11 Aliasing in time and in space

A camera is a sampler with $T = 1/f_s$, and a strobe lamp is the same device built from light. Nothing about the rule changes.

EXAMPLE 7.7 – A WHEEL THAT TURNS BACKWARDS

GIVEN A marked spoke turns at 9 revolutions per second. A camera records 10 frames per second.

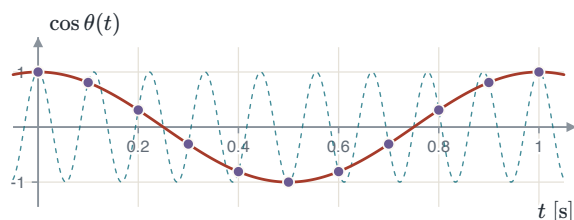
FIND The rotation the recording appears to show.

METHOD The angle is a signal at $\omega_0 = 18\pi$ rad/s and the camera samples it at $\omega_s = 20\pi$ rad/s.

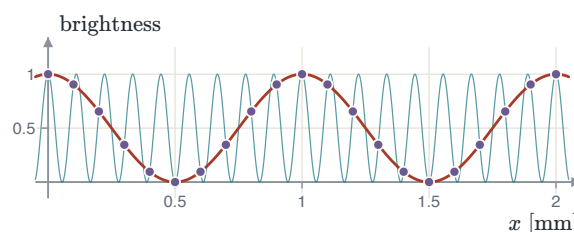
SOLUTION $\omega_s = 20\pi < 2\omega_0 = 36\pi$, so the rate is below the Nyquist rate and the surviving line sits at $|\omega_s - \omega_0| = 2\pi$ rad/s, that is 1 revolution per second. The surviving contribution comes from the copy of the negative-frequency component, so the apparent rotation is in the opposite direction.

CHECK $\cos(18\pi n/10) = \cos(2\pi n - 2\pi n/10) = \cos(2\pi n/10)$: the frames are exactly those a 1 Hz rotation would produce. Between frames the spoke advances 9/10 of a turn, which the eye reads as 1/10 of a turn backwards. At 20 frames per second, $40\pi > 36\pi$ and the recording is correct.

Position takes the place of time without any change to the arithmetic. A striped pattern of 9 cycles per millimetre recorded on a grid of 10 sample points per millimetre satisfies $2 \times 9 > 10$, so it is recorded as a pattern of $|10 - 9| = 1$ cycle per millimetre. The broad bands the grid produces are smooth, regular and convincing, and nothing in the samples marks them as false. Two regular patterns laid over one another give the same effect without any sampler: stripes at 1.0 mm and at 1.1 mm have spatial frequencies 1.000 and 0.909 cycles per millimetre, so the pattern they make together repeats every 11 mm.



Nine revolutions per second, dashed, ten frames per second as dots, and the one revolution per second the frames describe.



The same arithmetic in space: nine cycles per millimetre recorded as one.

7.12 Summary

1. Find ω_M , the highest angular frequency the signal carries. If there is none, the signal is not band-limited and must be filtered before anything else.
2. Choose ω_s strictly above $2\omega_M$, leaving a guard band $\omega_s - 2\omega_M$ wide enough for a real filter.
3. Compute $T = 2\pi/\omega_s$ and check $\omega_s T = 2\pi$. Quote $f_s = 1/T$ in hertz only when hertz was asked for.
4. Draw the copies at every multiple of ω_s , each $1/T$ times the original height, and mark the baseband $k = 0$.
5. Place the reconstruction filter: gain T , cutoff strictly inside $\omega_M < \omega_c < \omega_s - \omega_M$.

6. If the copies overlap, say which line came from which copy and give the alias frequency $|\omega_s - \omega_0|$ for each component that moved.

THE FOUR TRAPS, IN ONE PLACE

Saying the copies disappear below the Nyquist rate — they never do, they overlap. Treating $\omega_s = 2\omega_M$ as safe — the admissible cutoff interval is empty there. Reading $2\pi/T$ in hertz — that is rad/s, and the error is a factor of 2π . Calling a hold output the reconstructed signal — a hold is an approximation with a measurable error.

A continuous signal and a sequence of numbers are now the same object, provided one strict inequality holds. That equivalence is what lets every result of the earlier chapters be carried out on a machine.

7.13 Exercises

7.1 A signal is sampled at $\omega_s = 8000\pi$ rad/s. Give T and f_s , and verify the period with a unit check.

Answer: $T = 2\pi/\omega_s = 1/4000$ s = 0.25 ms and $f_s = 1/T = 4000$ Hz. Check: $\omega_s T = 8000\pi \times 2.5 \times 10^{-4} = 2\pi$. Note that 0.25 s fails the same check by a factor of 1000.

7.2 State in one sentence each what sampling does at every rate, and what happens only below the Nyquist rate.

Answer: At every rate sampling puts a copy of $X(j\omega)$ at every multiple of ω_s , scaled by $1/T$. Below the Nyquist rate those copies overlap and add, and that overlap is aliasing.

7.3 Show that the sampling theorem cannot be satisfied at $\omega_s = 2\omega_M$, and give a signal that demonstrates it.

Answer: The cutoff condition $\omega_M < \omega_c < \omega_s - \omega_M$ becomes $\omega_M < \omega_c < \omega_M$, which is empty. For $x(t) = \sin(\omega_M t)$ sampled at $\omega_s = 2\omega_M$ every sample is $\sin(\pi n) = 0$: the baseband and the $k = 1$ copy contribute equal and opposite weights at $\pm\omega_M$.

7.4 A triangular spectrum has peak $X_{\max} = 4000$ and is sampled with $T = 125$ μ s. Find the height of one copy, and say why the area $A = 8000\pi$ is not the number to use.

Answer: $X_{\max}/T = 4000 \times 8000 = 3.2 \times 10^7$. The area of the convolution and the peak of the spectrum differ by 2π , since squaring in time convolves in frequency and divides by 2π ; using A inflates every height by that factor.

7.5 $x(t) = \cos(\pi t) + \cos(3\pi t)$ is sampled with $T = 2/5$ s and reconstructed with $\omega_c = \omega_s/2$. Find $x_r(t)$ and name the component that moved.

Answer: $\omega_s = 5\pi < 6\pi$, so aliasing occurs. Surviving lines: π from the baseband and $5\pi - 3\pi = 2\pi$ from the $k = 1$ copy, giving $x_r(t) = \cos(\pi t) + \cos(2\pi t)$. The component at 3π moved to 2π ; the one at π passed untouched.

7.6 Explain why the anti-aliasing filter must precede the sampler, and state what the output of the whole chain is equal to.

Answer: After the sampler the intruding content has been added to the wanted content at the same frequency, and a filter cannot recover two numbers from their sum. Placed first, the filter removes that content while it is still at its own frequency. The output then equals the filtered signal exactly, not the original, and the difference is set by the chosen cutoff.

APPENDIX A

Summary of formulas

A.1 Energy and power

QUANTITY	CONTINUOUS TIME	DISCRETE TIME
Total energy	$E_\infty = \int_{-\infty}^{\infty} x(t) ^2 dt$	$E_\infty = \sum_{n=-\infty}^{\infty} x[n] ^2$
Average power	$P_\infty = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) ^2 dt$	$P_\infty = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x[n] ^2$
Energy signal	$E_\infty < \infty, P_\infty = 0$	same
Power signal	$E_\infty \rightarrow \infty, 0 < P_\infty < \infty$	same

A.2 Operations and periodicity

ITEM	STATEMENT
Shift	$x(t - t_0)$: delay if $t_0 > 0$, advance if $t_0 < 0$
Reversal	$x(-t), x[-n]$
Scaling	$y(t) = x(at)$: compressed if $a > 1$, stretched if $0 < a < 1$; support divided by a
Combination	$x(at - b)$: shift by b first, then scale by a
Periodicity	$x(t) = x(t + T); x[n] = x[n + N]$ with N an integer
Fundamental frequency	$\omega_0 = 2\pi/T_0 = 2\pi/N_0$
Even and odd	$\mathcal{E}\mathbf{v}\{x\} = \frac{1}{2}[x(t) + x(-t)], \mathcal{O}\mathbf{d}\{x\} = \frac{1}{2}[x(t) - x(-t)]$

A.3 Impulses and steps

ITEM	CONTINUOUS TIME	DISCRETE TIME
Step and impulse	$\delta(t) = \frac{d}{dt}u(t), u(t) = \int_{-\infty}^t \delta(\tau) d\tau$	$\delta[n] = u[n] - u[n - 1], u[n] = \sum_{k=0}^{\infty} \delta[n - k]$
Sampling	$x(t)\delta(t - t_0) = x(t_0)\delta(t - t_0)$	$x[n]\delta[n - n_0] = x[n_0]\delta[n - n_0]$
Sifting	$x(t_0) = \int_{-\infty}^{\infty} x(t)\delta(t - t_0) dt$	$x[n_0] = \sum_n x[n]\delta[n - n_0]$
Representation	$x(t) = \int x(\tau)\delta(t - \tau) d\tau$	$x[n] = \sum_k x[k]\delta[n - k]$

A.4 Complex exponentials

ITEM	STATEMENT
Continuous time	$x(t) = Ce^{at}$; with $C = Ae^{j\theta}$ and $a = r + j\omega_0$: $x(t) = Ae^{rt}e^{j(\omega_0 t + \theta)}$
Period	$T_0 = 2\pi/\omega_0$, always periodic for $\omega_0 \neq 0$
Discrete time	$x[n] = C\alpha^n$ with $\alpha = e^{\beta}$; growth boundary at $ \alpha = 1$
Period	$N = \frac{2\pi}{\omega_0}k$; periodic only if $\omega_0/2\pi \in \mathbb{Q}$
Frequency wrap-around	$e^{j(\omega_0 + 2\pi)n} = e^{j\omega_0 n}$

A.5 Systems and convolution

PROPERTY	GENERAL CRITERION	LTI CRITERION IN TERMS OF h
Memoryless	output at t uses only input at t	$h(t) = a\delta(t)$ or $h[n] = a\delta[n]$
Invertible	distinct inputs give distinct outputs	$h * g = \delta$ for some g
Causal	output uses only $\tau \leq t$	$h(t) = 0$ for $t < 0$; $h[n] = 0$ for $n < 0$
BIBO stable	bounded input gives bounded output	$\int h dt < \infty; \sum_k h[k] < \infty$
Time invariant	$x(t - t_0) \rightarrow y(t - t_0)$	—
Linear	$ax_1 + bx_2 \rightarrow ay_1 + by_2$	—

CONVOLUTION

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n - k]$$

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau) d\tau$$

Commutative, distributive and associative. Supports add. Total areas or total sums multiply.

A.7 Fourier series properties

A periodic signal of period T_0 with $\omega_0 = 2\pi/T_0$, or a periodic sequence of period N with $\omega_0 = 2\pi/N$. In both columns $x \leftrightarrow a_k$ and $y \leftrightarrow b_k$.

PROPERTY	CONTINUOUS TIME	DISCRETE TIME
Linearity	$Ax(t) + By(t) \leftrightarrow Aa_k + Bb_k$	$Ax[n] + By[n] \leftrightarrow Aa_k + Bb_k$
Time shift	$x(t - t_0) \leftrightarrow a_k e^{-jk\omega_0 t_0}$	$x[n - n_0] \leftrightarrow a_k e^{-jk(2\pi/N)n_0}$
Frequency shift	$e^{jM\omega_0 t} x(t) \leftrightarrow a_{k-M}$	$e^{jM(2\pi/N)n} x[n] \leftrightarrow a_{k-M}$
Conjugation	$x^*(t) \leftrightarrow a_{-k}^*$	$x^*[n] \leftrightarrow a_{-k}^*$
Time reversal	$x(-t) \leftrightarrow a_{-k}$	$x[-n] \leftrightarrow a_{-k}$
Time scaling	$x(\alpha t), \alpha > 0: a_k, \text{ period } T_0/\alpha$	$x_{(m)}[n]: \frac{1}{m}a_k, \text{ period } mN$
Periodic convolution	$\int_{T_0} x(\tau)y(t - \tau)d\tau \leftrightarrow T_0 a_k b_k$	$\sum_{r=\langle N \rangle} x[r]y[n - r] \leftrightarrow N a_k b_k$
Multiplication	$x(t)y(t) \leftrightarrow \sum_{\ell=-\infty}^{\infty} a_\ell b_{k-\ell}$	$x[n]y[n] \leftrightarrow \sum_{\ell=\langle N \rangle} a_\ell b_{k-\ell}$
Derivative or difference	$dx/dt \leftrightarrow jk\omega_0 a_k$	$x[n] - x[n - 1] \leftrightarrow (1 - e^{-jk(2\pi/N)})a_k$
Integral or running sum	$\int_{-\infty}^t x d\tau \leftrightarrow \frac{a_k}{jk\omega_0}, \text{ only if } a_0 = 0$	$\sum_{r=-\infty}^n x[r] \leftrightarrow \frac{a_k}{1 - e^{-jk(2\pi/N)}}, \text{ only if } a_0 = 0$
Real signal	$a_k = a_{-k}^*; a_k \text{ even, } \angle a_k \text{ odd}$	same
Real and even	a_k real and even	same
Real and odd	a_k purely imaginary and odd	same
Even-odd parts	$\mathcal{E}\{x\} \leftrightarrow \text{Re}\{a_k\}, \mathcal{O}\{x\} \leftrightarrow j \text{Im}\{a_k\}$	same
Parseval	$\frac{1}{T_0} \int_{T_0} x ^2 dt = \sum_k a_k ^2$	$\frac{1}{N} \sum_{n=\langle N \rangle} x ^2 = \sum_{k=\langle N \rangle} a_k ^2$

Three rows differ between the columns, and all three follow from one fact: discrete-time coefficients repeat, $a_k = a_{k+N}$.

Multiplication therefore sums over one period rather than over all integers; scaling carries a factor $\frac{1}{m}$, because the average is taken over m times as many samples; and a sequence has a first difference where a signal has a derivative. The two conditional rows fail at $k = 0$ for the same reason in both columns: a non-zero mean integrates or accumulates without bound.

A.8 Continuous-time Fourier transform

PROPERTY	STATEMENT
Linearity	$ax_1(t) + bx_2(t) \leftrightarrow aX_1(j\omega) + bX_2(j\omega)$
Time shift	$x(t - t_0) \leftrightarrow e^{-j\omega t_0} X(j\omega)$
Frequency shift	$e^{j\omega_0 t} x(t) \leftrightarrow X(j(\omega - \omega_0))$
Conjugation	$x^*(t) \leftrightarrow X^*(-j\omega)$
Time reversal	$x(-t) \leftrightarrow X(-j\omega)$
Time and frequency scaling	$x(at) \leftrightarrow \frac{1}{ a } X(j\omega/a)$
Convolution	$x(t) * h(t) \leftrightarrow X(j\omega)H(j\omega)$
Multiplication	$x(t)y(t) \leftrightarrow \frac{1}{2\pi} X(j\omega) * Y(j\omega)$
Differentiation in time	$\frac{d^n x}{dt^n} \leftrightarrow (j\omega)^n X(j\omega)$
Integration	$\int_{-\infty}^t x(\tau) d\tau \leftrightarrow \frac{1}{j\omega} X(j\omega) + \pi X(0)\delta(\omega)$
Differentiation in frequency	$t x(t) \leftrightarrow j \frac{dX(j\omega)}{d\omega}$
Duality	$X(t) \leftrightarrow 2\pi x(-\omega)$
Real signal	$X(-j\omega) = X^*(j\omega); X \text{ even, } \angle X \text{ odd}$
Real and even	$X(j\omega) \text{ real and even}$
Real and odd	$X(j\omega) \text{ purely imaginary and odd}$
Even-odd parts	$\mathcal{E}\{x\} \leftrightarrow \text{Re}\{X\}, \mathcal{O}\{x\} \leftrightarrow j \text{Im}\{X\}$
Parseval	$\int_{-\infty}^{\infty} x(t) ^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) ^2 d\omega$
SIGNAL	TRANSFORM
$\delta(t)$	1
$\delta(t - t_0)$	$e^{-j\omega t_0}$
$u(t)$	$\frac{1}{j\omega} + \pi\delta(\omega)$
$e^{-at}u(t), a > 0$	$\frac{1}{a + j\omega}$
$te^{-at}u(t), a > 0$	$\frac{1}{(a + j\omega)^2}$
$\frac{t^{n-1}}{(n-1)!} e^{-at}u(t), a > 0$	$\frac{1}{(a + j\omega)^n}$
$e^{-a t }, a > 0$	$\frac{2a}{a^2 + \omega^2}$
1 on $ t < T_1$, 0 beyond	$\frac{2 \sin(\omega T_1)}{\omega} = 2T_1 \text{sinc}(\omega T_1)$
$\frac{\sin(Wt)}{\pi t} = \frac{W}{\pi} \text{sinc}(Wt)$	1 on $ \omega < W$, 0 beyond
1	$2\pi\delta(\omega)$
$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$
$\cos \omega_0 t$	$\pi\delta(\omega - \omega_0) + \pi\delta(\omega + \omega_0)$
$\sin \omega_0 t$	$\frac{\pi}{j}\delta(\omega - \omega_0) - \frac{\pi}{j}\delta(\omega + \omega_0)$
$\sum_k a_k e^{jk\omega_0 t}$	$\sum_k 2\pi a_k \delta(\omega - k\omega_0)$
Square wave: 1 on $ t < T_1$, 0 on $T_1 < t \leq T/2$	$\sum_k \frac{2 \sin(k\omega_0 T_1)}{k} \delta(\omega - k\omega_0)$
$\sum_k \delta(t - kT)$	$\frac{2\pi}{T} \sum_k \delta(\omega - \frac{2\pi k}{T})$

Every sinc here is the unnormalised one, $\text{sinc}(\theta) = \sin \theta / \theta$. A table written in the normalised convention divides the argument by π , so the two look different while saying the same thing, and copying an argument between them loses a factor of π . The exponential pairs hold only for $a > 0$.

A.9 Discrete-time Fourier transform

PROPERTY	STATEMENT
Linearity	$ax_1[n] + bx_2[n] \leftrightarrow aX_1(e^{j\omega}) + bX_2(e^{j\omega})$
Time shift	$x[n - n_0] \leftrightarrow e^{-j\omega n_0} X(e^{j\omega})$
Frequency shift	$e^{j\omega_0 n} x[n] \leftrightarrow X(e^{j(\omega - \omega_0)})$
Conjugation	$x^*[n] \leftrightarrow X^*(e^{-j\omega})$
Time reversal	$x[-n] \leftrightarrow X(e^{-j\omega})$
Time expansion	$x_{(k)}[n] \leftrightarrow X(e^{jk\omega})$
Convolution	$x[n] * h[n] \leftrightarrow X(e^{j\omega})H(e^{j\omega})$
Multiplication	$x[n]y[n] \leftrightarrow \frac{1}{2\pi} \int_{2\pi} X(e^{j\theta})Y(e^{j(\omega - \theta)})d\theta$
Differencing in time	$x[n] - x[n - 1] \leftrightarrow (1 - e^{-j\omega})X(e^{j\omega})$
Accumulation	$\sum_{m=-\infty}^n x[m] \leftrightarrow \frac{X(e^{j\omega})}{1 - e^{-j\omega}} + \pi X(e^{j0}) \sum_k \delta(\omega - 2\pi k)$
Differentiation in frequency	$n x[n] \leftrightarrow j dX(e^{j\omega})/d\omega$
Real sequence	$X(e^{-j\omega}) = X^*(e^{j\omega}); X \text{ even, } \angle X \text{ odd}$
Real and even	$X(e^{j\omega}) \text{ real and even}$
Real and odd	$X(e^{j\omega}) \text{ purely imaginary and odd}$
Even-odd parts	$\mathcal{E}\{x\} \leftrightarrow \text{Re}\{X\}, \mathcal{O}\{x\} \leftrightarrow j \text{Im}\{X\}$
Parseval	$\sum_n x[n] ^2 = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) ^2 d\omega$
SEQUENCE	TRANSFORM
$\delta[n]$	1
$\delta[n - n_0]$	$e^{-j\omega n_0}$
$u[n]$	$\frac{1}{1 - e^{-j\omega}} + \pi \sum_k \delta(\omega - 2\pi k)$
$a^n u[n], a < 1$	$\frac{1}{1 - ae^{-j\omega}}$
$(n + 1)a^n u[n], a < 1$	$\frac{1}{(1 - ae^{-j\omega})^2}$
$\frac{(n+r-1)!}{n!(r-1)!} a^n u[n], a < 1$	$\frac{1}{(1 - ae^{-j\omega})^r}$
$a^{ n }, a < 1$	$\frac{1 - a^2}{1 - 2a \cos \omega + a^2}$
1 on $ n \leq N_1$, 0 beyond	$\frac{\sin(\omega(N_1 + \frac{1}{2}))}{\sin(\omega/2)}$
$\frac{\sin(Wn)}{\pi n}, 0 < W < \pi$	1 on $ \omega \leq W$, 0 on $W < \omega \leq \pi$
1	$2\pi \sum_k \delta(\omega - 2\pi k)$
$e^{j\omega_0 n}$	$2\pi \sum_k \delta(\omega - \omega_0 - 2\pi k)$
$\cos \omega_0 n$	$\pi \sum_k [\delta(\omega - \omega_0 - 2\pi k) + \delta(\omega + \omega_0 - 2\pi k)]$
$\sin \omega_0 n$	$\frac{\pi}{j} \sum_k [\delta(\omega - \omega_0 - 2\pi k) - \delta(\omega + \omega_0 - 2\pi k)]$
$\sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n}$	$2\pi \sum_k a_k \delta(\omega - \frac{2\pi k}{N})$
$\sum_k \delta[n - kN]$	$\frac{2\pi}{N} \sum_k \delta(\omega - \frac{2\pi k}{N})$

Every transform in this section is periodic in ω with period 2π , because time is discrete. A periodic sequence has a discrete set of coefficients and an aperiodic one a continuous spectrum: discreteness in one domain is periodicity in the other, and that single sentence organises both tables.

A.10 Symbols

SYMBOL	MEANING
$x(t), x[n]$	input signal, continuous and discrete time
$y(t), y[n]$	output signal
$h(t), h[n]$	impulse response of a linear time-invariant system
$\delta(t), \delta[n]$	unit impulse
$u(t), u[n]$	unit step
E_∞, P_∞	total energy and average power over all time
T_0, N_0	fundamental period
ω_0	fundamental angular frequency, rad/s or rad/sample
$*$	convolution
j	imaginary unit, $j^2 = -1$